

ATC 5301 v02.03

This is a connected intersection-ready Advanced Transportation Controller (ATC) Standard family developed by AASHTO, ITE, and NEMA

November 9, 2022

Advanced Transportation Controller (ATC) Cabinet Standard v02.03

(Working Group Draft)



Supported/Sponsored By: The United States Department of Transportation (USDOT)



U.S. Department
of Transportation

Foreword

The Advanced Transportation Controller Cabinet Standard is one of the four standards within the ATC family of standards.

More information on this standard effort can be found on the [ITE website](#).

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In addition to the SDOS, Co-Chairs, WG Members and SMEs, there were many others that contributed to the development of this standard and their input and assistance was critical to the final product. The following list includes those volunteers and others who gave their time to help both the consultant and the committee ensure that the resulting standard met their needs. The following is a more complete list of those who volunteered their time and travel to contribute to the input and review during the development of this standard: [The Standards Discussion Groups on ITE Community \(ATC Community\)](#).

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1 PURPOSE OF THE DOCUMENT

This document is a Standard for the Intelligent Transportation System (ITS) Cabinet Version 2 (v2) project under the United States Department of Transportation (USDOT) Work Order T-11001 (ITS Cabinet v2 Maintenance) for development of the ATC 5301 Advanced Transportation Controller Cabinet (ATCC) standard. The Standard is based on the Concept of Operations (ConOps) adapted from Section 8.4.5 of the "Systems Engineering Guidebook for ITS" (see Section 3 Referenced Documents) with system requirements derived from the user needs identified in the ConOps as a step in the process to create the ATC 5301 ATCC standard. ConOps development began with User Needs presented by the following End Users at the 2008 Workshop in Austin TX:

- Caltrans
- City of Houston TX
- Georgia DOT
- Harris County TX
- Kentucky DOT
- Los Angeles DOT
- Minnesota DOT
- New York City DOT
- San Francisco MTA

This workshop identified significant additions to User Needs since V1 was published, including the following:

- USDOE ban on the sale of incandescent bulbs used in traffic signal heads
- EPA mercury waste regulations affecting high-capacity mercury relays
- State legislation to eliminate shock hazard inside electrical cabinets
- State legislation to eliminate shock hazard of field wires
- State legislation to eliminate arc flash hazard under NFPA 70E
- Need to replace internal components while in flash for motorist safety
- Low power features to facilitate alternate power sources such as solar panels

This Standard provides the detailed description of the elements, interfaces, and communications protocols of the ATC 5301 Advanced Transportation Controller Cabinet standard, traceable to the Standards Requirements Specification for the following:

- a) The USDOT Joint Program Office (JPO) who is sponsoring the work
- b) The Standard Development Organizations (SDOs) overseeing the development
- c) The consultants, manufacturers, and public transportation professionals who participate in the committees and working groups (WGs)

2 SCOPE OF PROJECT

2.1 Organization and Development Process

The ITS Cabinet V2 project is sponsored by the USDOT JPO as part of an ITS Standards Development Program. The project is performed under the direction of the Advanced Transportation Controller (ATC) Joint Committee (JC). The ATC JC is made up of representatives from three SDOs: the American Association of State Highway and Transportation Officials (AASHTO), the Institute of Transportation Engineers (ITE) and

the National Electrical Manufacturers Association (NEMA). The development is carried out by the ITS Cabinet WG, a technical subcommittee of the ATC JC, and a paid consultant team.

The ITS Cabinet Standard Version 1.02.17b (also known as Version 1 or V1, (see RD[1]) was published in 2006. It defined a transportation field cabinet system (TFCS) that was highly modular and expandable but the standard was missing formalized user needs and requirements that come from the rigor of a Systems Engineering Process (SEP). The objectives of the ITS Cabinet V2 project are as follows:

- 1) Develop ATC 5301 Advanced Transportation Controller Cabinet standard by assessing issues and integrating lessons learned from current deployments of the ITS Cabinet Standard V1 into a Concept of Operation, Requirements, and Design. User needs to be considered, but are not limited to the following: low-power features, items referred to as "B-List" items by the ITS Cabinet WG, and mercury relay replacement. These items along with all others solicited were introduced into the Systems Engineering Process (see objective #2) that examines their relevancy.
- 2) Use a systems engineering process to ensure the completeness and correctness of ATC 5301 Advanced Transportation Controller Cabinet standard and associated documents. The standard must be traceable and logically consistent.
- 3) Develop a detailed conformance statement that addresses backwards compatibility and provides clear and unambiguous instruction on how to extend the standard.

2.2 User Needs

ITS Standard Specification for Roadside Cabinets v01.02.17b (known as V1) was published on 9/26/2006 after successful ballot by AASHTO, ITE, and NEMA. Adopted standards are either reaffirmed or revisited after five years. In this case, V1 was revisited due to significant additions to User Needs including the following:

- USDOE ban on the sale of incandescent bulbs
- EPA mercury waste regulations
- State legislation to eliminate shock hazard inside electrical cabinets
- State legislation to eliminate shock hazard of field wires
- State legislation to eliminate arc flash hazard under NFPA 70E
- Need to replace internal subassemblies while in flash for worker safety
- Low power features to facilitate alternate power sources such as solar

2.3 Project Timeline

Beginning in 2007, a Systems Engineering Process (SEP) is underway to develop the ITS Cabinet Version 2 standard (known as V2) from the User Needs workshop of ITS owner/operators held in Austin TX. Prior effort resulted in the completion of User Needs,

Concept of Operations, Requirements and High Level Design before the project was suspended at ID 50 (Develop Design Content) of the Project Management Plan (PMP).

In September 2015, the project was restarted to complete the remaining tasks of the original PMP, but with updated milestone dates. The ITS cabinet working group, operating under the direction of the ATC Joint Committee, held working group meetings and teleconferences to complete this work. Siemens Industry, Inc. has been retained under contract to ITE to provide consultant services to the working group for preparation of the V2 standard. This remaining effort picked up the prior work completed by ConSysTec (Bruce Eisenhart), Pillar Consulting (Ralph Boaz), and Adaptive Solutions, Inc. (James Kinnard) at Develop Design Content (ID 51) of the PMP, through the completion of the V2 standard.

While the project was suspended, volunteer efforts by several manufacturers have resulted in the availability of Commercial Off-The-Shelf (COTS) subassemblies that were considered in the remaining standards development and adopted when found to meet the project Requirements traceable to the original User Needs. Multiple but differing COTS subassemblies were evaluated using a prior survey of ITS Owner/Operator Non-Functional Requirement (NFR) rankings, such as Reliability, Operational Safety, Modularity, etc.; these were included as Section 15 and Table 78. COTS subassemblies with higher NFR rankings were adopted over similar but differing COTS subassemblies.

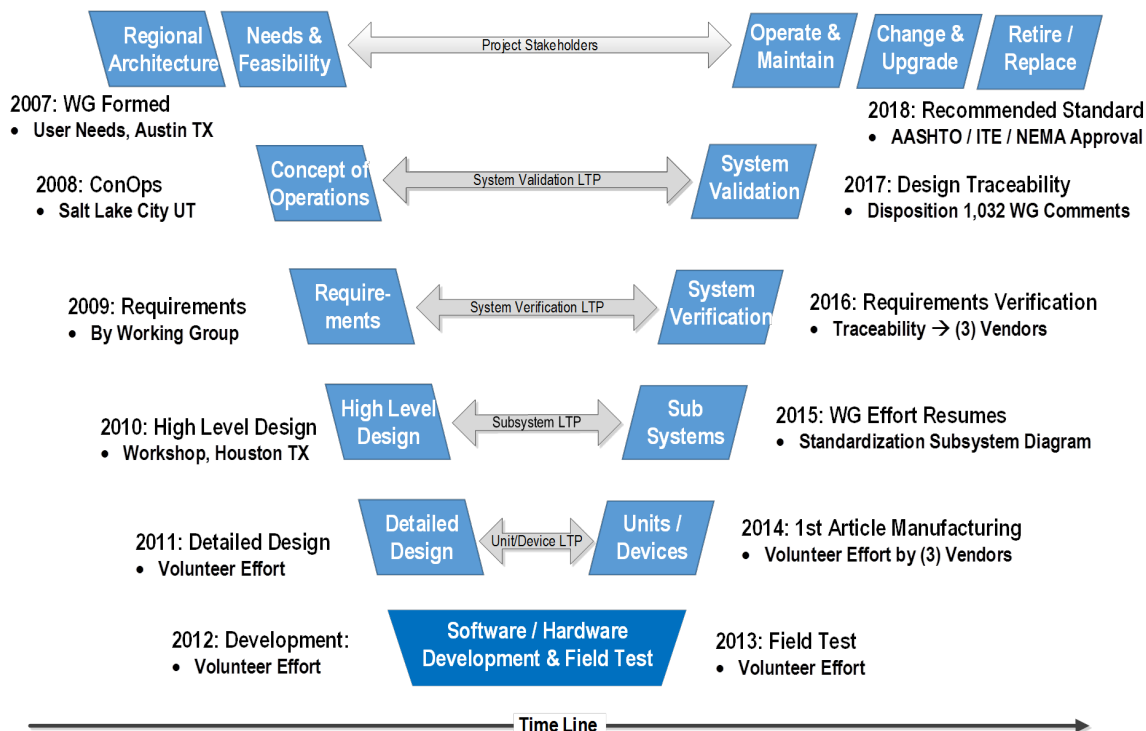


Figure 1: ITS Cabinet V2 Major Milestones)

<https://www.ite.org/technical-resources/standards/its-cabinet/version-2/>

The project timeline and major milestones are depicted in Figure 1:

- 2007: Working Group (WG) is formed including public agency end users and private sector manufacturers. The User Needs Workshop was conducted in Austin TX with presentations by end user Owner / Operators of signal control equipment and software.
- 2008: Concept of Operation (ConOps) is developed to fulfill the User Needs. ConOps Workshop is conducted in Salt Lake City UT with end user Owner / Operators. Feedback from that workshop results in an updated ConOps document posted on the ITE website.
- 2009: WG is developed Requirement based on User Needs and ConOps. After WG review, the Requirements document is posted on the ITE website.
- 2010: WG conducts a week-long High Level Design workshop in Houston TX. A survey of end user, owner/operators is conducted to rank the importance of Non-Functional Requirements, such as product safety, cost, serviceability, and others.
- 2011: Project funding ends but volunteer effort continued to update the Requirements document.
- 2012: Volunteer effort continues to develop and test key missing subsystems, such as low voltage, high density switch packs.
- 2013: Volunteer effort continues to include field test of prototypes, such as field test of working systems operating off-grid on alternative power sources.
- 2014: First article manufacturing of commercially-available equipment by three private sector WG members is completed.
- 2015: Funding is available to resume the standards development process. The High Level design is captured in High Level block diagrams that identify subassemblies and interfaces that are controlled by the standard versus those that are controlled by end user and manufacturer specifications.
- 2016: Instead of the normal workflow to build prototypes based on the Requirements, the WG verified each requirement to the existing equipment developed voluntarily by three manufacturers. Conflicting designs were resolved by standardizing the design elements best meeting the Non-Functional requirements, or by identifying design elements that should differ to meet differing local needs. Requirements management process enforces a Requirements update with approval of WG members. The Needs to Requirement Traceability Workbook is approved by WG, included as Section 14.
- 2017: User Comment Draft is created and reviewed by the wider transportation sector, resulting in disposition of 1,032 comments while tracing the design elements back to the Needs and Requirements. The User Comment Draft is updated to become the Recommended Standard document.
- 2018: The Needs to Requirements Traceability Workbook of Section 14 was updated to include traceability to Related Design Elements. The Recommended Standard is sent to the Joint Committee for ballot by AASHTO, ITE, and NEMA.

3 REFERENCED DOCUMENTS

- [1] "ATC 5201 v06A.37, ATC JC, 29 July 2020. Available from Institute of Transportation Engineers
- [2] "Cabinets, Racks, Panels and Associated Equipment," EIA-310, Electronic Industries Alliance
- [3] "Systems Engineering for Intelligent Transportation Systems," January 2007, available from FHWA

- [4] Transportation Electrical Equipment Specification, Caltrans, dated 2009
- [5] NEMA Standards Publication TS 2-2016 v03.07 Traffic Controller Assemblies with NTCIP Requirements. Available from the National Electrical Manufacturers Association.

4 CONVENTIONS USED IN THIS DOCUMENT

- RD [nn] indicates Reference Document number nn of REFERENCED DOCUMENTS above
- Rectangular symbol indicates a physical cabinet element
- Oval symbol indicates an interface used to connect physical cabinet elements
- Hatched symbol with dashed border indicates standard does not control mechanical dimensions
- Solid symbol with solid border indicates standard controls the mechanical dimensions for plug-in interchangeability
- Electrical signals, signal names, and signal functions are controlled by the standard in all cases
- Solid connecting line indicates the standard controls the connector and pin assignments for plug-in interchangeability
- Dashed connecting line indicates the standard does not control the connector type and pin assignments. The signals are electrically identical among manufacturers, but may require an adapter cable for interchangeability among manufacturers.

5 HIGH LEVEL BLOCK DIAGRAM

5.1 ATCC Versions

The ATCC standard defines the following two versions:

- High Voltage (HV) version fulfills the need for operation on traditional 120 VAC service voltage to control 120 VAC low power (non-incandescent) signal heads
- Low Voltage (LV) version fulfills the need for operation on 48 VDC alternate power sources, such as battery and solar, to control low voltage DC signal heads.

The ATCC standard reserves attributes for future version:

- Very High Voltage (VHV) version fulfills the need for operation on 240 VAC service voltages to control 240 VAC low power (non-incandescent) signal heads, typically used outside of the USA and Canada

Refer to Figure 2: High Level Functional Block Diagram, High Voltage Version and to Figure 3: High Level Design Functional Block Diagram, Low Voltage . Signal naming conventions shall conform to Section 16.

5.2 Traceability

To support traceability from High Level Design to this document, each of the elements and interfaces shown in the Block Diagrams is described in a corresponding section of this standard below.

Figure 2: High Level Functional Block Diagram, High Voltage Version

Figure 3: High Level Design Functional Block Diagram, Low Voltage Version

6 COMPONENTS

6.1 Model 2218 Serial Interface Unit (SIU)

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

6.1.1 General

The Model 2218 functions as the cabinet communications and control unit. The Model 2218 communicates with the ATC through Serial Bus 1 (SB#1) and Serial Bus 2 (SB#2). One Model 2218 is needed for each input functionality and each output functionality connected to the ATC.

The Model 2218 is capable of processing 54 input/output pins and four optical input pins. When installed in an Input Functionality, it processes 24 detector outputs, 24 detector status outputs, and provides six detector reset signals. In advanced detectors, the status outputs may be converted to per channel resets. When installed in an Output Functionality, the Model 2218 controls 16 switch pack units (48 outputs) and four optical inputs through the CDC socket.

6.1.2 Model 2218 Functions

6.1.2.1 Time Reference

The Model 2218 shall include a 1 Kilohertz time reference to provide system response time stamps. The 1 Kilohertz time reference shall maintain a frequency accuracy of ± 0.01 percent (± 0.1 counts per second). A 32-bit millisecond counter shall be provided for time stamping. A timestamp rollover flag set by millisecond counter (MC) rollover shall be cleared only on command.

6.1.2.2 Clocking

The SIU shall not depend upon receiving a continuous SDLC clock signal on SB#1 or SB#2. Incoming clock signals shall only be used to read the corresponding data signals.

6.1.3 Watchdog

Watchdog functionality shall be provided. The Model 2218 shall power up with the watchdog enabled. Within the first watchdog time period, the watchdog value shall be set to 200 milliseconds $\pm$ 100 milliseconds. The watchdog state shall be reported in the model 2218 status byte as an indication that a watchdog has occurred, which will

remain until cleared in the Request Module Status command, see 9.1.6.1. Failure of the Model 2218 to reset the watchdog timer shall result in hardware reset.

6.1.4 Power Up Initialization

At Power Up, the Model 2218 shall be fully functional within 1000 milliseconds and all outputs shall be initialized OFF. The Request Module Status Response frame shall indicate a communications loss until the user program sends the Request Module Status message to reset the "E" bit (see 9.1.6.1) and a subsequent Set Outputs command is processed (see 9.1.6.7).

6.1.5 Input Functions

The Model 2218 shall control fifty-four input/output lines using ground-true logic.

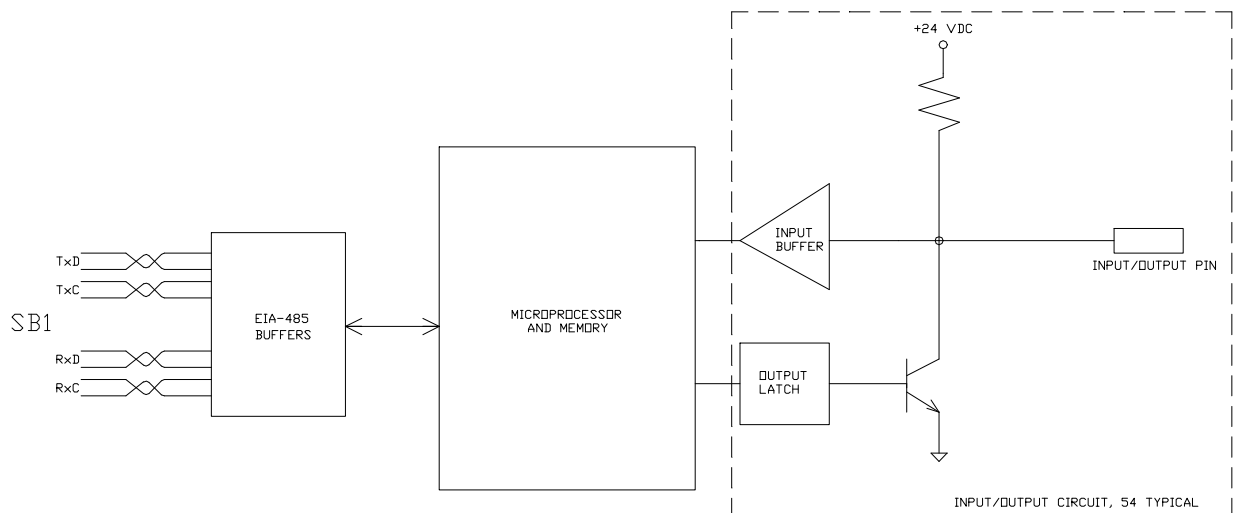


Figure 4: SIU Parallel Inputs and Outputs

At Power Up, the Model 2218's 54 outputs shall be initialized OFF and 54 inputs shall be available. Each output must be able to be read back as an input in order to check integrity.

6.1.5.1 Input Section

Each input shall be read logic "1" (ON) when the input voltage at its field connector input is less than 8 VDC and shall be read logic "0" (OFF) when the input voltage exceeds 16 VDC. Each input shall have an internal pull-up to +24 VDC of 10K Ohms +/- 10 percent and shall not deliver greater than 10 milliamperes to a short circuit to ground.

The Model 2218 IN INPUT / OUTPUT TRANSISTOR is continually OFF (passively pulled up to 24 VDC). The Sensor plugged into the IN SLOT asserts the INPUT / OUTPUT by actively pulling the INPUT / OUTPUT pin to 0 volts (Ground True). The

Sensor de-asserts the model 2218 IN signal from the Sensor in IN SLOT by releasing the signal (OFF), allowing the resistor to actively pull the signal to +24 VDC.

6.1.5.2 Optical-Coupled Inputs

6.1.5.2.1 Opto Common

The OPTO COMMON input shall be the common reference pin for the four opto inputs.

6.1.5.2.2 Opto Inputs

The OPTO inputs are intended to provide optical isolation for pedestrian detection, internal cabinet functions, remote interconnects or other auxiliary inputs. Opto Inputs 1 to 4 are assigned as I56 to I59 respectively.

1. The OPTO inputs shall provide electrical isolation of 10 Megohms minimum resistance and 1000 VAC RMS minimum breakdown to all connector pins except the OPTO COMMON pin, at a maximum breakdown leakage current of 1 milliamperes RMS.
2. The OPTO inputs shall exhibit nominal impedance to the OPTO COMMON pin of 5000 Ohms, ± 10 percent.
3. The OPTO inputs shall not recognize 3 Volts RMS (AC sinusoid or DC) or less relative to the OPTO COMMON input.
4. The OPTO inputs shall recognize 8 Volts RMS (AC sinusoid or DC) or more relative to the OPTO COMMON input.
5. Any steady state voltage applied between an OPTO input and the OPTO COMMON input shall not exceed 35 VAC RMS.
6. The OPTO inputs shall not be acknowledged when active for 25 milliseconds or less and shall be acknowledged when active for 50 milliseconds or more.
7. The OPTO inputs are intended to connect through external 27K Ohm, 1-Watt resistors for 120 VAC operations, external 6.8K Ohm 0.5-Watt resistors for 48 Vdc operation, and are intended for direct connection to 12 VAC or 24 VDC (Low True) for pedestrian detector applications.

6.1.5.3 NRESET Input Signal

The Model 2218 uses the EIA-485 NRESET input for model 2218 shut down/turn on operations matching the ATC. The Model 2218 shall be fully initialized and providing specified operation upon NRESET Line going HIGH (Power Up). The NRESET operation shall cause the model 2218 program to restart. No prior message operation data retention is required. The Request Module Status response may report this restart as either a Power On or Watchdog. ATC LINESYNC is used as a system time reference. The microprocessor/controller unit shall be reset by any of the following:

1. Pushbutton Reset
2. NRESET Signal
3. Internal voltage supply out of regulation
4. Microprocessor/controller unit watchdog

6.1.5.4 LINESYNC Input Signal

The EIA-485 LINESYNC interrupt shall be generated by both the 0-1 and 1-0 transitions of the LINESYNC signal. The LINESYNC interrupt shall monitor the millisecond counter interrupt and set the millisecond counter error flag if there has not been an interrupt from the 1 Kilohertz source for 0.5 seconds. The LINESYNC interrupt shall synchronize the 1 KHz time reference with the 0-1 transition of the LINESYNC signal once a second. A LINESYNC error flag shall be set if the LINESYNC interrupt has not successfully executed for 0.5 seconds or longer. The absence of the LINESYNC signal shall only affect the SIU output phase synchronization with the AC Line and shall not prevent the SIU from otherwise operating.

6.1.5.5 SB#1 Address Select Inputs

The SB#1 address select input bits shall define the SB#1 address of each SIU. This address is related to the SIU position and function in the cabinet. No connection shall be logical False, while a connection to LOGIC GROUND shall be a logical True. There shall be sixteen unique address positions selected with a binary code, using bit 0 as least significant and bit 3 as most significant.

Table 1: Address Select Inputs

Address s	Functionality	2218 Address			
		A3	A2	A1	A0
0	Reserved	0	0	0	0
1	Output Functionality 1	0	0	0	1
2	Reserved	0	0	1	0
3	Output Functionality 2	0	0	1	1
4	Reserved	0	1	0	0
5	Reserved	0	1	0	1
6	Reserved	0	1	1	0
7	Manufacturer-Specific	0	1	1	1
8	Manufacturer-Specific	1	0	0	0
9	Input Functionality #1	1	0	0	1
10	Input Functionality #2	1	0	1	0
11	Input Functionality #3	1	0	1	1
12	Input Functionality #4	1	1	0	0
13	Input Functionality #5	1	1	0	1
14	Reserved	1	1	1	0
15	CMU #1	1	1	1	1

Note 1: A0 to A3 are Input to model 2218 with Logic Ground as common.

Note 2: 0 = open or ground false. 1= closed or ground true (shunted)

Note 3: 'Reserved' functionality addresses shall not be used under any circumstances.

Note 4: 'Manufacturer-Specific' functionality addresses may be used for non-standard devices (not specified by this standard), but must utilize standard SIU protocol messaging and functionality and must not interfere with the normal operation of standard installed SIUs.

6.1.6 Output Functions

6.1.6.1 Output Section

Each output written as a logic "1" (ON) shall have a voltage at its field connector output of less than 4.0 VDC. Each output written as logic "0" (OFF) shall provide an open circuit (1 Megohm or more) at its field connector output. Each output shall consist of an open-collector capable of driving 40 VDC minimum and sinking 100 milliamperes minimum. Each output circuit shall be capable of switching from logic "1" to logic "0" within 100 microseconds when connected to a load of 100K Ohms minimum. Each output circuit shall be protected from transients of 10 microseconds (± 2 microseconds) duration, ± 300 VDC from a 1K Ohm source, with a maximum rate of 1 pulse per second.

Each output shall latch the data written and remain stable until either new data is written or the active-low reset signal. Upon an active-low reset signal, each output shall latch a logic "0" and retain that state until a new writing. The state of all output circuits at the time of Power Up or in Power Down state shall be OFF. It shall be possible to simultaneously assert all outputs within 100 microseconds of each other. An output circuit state not changed during a new writing shall not glitch when other output circuits are updated.

The Model 2218 OUT INPUT / OUTPUT PIN is asserted when the OUTPUT TRANSISTOR is ON (0 volts). The model 2218 OUT INPUT / OUTPUT PIN is unasserted when the OUTPUT TRANSISTOR is OFF (passively pulled up to 24 VDC).

6.1.6.2 Functionality Address Output

The Model 2218 shall sense the rack address block and generate a square wave ($\pm 15\%$) on the ASSEMBLY ADDRESS signal as follows:

Table 2: Rack Address Frequency

Input Functionality	Address Frequency (Hz)
1	60
2	30
3	15
4	7.5
5	3.75
Others	Don't Care

The Address Frequencies shown in Table 2 are not synchronized to service voltage.

6.1.7 Model 2218 Operational Functions

6.1.7.1 Input Transition Buffers

A transition buffer shall be provided capable of holding a minimum of 1024 recorded entries. The transition buffer shall default to empty. The following are the two entry types: transition and rollover. The inputs shall be monitored for state transition. At each transition (if the input has been configured to report transition), a transition entry shall be added to the transition buffer. If multiple inputs change state during one input sample, these transitions shall be entered into the input transition buffer by increasing input number.

The millisecond counter shall be monitored for rollover. At each rollover transition (\$xxxx FFFF - \$xxxx 0000), a rollover entry shall be added to the transition buffer. For rollover entries, all bits of byte 1 are set to indicate that this is a rollover entry.

A new entry shall be discarded when storage is not available for the new entry. Transition buffer blocks are sent to the ATC upon command. Upon confirmation of their reception, the blocks shall be removed from the transition buffer.

The entry types are as follows:

Table 3: Input Transition Entry

Description	msb lsb								Byte Number
Transition Entry Identifier	S	Input Number (I0 – I59)							1
Timestamp NLSB	X	X	X	X	X	X	X	X	2
Timestamp LSB	X	X	X	X	X	X	X	X	3

Table 4: Millisecond Counter Rollover Entry

Description	msb lsb								Byte Number
Rollover Entry Identifier	1	1	1	1	1	1	1	1	1
Timestamp MSB	X	X	X	X	X	X	X	X	2
Timestamp NMSB	X	X	X	X	X	X	X	X	3

6.1.7.2 Input Section Function

Input scanning shall begin at I0 (bit 0) and proceed to the highest numbered input, ascending from LSB to MSB. Each complete input scan shall finish within 100 microseconds. Once sampled, the logic state of an input shall be held until the next input scan. Each input shall be sampled 1,000 times per second. The time interval between samples shall be 1 milliseconds (± 100 microseconds). The millisecond counter reported shall indicate the value at the completion of the input scan.

6.1.7.3 Input Data Filtering

If configured, the inputs shall be filtered by the Model 2218 to remove signal bounce. The filtered input signals shall then be monitored for changes as noted. The filtering parameters for each input shall consist of ignore input flag and the ON and OFF filter values. If the ignore input flag is set, no input transition entries shall be placed into the input transition buffer. The ON and OFF filter values shall determine the number of consecutive samples an input must be ON and OFF, respectively, before a change of state is recognized. If the change of state is shorter than the specified value, the change of state shall be ignored. The ON and OFF filter values shall be in the range of 0 to 255. A filter value of 0, for either or both values, shall result in no filtering for this input. The default values for input signals after reset shall be as follows:

1. Filtering: Enabled
2. On and off filter values shall be set to: 5
3. Transition monitoring: Disabled (Timestamps are not logged)

6.1.7.4 Output Function

Simultaneous assertion of all outputs shall occur within 100 microseconds. Each output shall be capable of being individually configured in state to ON, OFF, or a state synchronized with either phase of LINESYNC. The condition of the outputs shall only be "ON" if the Model 2218 continues to receive active communications from the ATC. If there is no valid communications with the ATC for 2.0 seconds, all outputs shall revert to the OFF condition, and the Model 2218 status byte shall be updated to reflect the loss of communication from the ATC. The data and control bits in the ATC -model 2218 frame protocol shall control each output as follows:

Table 5: Output Bit Translation

Case	Data Bit	Control Bit	Function
A	0	0	Output in the OFF state
B	1	1	Output is a square wave, synchronized to the LINESYNC signal. When LINESYNC is ON (1), the output is OFF, and When LINESYNC is OFF (0), the output is ON.
C	0	1	Output is a square wave, synchronized to the LINESYNC signal. When LINESYNC is ON (1), the output is ON, and When LINESYNC is OFF (0), the output is OFF
D	1	0	Output is in the ON state.

In Case A above, the corresponding output shall be turned OFF if previously ON and if previously OFF remain OFF until otherwise configured. For half-cycle switching (cases B and C), all outputs to be changed shall be changed within 50 microseconds after the corresponding LINESYNC transition and shall remain in the same state during the entire half cycle. In Case D above, the corresponding output shall be turned ON if previously

OFF and if previously ON remain ON until otherwise configured. All outputs shall never change state unless configured to do so.

6.1.8 Power Requirements

The Model 2218 requires a nominal supply voltage of 24 VDC (± 2 VDC) with a maximum of 30 VDC. A voltage of 16 VDC or less shall be considered loss of power and a voltage of 18 VDC or greater shall be considered adequate for operation. The Model 2218 shall not require more than 300 milliamperes over the voltage range of 16 VDC to 30 VDC and the Inrush current shall be limited to a maximum of 1.5 Amperes from initial application of DC power. The Model 2218 shall not be damaged by insertion to, or removal from, powered input or output assemblies. The Model 2218 shall operate normally a maximum of 1000 milliseconds after power is restored.

6.1.9 Communications Ports

The Model 2218 shall have a minimum of four communications ports, identified as Model 2218 Ports SB#1, SB#2, INBUS and a front panel Terminal port. Communications port SB#1 and the Terminal port are connected to the model 2218 microprocessor/controller unit, while Serial ports SB#2 and INBUS provide a buffered communications path from the ATC to the intelligent devices in the Input Functionality.

6.1.9.1 Model 2218 Port SB#1

Port SB#1 shall interface the Model 2218 to the CU via Serial Bus 1 (SB#1) of the cabinet bus. See the SB#1 Protocol section 9.1 of this standard for the protocol requirements. The SB#1 frame address assignments of multiple SIUs are as defined in section 6.1.5.5.

6.1.9.2 Model 2218 Port SB#2

Port SB#2 shall interface the INBUS port of the Model 2218 to the CU via Serial Bus 2 (SB#2) of the cabinet bus. The SB#2 protocol and Data Link parameters to the model 2218 are unspecified and are application specific. INBUS provides a buffered serial interface to intelligent devices located in the Input functionality.

6.1.9.2.1 Synchronous Operation

If the CU is communicating via logical port SP3S, the Model 2218 SB#2 shall communicate in HDLC format and protocol, and the hardware requirements shall match serial bus 2 (synchronous TX/RX using Tx/C from the CU CPU for common clocking) per ATC 5201 standard.

6.1.9.2.2 Asynchronous Operation

If the CU is communicating via logical port SP3, Model 2218 SB#2 shall communicate in an asynchronous START BIT / STOP BIT format and protocol per ATC 5201 standard.

6.1.9.3 Terminal Port Operation

The SIU Terminal port shall be provided for communication to a personal computer using EIA-694 signaling. The model 2218 Terminal port protocol shall be defined by the vendor and operate with vendor-supplied software. The pin assignments of Model 2218 Terminal Port shall be defined by section 6.1.10.5.

6.1.9.4 INBUS Operation

INBUS is a legacy SIU function that is not required for conformance to the standard. If INBUS is provided, INBUS functions as described in this section. If INBUS is not provided, the INBUS pins are not connected within the SIU.

The Model 2218 INBUS consists of detector rack signal INBUS TxD, INBUS RxD, INBUS TxC, and INBUS RxC and shall conform to the electrical standards of EIA-485, single-ended. In this scheme, the RxD- and RxC- inputs of the EIA-485 receivers are connected to 2.5 Volts, while the TxD- and TxC- outputs of the EIA-485 drivers are not used. Model 2218 INBUS receivers shall withstand ± 25 Volts, suitable for reception of EIA-694 bipolar signals. All four INBUS signals shall be terminated at each receiver with impedance of 6,800 Ohms (± 5 percent), connected from signal to Logic Ground on the SIU. The protocol shall be defined by the receiving device.

The Model 2218 shall ensure that an SB#2 MARK equates to an INBUS MARK. The model 2218 shall provide an LED indicator for TxD and RxD, such that it is illuminated during a MARK (START Bit, for example) and extinguished during a SPACE (STOP Bit, for example).

6.1.9.4.1 Synchronous Operation

If the CU is communicating to devices via logical port SP3S, the Model 2218 INBUS buffers shall convert model 2218 SB#2 TxD+ and TxD- to EIA-485, which shall then be transmitted to the devices via INBUS TxD. Likewise, the model 2218 INBUS buffers shall convert Model 2218 SB#2 TxC+ and TxC- to EIA-485, which shall then be transmitted to the devices via INBUS TxC.

If the CU is communicating to detectors via logical port SP3S, the Model 2218 INBUS buffers shall convert INBUS RxD from EIA-485, which shall then be transmitted to the ATC via Model 2218 SB#2 RxD+ and RxD-. Likewise, the Model 2218 INBUS buffers shall convert INBUS RxC from EIA-485, which shall then be transmitted to the ATC via Model 2218 SB#2 RxC+ and RxC-.

6.1.9.4.2 Asynchronous Operation

If the CU is communicating to detectors via logical port SP3, the Model 2218 INBUS buffers shall convert Model 2218 SB#2 TxD+ and TxD- to EIA-485 which shall be transmitted to the detectors via INBUS TxD.

If the CU is communicating to detectors via logical port SP3, the Model 2218 INBUS buffers shall convert INBUS RxD from EIA-485, which shall be transmitted to the ATC via model 2218 SB#2 RxD+ and RxD-.

Asynchronous operation shall not use SB#2 TxC+, TxC-, RxC+, RxC-, nor INBUS TxC, nor INBUS RxC.

6.1.9.4.3 INBUS RTS Input

The INBUS RTS input shall be pulled to +24 Volts via a 10K Ohm resistor on the SIU. In systems using legacy devices that do not use INBUS RTS, this line shall not be used (no connection). Devices equipped with INBUS RTS shall drive this line low when transmitting data from that detector to the Model 2218 via INBUS. When not transmitting data, this line shall not be driven low and is pulled to +24V via the 10K Ohm resistor.

6.1.9.4.4 Serial Bus 2 Control

The controller transmits a message on SB#2 which shall be received by each device via the Model 2218 INBUS TxD and INBUS TxC. If the device is asynchronous, INBUS TxC shall be ignored. Each detector shall compare the address field of the message with its own slot address and assembly address. If the address matches, that device shall respond with data on INBUS RxD and INBUS RxC. If the detector is asynchronous, INBUS RxC shall not be used. The Model 2218 of the responding detector shall enable its EIA-485 line drivers to transmit the response from INBUS to SB#2. This driver shall be enabled by any of the three following conditions:

- Activity on INBUS RxD
- Activity on INBUS RxC
- INBUS RTS at low (true)

This driver shall be disabled by either of the following two conditions:

- Lack of activity on both INBUS RxD and RxC for 1.5 milliseconds
- INBUS RTS transitions from low (true) to high (false)

6.1.9.4.5 NEMA TS-2 Port 1

NEMA TS-2 Port 1 can operate on SB#2 as an option, using TxD, RxD, TxC and RxC signal pairs of Table 7 in Section 6.1.10.6. The communication functions operate per the BIU Port 1 section of the NEMA TS-2 standard.

6.1.9.5 Sensors with Serial Ports

Circuitry shown in

Figure 5 shall be included in each SIU, providing a direct serial connection from the controller to each individual serial detector. This serial connection shall be in addition to all of the NEMA CALL and STATUS lines are shown:

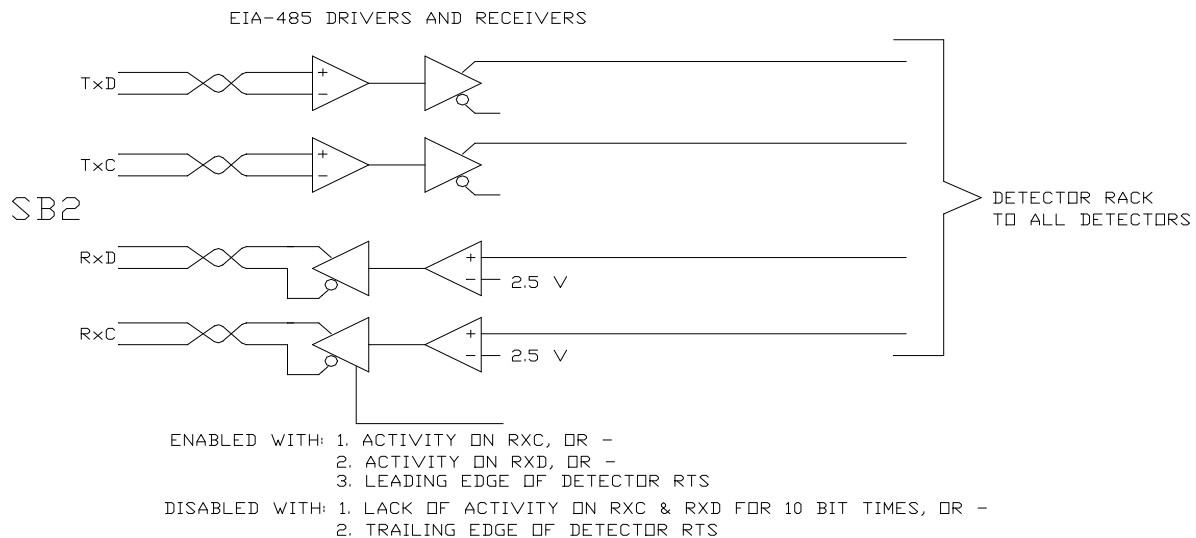


Figure 5: Serial Sensors

Detectors and other sensor types that include serial ports can be accessed directly by the Controller Unit via Serial Bus 2 as shown above. The protocol is not dependent upon the SIU. For example, serial detector protocol is installed in the CU, which then reads the detector CALLs directly from the detector, and the detectors are configured directly from the CU. The model 2218 circuitry translates the EIA-485 balanced differential signals from the CU into single-ended signals compatible with modern detectors. In addition, the detectors can be tuned and monitored via the serial bus.

6.1.10 Model 2218 Mechanical Details

6.1.10.1 Form Factor

The Model 2218 module shall be physically composed of a printed circuit board, 4.5 inches high by 6.5 inches long, a front panel 1.42 inches wide by 4.5 inches high with a DIN 96-pin connector on the connector end (opposite the front panel).

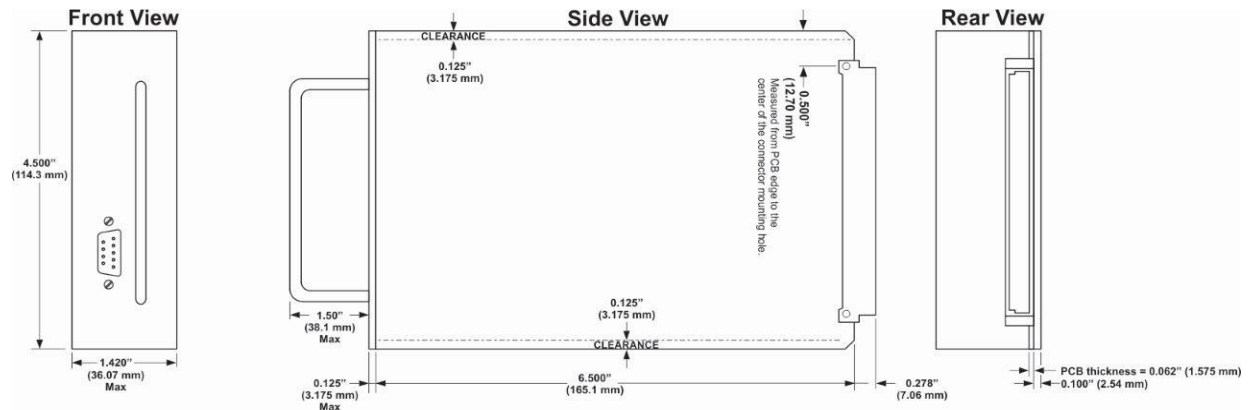


Figure 6: Model 2218 Mechanical Dimensions

6.1.10.2 Model 2218 Insertion and Extraction

A “U” handle shall be mounted on the front panel for insertion/extraction.

6.1.10.3 LED Indicators

Four LED waterclear indicators shall be provided on the front panel, as follows:

- Active
- Power
- Serial Bus #1 TxD
- Serial Bus #1 RxD
- Serial Bus #2 TxD (if INBUS is provided)
- Serial Bus #2 RxD (if INBUS is provided)

The Model 2218 power LED shall indicate that the +24 VDC power supply is within regulation. The Model 2218 active LED shall be controlled via Model 2218 I/O 55. The Serial Bus #1 indicators shall pulse when a frame is correctly received or transmitted.

6.1.10.4 Reset Button

The Model 2218 front panel shall provide a recessed Reset pushbutton that shall provide a hardware reset to the microprocessor/controller unit.

6.1.10.5 9-Position Subminiature D-Type Connector

The front panel EIA-694 connector is a 9 pin “DE9S” subminiature type connector. The connector shall provide #4-40 mating side jack posts for securing an attached cable. Because the port is configured as a DTE device, a null modem cable is required to connect directly to a personal computer COM port.

Table 6: Model 2218 EIA-694 Port

Pin #	Function	I/O
1	RESERVED	-
2	RECEIVE DATA	I
3	TRANSMIT DATA	O
4	RESERVED	-
5	SIGNAL GROUND	X
6	RESERVED	-
7	RESERVED	-
8	RESERVED	-
9	RESERVED	-

6.1.10.6 Model 2218 Connector

The Model 2218 main connector is a three row DIN 41612 (96-pin) Header Type. Signal names are with respect to the SIU.

Table 7: Model 2218 Connector Assignments

Pin	Description	Pin	Description	Pin	Description
A1	+24 VDC IN	B1	+24 VDC IN	C1	INPUT / OUTPUT 47
A2	INPUT / OUTPUT 0	B2	INPUT / OUTPUT 1	C2	INPUT / OUTPUT 48
A3	INPUT / OUTPUT 2	B3	INPUT / OUTPUT 3	C3	INPUT / OUTPUT 49
A4	INPUT / OUTPUT 4	B4	INPUT / OUTPUT 5	C4	INPUT / OUTPUT 50
A5	INPUT / OUTPUT 6	B5	INPUT / OUTPUT 7	C5	INPUT / OUTPUT 51
A6	INPUT / OUTPUT 8	B6	INPUT / OUTPUT 9	C6	INPUT / OUTPUT 52
A7	INPUT / OUTPUT 10	B7	INPUT / OUTPUT 11	C7	INPUT / OUTPUT 53
A8	INPUT / OUTPUT 12	B8	INPUT / OUTPUT 13	C8	SB#1 TXD +
A9	INPUT / OUTPUT 14	B9	INPUT / OUTPUT 15	C9	SB#1 TXD -
A10	INPUT / OUTPUT 16	B10	INPUT / OUTPUT 17	C10	SB#1 RXD +
A11	INPUT / OUTPUT 18	B11	INPUT / OUTPUT 19	C11	SB#1 RXD -
A12	INPUT / OUTPUT 20	B12	INPUT / OUTPUT 21	C12	SB#1 TXC +
A13	INPUT / OUTPUT 22	B13	INPUT / OUTPUT 23	C13	SB#1 TXC -
A14	INPUT / OUTPUT 24	B14	INPUT / OUTPUT 25	C14	SB#1 RXC +
A15	INPUT / OUTPUT 26	B15	INPUT / OUTPUT 27	C15	SB#1 RXC -
A16	INPUT / OUTPUT 28	B16	INPUT / OUTPUT 29	C16	LINESYNC +
A17	INPUT / OUTPUT 30	B17	INPUT / OUTPUT 31	C17	LINESYNC -
A18	INPUT / OUTPUT 32	B18	INPUT / OUTPUT 33	C18	NRESET +
A19	INPUT / OUTPUT 34	B19	INPUT / OUTPUT 35	C19	NRESET -

Pin	Description	Pin	Description	Pin	Description
A20	INPUT / OUTPUT 36	B20	INPUT / OUTPUT 37	C20	ASSEMBLY ADR
A21	INPUT / OUTPUT 38	B21	INPUT / OUTPUT 39	C21	INBUS RTS
A22	INPUT / OUTPUT 40	B22	INPUT / OUTPUT 41	C22	SB#2 TXD +
A23	INPUT / OUTPUT 42	B23	INPUT / OUTPUT 43	C23	SB#2 TXD -
A24	INPUT / OUTPUT 44	B24	INPUT / OUTPUT 45	C24	SB#2 RXD +
A25	INPUT / OUTPUT 46	B25	OPTO INPUT 1	C25	SB#2 RXD -
A26	OPTO INPUT 2	B26	OPTO INPUT 3	C26	SB#2 TXC +
A27	OPTO INPUT 4	B27	OPTO COMMON	C27	SB#2 TXC -
A28	ADDRESS – 0	B28	ADDRESS – 1	C28	SB#2 RXC +
A29	ADDRESS – 2	B29	ADDRESS – 3	C29	SB#2 RXC -
A30	INBUS TXD	B30	INBUS RXD	C30	INBUS TXC
A31	EQUIPMENT GROUND	B31	RESERVED	C31	INBUS RXC
A32	24 VDC GROUND	B32	24 VDC GROUND	C32	RESERVED

6.2 Model 2202 High Density Switch Pack / Flasher Unit (HDSP/FU)

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

6.2.1 General

The Model 2202 High-Density Switch Pack / Flasher Unit (HDSP/FU) shall be a modular PCB-based plug-in device providing six solid-state switches (two channels) to drive field signal loads. Each switch shall open or close a connection between applied signal power and external load. The Model 2202 shall operate in either the switch pack mode (HDSP) or the flasher mode (HDFU) depending on the applied SB#3 bus address.

This standard defines both high voltage AC output (2202-HV) and low voltage DC output (2202-LV) units. Unless otherwise specified, all HDSP/FU requirements apply to both output types.

A common form factor, including connector pin-out, is used for both defined units.

6.2.2 HDSP/FU Functions

6.2.2.1 Field Output Drivers

The HDSP shall provide two channels consisting of three individual field outputs per channel. The HDFU shall provide two channels consisting of two individual field outputs per channel.

6.2.2.2 Field Signal Voltage Sense Inputs

The 2202 HDSP/FU shall monitor the voltages at each of its six sense terminals. All HV and VHV voltage measurements shall be made over an RMS period of two line cycles of the HV SIGNALS input. All LV voltage measurements shall be made over a period of 33.3 milliseconds. All measurements shall be performed at a minimum sampling rate of 1920 samples per second.

A true RMS voltage measurement shall be made regardless of phase or wave-shape. All 2202-HV HDSP/FU voltage measurements shall be accurate to $\pm 2\%$ of full scale. All 2202-HV HDSP/FU voltage measurements shall include both positive and negative half wave sinusoids, over the voltage range of 0 to 135 Volts RMS. All 2202-LV HDSP/FU voltage measurements shall be accurate to $\pm 2\%$ of full scale over the voltage range of 0 to 64 Volts RMS.

The 2202 HDSP/FU shall be designed such that unused signal inputs are not sensed by the CMU as active signals. The 2202-HV HDSP/FU shall report an unused input at less than 15 Volts RMS when connected to HV+ SIGNAL through 1500 picofarads.

6.2.2.3 Field Load Current Monitor

The 2202 HDSP/FU shall sense the output current independently for each of the six outputs. HV and VHV current measurements shall be made over an RMS period of two line cycles of the HV SIGNALS input. All LV voltage measurements shall be made over a period of 33.3 milliseconds. All measurements shall be performed at a minimum sampling rate of 1920 samples per second.

All 2202 HDSP/FU RMS current measurements shall be accurate to $\pm 2\%$. A true RMS current measurement shall be made regardless of phase or wave-shape.

6.2.2.4 Diagnostic

If the 2202 HDSP/FU is inoperative due to an internal malfunction, it shall go to a Diagnostic state and force the field outputs to the Off state. A DIAGNOSTIC indicator shall illuminate when the 2202 HDSP/FU is inoperative as a result of a diagnostic fault. The definition of Diagnostic fault is not controlled by the standard, but the Diagnostic state is reported as a bit in the Status byte of the SB#3 response frame, see Table 66 and Table 68.

6.2.3 Input Signals

6.2.3.1 Field Signal Control Inputs

A Ground True input (0 VDC to 6 VDC) shall cause the output switch to conduct (ON), and a Ground False (16 VDC or more) shall cause it to not conduct (OFF). State transition shall occur between 6 VDC and 16 VDC. Each input shall not sink more than 20 milliamperes nor be subjected to more than 30 VDC. Each input shall have reverse polarity protection.

6.2.3.2 2202 HDSP/FU Slot Address

The address select input pins ADDRESS 4:0 define the Serial Bus #3 address of the HDSP mode as shown in

Table 8 and the HDFU mode shown in Table 9. The pins are left open for a logical False, and are connected to ADDRESS COMMON for a logical True. ADDRESS COMMON shall be connected only to the ADDRESS 4:0 pins of the same slot. The Serial Bus #3 address is the binary value of the ADDRESS 4:0 inputs plus 1.

Table 8: 2202 HDSP Slot Address

ADDR 4	ADDR 3	ADDR 2	ADDR 1	ADDR 0	SB#3 Address	Function	
False	False	False	False	False	0x01	Slot #1	Ch 1,2
False	False	False	False	True	0x02	Slot #2	Ch 3,4
False	False	False	True	False	0x03	Slot #3	Ch 5,6
False	False	False	True	True	0x04	Slot #4	Ch 7,8
False	False	True	False	False	0x05	Slot #5	Ch 9,10
False	False	True	False	True	0x06	Slot #6	Ch 11,12
False	False	True	True	False	0x07	Slot #7	Ch 13,14
False	False	True	True	True	0x08	Slot #8	Ch 15,16
False	True	False	False	False	0x09	Slot #9	Ch 17,18
False	True	False	False	True	0x0A	Slot #10	Ch 19,20
False	True	False	True	False	0x0B	Slot #11	Ch 21,22
False	True	False	True	True	0x0C	Slot #12	Ch 23,24
False	True	True	False	False	0x0D	Slot #13	Ch 25,26
False	True	True	False	True	0x0E	Slot #14	Ch 27,28
False	True	True	True	False	0x0F	Slot #15	Ch 29,30
False	True	True	True	True	0x10	Slot #16	Ch 31,32

Table 9: 2202 HDFU Slot Address

ADDR 4	ADDR 3	ADDR 2	ADDR 1	ADDR 0	SB#3 Address	Function
True	False	False	False	False	Reserved	--
True	False	False	False	True	Reserved	--
True	False	False	True	False	Reserved	--
True	False	False	True	True	Reserved	--
True	False	True	False	False	Reserved	--
True	False	True	False	True	Reserved	--
True	False	True	True	False	Reserved	--
True	False	True	True	True	Reserved	--
True	True	False	False	False	Reserved	--
True	True	False	False	True	Reserved	--
True	True	False	True	False	Reserved	--
True	True	False	True	True	Reserved	--
True	True	True	False	False	Reserved	--
True	True	True	False	True	0x1E	HDFU #1
True	True	True	True	False	0x1F	HDFU #2
True	True	True	True	True	Reserved	--

6.2.4 Output Signals

6.2.4.1 Field Signal Outputs

The 2202 HDSP/FU outputs shall be capable of proper operation with signal power over the Operational Power voltage ranges. In the HDFU mode the flashing outputs shall provide not less than 50 nor more than 60 flashes per minute with an On period of $50 \pm 5\%$. The flash rate shall not exceed these limits even with variations of the AC Line frequency.

Each individual 2202-HV HDSP/FU output shall be capable of supplying an output current of a minimum of 5 milliamperes to a maximum of 2.0 Amperes RMS to the load. In the 2202-HV HDSP mode, the total output current shall be limited to 1.0 Amperes continuous on all six outputs. In the 2202-HV HDFU mode, the total output current shall be limited to 2.0 Amperes at 50 percent duty cycle on all four flasher outputs, and the Ch 1 and Ch 2 Aux outputs shall be limited to 1.0 Ampere continuous. Each 2202-HV HDSP/FU output shall have a peak stand-off voltage of 500 volts or greater.

Each individual 2202-LV HDSP/FU output shall be capable of supplying an output current of a minimum of 5 milliamperes to a maximum of 3.0 Amperes RMS to the load. In the 2202-LV HDSP mode, the total output current shall be limited to 2.0 Amperes continuous on all six outputs. In the 2202-LV HDFU mode, the total output current shall be limited to 3.0 Amperes at 50 percent duty cycle on all four flasher outputs, and the Ch 1 and Ch 2 Aux outputs shall be limited to 2.0 Amperes continuous. Each 2202-LV HDSP/FU output shall have a peak stand-off voltage of 100 volts or greater.

Each switch shall have an OFF state dv/dt rating of 100 Volts per microsecond or better. Each switch shall provide isolation between control inputs and field outputs of at least 2000 VDC and at least 100 megohms resistive.

6.2.4.1.1 Off-State Leakage

When the switch is in the OFF state, the output current through the load shall not exceed 1 milliampere maximum.

6.2.4.1.2 Zero-Cross

The 2202-HV HDSP/FU output shall turn On within 10 degrees of the zero voltage point of the AC Line sinusoid on the first turn-on cycle. All subsequent cycles shall turn On within 5 degrees of the zero voltage point of the AC line sinusoid. The output shall turn On within one half cycle following the application of the input signal. The output shall turn Off within 5 degrees of the zero voltage point of the AC line sinusoid and within one half cycle following the removal of the input signal.

6.2.5 Power Requirements

6.2.5.1 Operational Power

The 2202 HDSP/FU shall commence normal operation within 500 ms of application of nominal power. The Field Outputs shall be held in the OFF state and not glitch to the ON state during this period. Inrush current drawn from the +24VDC source during power-up shall not exceed 100 mA peak. Inrush current drawn from the HV+MAINS or LV+MAINS source during power-up shall not exceed 2000 mA peak.

6.2.5.1.1 HV+MAINS

The HV+MAINS input provides operational power to the 2202-HV HDSP/FU. The 2202-HV HDSP/FU shall be operational over the voltage range of 60 Volts AC to 135 Volts AC. Nominal operating current shall not exceed 40 mA."

6.2.5.1.2 HV+SIGNAL

The HV+SIGNAL input provides field signal power to the 2202-HV HDSP/FU. This input shall be driven by the output of the Main Contactor secondary. The 2202-HV HDSP/FU field signal outputs shall be operational over the voltage range of 60 Volts AC to 135 Volts AC.

6.2.5.1.3 LV+MAINS

The LV+MAINS input provides operational power to the 2202-LV HDSP/FU. The 2202-LV HDSP/FU shall be operational over the voltage range of 18 Volts RMS to 60 Volts RMS. . Nominal operating current shall not exceed 60 mA."

6.2.5.1.4 LV+SIGNAL

The LV+SIGNAL input provides field signal power to the 2202-LV HDSP/FU. This input shall be driven by the output of the Main Contactor secondary. The 2202-LV HDSP/FU field signal outputs shall be operational over the voltage range of 18 Volts RMS to 60 Volts RMS.

6.2.5.2 +24VDC Power Supply

The 2202 HDSP shall sense the state of the +24VDC input. The +24VDC circuitry is with respect to DC (LOGIC) GROUND and electrically isolated from the MAINS and SIGNAL referenced circuitry. The HDSP shall report the status of the 24VDC input voltage in the response frame byte Status. A value less than 18 Vdc shall be reported as Low in the response frame Status byte. A value greater than 22 Vdc shall not be reported as Low in the response frame Status byte. A value between 18 Vdc and 22 Vdc may or may not be reported as Low.

The maximum current into the +24 VDC input pin over the voltage range of 0 VDC to 30 VDC shall be less than 60 milliamperes.

6.2.6 Front Panel Indicators

All indicators shall be clear LEDs. Clear LEDs shall not depend on a reflector or diffusion as part of its design. Clear LEDs shall not appear to be ON when exposed to ambient light. The following indicators shall be provided:

6.2.6.1 Power

A green POWER indicator shall illuminate to indicate Operational input voltage (HV+MAINS or LV+MAINS) is adequate for operation.

6.2.6.2 Diagnostic

A red DIAGNOSTIC indicator shall illuminate when the 2202 HDSP/FU is inoperative as a result of a diagnostic condition.

6.2.6.3 SB#3 Rx

A yellow indicator shall pulse ON each time the 2202 HDSP/FU correctly receives a frame on Serial Bus #3 from the CMU-2212.

6.2.6.4 SB#3 Tx

A yellow indicator shall pulse ON each time the 2202 HDSP/FU transmits a frame on Serial Bus #3 to the CMU-2212.

6.2.6.5 Field Signal Control Input

An individual red, yellow, and green LED indicator shall illuminate when the Field Signal Control input is active. There shall be one indicator for each Control input of each channel.

6.2.6.6 Channel ID Status (HDSP mode)

Two blue LED indicators shall be provided for each channel and shall illuminate based on the command Frame #01 data from the CMU-2212. These Channel ID Status indicators may be used by the CMU-2212 to indicate various status conditions based on an 2202 HDSP/FU channel.

6.2.6.7 Flasher Channel Status (HDFU mode)

Two LED indicators shall be provided for each flasher channel and shall illuminate to indicate an active Flasher output. The voltage level of the FL#x-x SENSE input is used to determine the active status of the LEDs such that they reflect actual flasher status to the field.

6.2.6.7.1 Flasher Channel Status Thresholds (HDFU-HV mode)

The Flasher output status shall be On when the RMS voltage exceeds 70 Vrms. The Flasher output status shall be Off when the RMS voltage is less than 50 Vrms.

6.2.6.7.2 Flasher Channel Status Thresholds (HDFU-LV mode)

The Flasher output status shall be On when the RMS voltage exceeds 44 Vrms. The Flasher output status shall be Off when the RMS voltage is less than 40 Vrms.

6.2.7 HDSP/FU Connector

The 2202 HDSP/FU shall connect to the OA2 backplane using a male DIN 41612 Type E series, 48-pin connector. The Rx and Tx signal names are with respect to the 2202 HDSP/FU. The connector pin assignments for the HDSP mode are shown in Table 10:

Table 10: HDSP Mode Connector Pin Assignments

Pin #	A (Bottom Row)	C (Middle Row)	E (Top Row)
2	Ch 1 Red In	Ch 1 Yellow In	Ch 1 Green In
4	Ch 2 Red In	Ch 2 Yellow In	Ch 2 Green In
6	+24VDC	VDC Ground	Address 4
8	Equipment Ground	MAINS Ground	MAINS Ground
10	SB#3 Rx+	SB#3 Tx+	Address Common
12	SB#3 Rx-	SB#3 Tx-	Address 3
14	Address 0	Address 1	Address 2
16	Ch 1 Red Sense	Ch 1 Red Out	Ch 1 Red Out
18	Ch 1 Yellow Sense	Ch 1 Yellow Out	Ch 1 Yellow Out
20	Ch 1 Green Sense	Ch 1 Green Out	Ch 1 Green Out
22	Ch 2 Red Sense	Ch 2 Red Out	Ch 2 Red Out
24	Ch 2 Yellow Sense	Ch 2 Yellow Out	Ch 2 Yellow Out
26	Ch 2 Green Sense	Ch 2 Green Out	Ch 2 Green Out
28	LV+ Signal	LV+ Signal	LV+ Signal
30	HV+ Signal	HV+ Signal	HV+ Signal
32	LV+ MAINS	MAINS Ground	HV+ MAINS

The connector pin assignments for the HDFU mode are shown in Table 11

Table 11: HDFU Mode Connector Pin Assignments

<i>Pin #</i>	<i>A (Bottom Row)</i>	<i>C (Middle Row)</i>	<i>E (Top Row)</i>
2	Reserved	Reserved	Ch 1 Aux In
4	Ch 2 Aux In	Reserved	Reserved
6	+24VDC	VDC Ground	Address 4
8	Equipment Ground	MAINS Ground	MAINS Ground
10	SB#3 Rx+	SB#3 Tx+	Address Common
12	SB#3 Rx-	SB#3 Tx-	Address 3
14	Address 0	Address 1	Address 2
16	FL#1-1 Sense	FL#1-1 Out	FL#1-1 Out
18	FL#1-2 Sense	FL#1-2 Out	FL#1-2 Out
20	Ch 1 Aux Sense	Ch 1 Aux Out	Ch 1 Aux Out
22	Ch 2 Aux Sense	Ch 2 Aux Out	Ch 2 Aux Out
24	FL#2-1 Sense	FL#2-1 Out	FL#2-1 Out
26	FL#2-2 Sense	FL#2-2 Out	FL#2-2 Out
28	LV+ Signal	LV+ Signal	LV+ Signal
30	HV+ Signal	HV+ Signal	HV+ Signal
32	LV+ MAINS	MAINS Ground	HV+ MAINS

6.2.8 2202 HDSP/FU Mechanical

The 2202 HDSP/FU shall be 1.12" W x 4.50" H x 6.50" D. Mechanical details are illustrated Figure 7.

The 2202 HDSP/FU shall slide into two card guides in the High-Density Output Functionality (OA2), each having a nominal slot width of 0.075" and a maximum slot depth of 0.125". The nominal thickness of the 2202 HDSP/FU PCB shall be $0.062" \pm 10\%$. The maximum height of components used on the 2202 HDSP/FU shall be such that they do not extend beyond the designated width envelope of the module. The handle shall not extend further than 1.50" from the front panel surface.

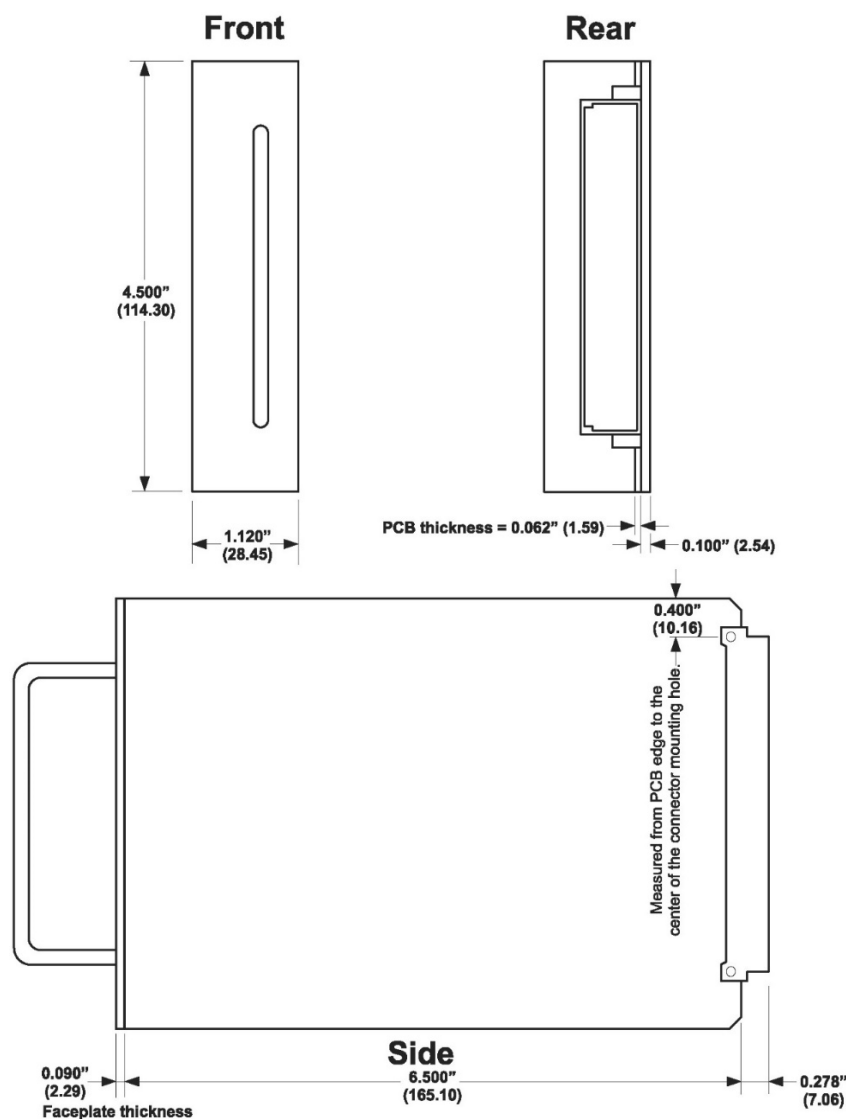


Figure 7: 2202 HDSP/FU Mechanical Dimensions

6.3 Model 2212 Cabinet Monitor Unit

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

6.3.1 General

The Cabinet Monitor Unit (CMU) is the principle part of the ATC Cabinet monitoring system. The role of the CMU is to query various cabinet conditions and, if the application requires action, the CMU will transfer control from the ATC to a flashing control mode.

The Model 2212 CMU is composed of a microprocessor, memory devices including non-volatile memory, communications circuitry to interface with Serial Buses #1 and #3, front panel indicators, front panel communication connector and a serial memory key device. The operating program shall be resident in the non-volatile memory and shall be user upgradeable via a front panel communication port. The serial memory key shall contain all of the conditions and function selections of the CMU.

This standard defines both high voltage AC powered (2212-HV) and low voltage DC powered (2212-LV) units. Unless otherwise specified, all CMU requirements apply to both output types.

6.3.2 Failed State Action (FSA)

Only Unit Reset shall reset the CMU from a LATCHED FAILED STATE ACTION (LFSA). Only a Unit Reset or a CMU Power Fail shall reset a LATCHED RESETTABLE FAILED STATE ACTION (LFSA-R).

A NONLATCHED FAILED STATE ACTION (NFSA) shall be reset if the fault conditions causing the NFSA have been removed. An NFSA shall last for the programmed Minimum Flash time at a minimum. Only one LFSA, LFSA-R or NFSA fault state shall be set at any time.

6.3.3 Unit Reset

The CMU shall be reset from a FSA as a result of the front panel reset button or the EXTERNAL TEST RESET input. This reset command shall be a one-time event, such that a continuous reset command does not prevent the CMU from operating.

The EXTERNAL TEST RESET input shall be True when the voltage is less than 8 VDC. The EXTERNAL TEST RESET input shall be False when the voltage is greater than 16 VDC. A minimum True pulse width on the EXTERNAL TEST RESET input of 100

milliseconds shall be required for a Unit Reset. The EXTERNAL TEST RESET input shall be referenced to VDC GROUND.

6.3.4 Exit From Failed State Action

Prior to the CMU transferring the output relay contacts from the Fault state to the No Fault state, a transition period of 500 milliseconds shall occur. During the transition period the output relay contacts shall be in the Fault state and the CMU shall set the Start-Up Flash Call bit in the Type 195, 209 and 211 Frames to 1. At all other times the Start-Up Flash Call bit of the Type 195, 209 and 211 Frames shall be set to 0.

6.3.5 Monitor Functions

6.3.5.1 Cabinet Power Supply

The CMU shall sense the Cabinet +48VDC, +24 VDC, and +12 VDC power supply sources to ± 5 percent accuracy. The CMU shall also sense the Cabinet +24 VDC state in each Output Functionality as reported by each HDSP. A +48 VDC, +24 VDC or +12 VDC failure during the programmed Minimum Flash time or during a CMU Power Failure shall not cause a LFSA. The CMU shall report the value of the +24 VDC and +12 VDC power supply sources in the Type 209 and Type 211 response frame.

6.3.5.1.1 48VDC Monitoring (2212-HV CMU Only)

Voltages equal to or greater than +46 VDC shall not cause a LFSA. Voltages at or less than +44 VDC for 500 milliseconds or longer shall cause a LFSA. If the sensed voltage is less than +44 VDC for 200 milliseconds or less, the CMU shall not cause a LFSA. All other timing or voltage conditions may or may not cause LFSA. There shall be programming in the serial memory key to disable +48 VDC power supply monitoring for legacy ATC Cabinets that do not provide this 48VDC monitor input to the CMU.

6.3.5.1.2 24VDC Monitoring

Voltages equal to or greater than +22 VDC shall not cause a LFSA. Voltages at or less than +18 VDC for 500 milliseconds or longer shall cause a LFSA. If the sensed voltage is less than +18 VDC for 200 milliseconds or less, the CMU shall not cause a LFSA. All other timing or voltage conditions may or may not cause LFSA.

6.3.5.1.3 12VDC Monitoring

Voltages equal to or greater than +11 VDC shall not cause a LFSA. Voltages at or less than +9 VDC for 500 milliseconds or longer shall cause a LFSA. If the sensed voltage is less than +9 VDC for 200 milliseconds or less, the CMU shall not cause a LFSA. All other timing or voltage conditions may or may not cause LFSA. There shall be programming in the serial memory key to disable +12 VDC power supply monitoring.

6.3.5.2 Conflicting Channels

For purpose of conflict determination, an active signal on either of the Green/Walk or Yellow inputs associated with any of the 32 channels shall be considered as that channel being active. The serial memory key shall contain the permissive channel pair programming. When any conflicting channels are detected as concurrently active for less than 200 milliseconds, the CMU shall not cause a LFSA. When any conflicting channels are detected as concurrently active for 500 milliseconds or more, the CMU shall cause a LFSA. When any conflicting channels are detected as concurrently active for more than 200 milliseconds but less than 500 milliseconds, the CMU may or may not cause a LFSA. The time interval between the beginning of the concurrently conflicting channels and the transfer to the LFSA shall not exceed 500 milliseconds.

6.3.5.3 Serial Bus Error

The CMU communicates using both Serial Buses (SB) #1 and #3. On SB#1, the CMU is a secondary, polled by the CU, which is the primary. On SB#1, the CMU shall respond to the SB#1 address defined by the ADDRESS 0 and ADDRESS 1 pins. On SB#3, the CMU is the primary, polling each high density switch pack (HDSP) and high density flasher unit pack (HDFU) secondary device.

6.3.5.3.1 Serial Bus #1 Error

The CMU shall cause a FSA when a Type 67, Type 81, or a Type 83 frame has not been received from the ATC for greater than 600 milliseconds. The first and second failures in a 24-hour period shall be a NFSA. The third failure in a 24-hour period shall be a LFSA-R. If a CMU Power Fail resets the LFSA-R, the SB#1 timeout count shall be reset to 2 such that the next SB#1 timeout results in a LFSA-R. At least one of the frames must be received every 100 milliseconds for 10 consecutive 100 millisecond periods to recover from the fault.

An SB#1 timeout failure during the programmed minimum flash time or during a CMU power failure shall not cause a FSA. The SB#1 timeout function shall be disabled if the SB#1 DISABLE input is at a True (Low) state. The SB#1 DISABLE input shall be True when the voltage is less than 8 VDC. The SB#1 DISABLE input shall be False when the voltage is greater than 16 VDC. The SB#1 DISABLE input shall be referenced to VDC GROUND.

6.3.5.3.2 Serial Bus #3 Error

The CMU shall cause a FSA when a Type 129 or Type 130 Frame has not been received from each HDSP for greater than 300 milliseconds. The first and second failures in a 24-hour period shall be a NFSA. The third failure in a 24-hour period shall be a LFSA-R. If a CMU power fail resets the LFSA-R, the SB#3 timeout count shall be reset to 2 such that the next SB#3 timeout results in a LFSA-R. A SB#3 timeout failure during the programmed minimum flash time or during a CMU power failure shall not cause a FSA.

6.3.5.4 Type 62 – Send to Local Flash Command Message

If the “N” bit is set in a Type 62 – Send to Local Flash Command message, the CMU shall react by causing a NFSA. The NFSA shall remain until the receipt of a Type 62 – Send to Local Flash Command message with the “N” bit cleared or until the CMU is reset by a unit reset or CMU power fail. The NFSA shall last for the programmed minimum flash time at a minimum. If the “L” bit is set in a Type 62 – Send to Local Flash Command message, the CMU shall react by causing a LFSA.

6.3.5.5 Diagnostics

6.3.5.5.1 Diagnostic Error

The CMU shall be provided with a resident series of self-check diagnostic capabilities. The CMU shall contain provisions to verify all memory elements on power-up. When a fault is detected, the LFSA-R shall be set and the DIAGNOSTIC indicator illuminated.

6.3.5.5.2 RAM Diagnostic

This test shall verify that all RAM elements are operating correctly at power-up or following a Unit Reset. Patterns shall be written to RAM. Each write shall be followed by a read to verify that it contains the written pattern.

6.3.5.5.3 Nonvolatile Diagnostic

This test shall verify that the nonvolatile ROM(s) contain the proper program. The routine shall perform a check on each ROM and make a comparison with a preprogrammed check value. This test shall be performed at power-up and at a minimum rate of 1024 bits per second during operation.

6.3.5.5.4 Serial Memory Key Memory Diagnostic

This test shall verify whether the non-volatile serial memory key contains valid data. The routine shall perform a check on each nonvolatile memory element at power-up and whenever read and make a comparison with a preprogrammed check value as described in section 6.3.14. The serial memory key not present shall cause a LFSA if the DOOR SWITCH FRONT input is sensed as not active (door closed).

6.3.5.5.5 Internal MPU Monitor

The CMU shall monitor the operation of its microprocessor with an independent circuit. At a minimum, the monitoring circuit shall receive logic state transitions at least once every 50 milliseconds from the microprocessor. When the logic state transition is not received for 500 milliseconds the monitor circuit shall force a LFSA-R and illuminate the DIAGNOSTIC indicator.

6.3.5.6 Multiple Input

The CMU shall be capable of monitoring for the presence of an active signal on two or more inputs of a channel. When the presence of an active signal on two or more inputs of a channel is detected for less than 200 milliseconds, the CMU shall not cause a LFSA. When the presence of an active signal on two or more inputs to a channel is detected for 450 milliseconds or more, the CMU shall cause a LFSA. When the presence of an active signal on two or more inputs to a channel is detected for more than 200 milliseconds but less than 450 milliseconds, the CMU may or may not cause a LFSA. Multiple Indication monitoring shall be disabled when the MAIN CONTACTOR COIL STATUS input is not active. There shall be programming in the serial memory key to disable Multiple Input monitoring on a color combination basis (G+Y, Y+R, G+R).

6.3.5.7 Lack of Signal Inputs

The CMU shall be capable of monitoring for the absence of any required signal voltage OR signal load current (CSU) on the inputs of a channel. For voltage purposes a required signal on the Green or Yellow or Red inputs associated with a channel shall be considered as that channel being active. For current purposes, a load current value that exceeds the programmed threshold shall be considered as that output being active.

When an absence of an active channel (voltage OR output current) is detected for less than 1200 milliseconds, the CMU shall not cause a LFSA. When an absence of an active channel is detected for 1500 milliseconds or more, the CMU shall cause a LFSA. When an absence of an active channel is detected for more than 1200 milliseconds but less than 1500 milliseconds, the CMU may or may not cause a LFSA. Lack of Signal Input monitoring shall be disabled for all channels when the MAIN CONTACTOR COIL STATUS input is not active. There shall be programming in the serial memory key to disable Lack of Signal Input monitoring on a per channel basis.

Lack of Signal Input monitoring shall also be disabled for any channel which has the DARK CHANNEL MAP bit set to "1" in the serial memory key programming for the DARK CHANNEL MAP addressed by the DARK CHANNEL MAP SELECT bits in a Type 81 message.

6.3.5.8 Minimum Yellow Clearance

The CMU shall verify that the Yellow Change interval is at least 2.7 seconds (± 0.1 seconds). When the minimum Yellow Change interval is not satisfied, the CMU shall cause a LFSA. The CMU shall report a Skipped Yellow Clearance when the Yellow Change interval is less than 100 milliseconds. The CMU shall report a Short Yellow Clearance when the Yellow Change interval is less than 2.7 seconds (± 0.1 seconds) and greater than 100 milliseconds.

Minimum Yellow Change interval monitoring shall be disabled when the MAIN CONTACTOR COIL STATUS input is not active. There shall be programming in the

serial memory key to disable Minimum Yellow Change interval monitoring on a per channel basis.

6.3.5.9 Minimum Yellow Plus Red Clearance

The CMU shall verify that the Yellow Change plus Red Clearance interval between the end of an active GREEN signal and the beginning of the next conflicting GREEN signal is at least 2.7 (± 0.1 seconds). When the minimum Yellow Change plus Red Clearance interval is not satisfied, the CMU shall cause a LFSA.

Minimum Yellow Change plus Red Clearance monitoring shall be disabled when the MAIN CONTACTOR COIL STATUS input is not active. There shall be programming in the serial memory key to disable Minimum Yellow Change plus Red Clearance interval monitoring on a per channel basis.

6.3.5.10 Flashing Yellow Arrow

A minimum of six Flashing Yellow Arrow channel pairs shall be supported by the CMU. Programming shall be provided in the serial memory key to specify the Overlap channel, Protected Left Turn channel, and the Opposing Thru channel. Any of the 32 CMU channels may be selected for the FYA operation including Virtual channels.

6.3.5.11 Local Flash Status

The CMU shall monitor the LF STATUS input. When this signal is sensed as not active for greater than 350 milliseconds, the CMU shall cause a NFSA. When this signal is sensed as not active for less than 200 milliseconds, the CMU shall not cause a NFSA.

6.3.5.11.1 Local Flash Status Recovery

Recovery from Local Flash Status NFSA shall occur when this signal is sensed as active for greater than 350 milliseconds. When this signal is sensed as active for less than 200 milliseconds, the CMU shall not cause recovery from Local Flash Status NFSA.

6.3.5.12 Circuit Breaker Trip Status

The CMU shall monitor the CB TRIP STATUS input. When this signal is sensed as not active for greater than 400 milliseconds, the CMU shall cause a LFSA. When this signal is sensed as not active for less than 200 milliseconds, the CMU shall not cause a LFSA.

6.3.5.13 Flasher Unit Output Failed

The CMU shall monitor the FLASHER 1-1, FLASHER 1-2, FLASHER 2-1 and FLASHER 2-2 voltage states reported by each HDFU. When a transition from the inactive state to the active state or a transition from the active state to the inactive state is absent for greater than 2600 milliseconds, the CMU shall set a status bit in the Type 195, Type 209 and Type 211 frames. This alarm condition shall not cause a FSA. It

should cause the appropriate response in the ATC. This status is non-latching, such that once a status bit has been set, the sensing of five valid transitions of the input shall clear the status bit.

6.3.5.13.1 HDFU Flasher Thresholds

For a 2212-HV CMU these inputs shall be considered active when the input voltage exceeds 70 Volts RMS. These inputs shall not be considered active when the input voltage is less than 50 Volts RMS. Signals between 70 Volts RMS and 50 Volts RMS may or may not be considered active.

For a 2212-LV CMU these inputs shall be considered active when the input voltage exceeds 44 Volts RMS. These inputs shall not be considered active when the input voltage is less than 40 Volts RMS. Signals between 44 Volts RMS and 40 Volts RMS may or may not be considered active.

6.3.5.14 CMU Power Failure

The CMU shall monitor the HV/LV MAINS POWER inputs (2212-HV pin B30, 2212-LV pin B29) and the NRESET and POWERDOWN cabinet control inputs to determine a CMU power failure response. The POWERDOWN signal in the low state indicates loss of MAINS in the controller unit. A CMU power failure shall cause a NFSA.

A model 2212-HV CMU power failure shall be recognized when both the POWERDOWN and NRESET signals are active low for greater than 100 milliseconds or the HV MAINS POWER voltage is less than 82 Volts RMS (± 2 Volts RMS) for 700 milliseconds.

A model 2212-LV CMU power failure shall be recognized when both the POWERDOWN and NRESET signals are active low for greater than 100 milliseconds or the LV MAINS POWER voltage is less than 42 Volts RMS (± 1 Volts RMS) for 700 milliseconds.

6.3.5.14.1 MAINS POWER Level Sense

The CMU shall monitor the MAINS POWER input and MAINS POWER inputs reported by each HDSP. For a 2212-HV CMU, when any MAINS POWER voltage is less than 82 Volts RMS (± 2 Volts RMS) for greater than 700 milliseconds (± 100 milliseconds) the CMU shall cause a NFSA. Once NFSA has been set, the POWERDOWN and NRESET signals shall not be monitored until all MAINS POWER voltages have exceeded 87 Volts RMS (± 2 Volts RMS) for 150 milliseconds.

For a 2212-LV CMU, when any MAINS POWER voltage is less than 40 Volts RMS (± 1 Volts RMS) for greater than 700 milliseconds (± 100 milliseconds) the CMU shall cause a NFSA. Once NFSA has been set, the POWERDOWN and NRESET signals shall not be monitored until all MAINS POWER voltages have exceeded 46 Volts RMS (± 1 Volts RMS) for 150 milliseconds.

6.3.5.14.2 Power Interruption

The CMU shall disable monitoring of the +48 VDC, +24VDC and +12VDC power supply inputs when either the POWERDOWN or NRESET input is in the low state. When the POWERDOWN and NRESET signals are both in the low state the CMU shall cause a NFSA.

6.3.5.14.3 Power Recovery

When the POWERDOWN input is high and the NRESET signal goes from low to high the CMU shall begin timing the programmed Minimum Flash Interval. During the Minimum Flash Interval, the CMU shall be in NFSA.

6.3.5.14.4 Power Up

Following initial application of MAINS Power voltage, the CMU shall maintain a NFSA until the POWERDOWN input is high and the NRESET signal goes from low to high. The CMU shall then begin timing the programmed Minimum Flash Interval. During the Minimum Flash Interval, the CMU shall be in NFSA.

6.3.5.15 Minimum Flash Interval

The CMU shall be in NFSA during the Minimum Flash Interval. The Minimum Flash Interval shall be programmed in the serial memory key between the limits of 6 seconds to 16 seconds with an incremental adjustment of 1 second. The CMU shall not set a FSA during the Minimum Flash Interval. External reset sources may manually shorten the minimum flash interval for test purposes ONLY.

6.3.5.16 Field Output Check

6.3.5.16.1 Field Check Fault

The CMU shall compare the active states of the field signals with the states reported by the ATC in the Type 81 or Type 67 frame. When a mismatch is detected for less than 1400 milliseconds the CMU shall not cause a LFSA. When a mismatch is detected for 1600 milliseconds or more, the CMU shall cause a LFSA. When a mismatch is detected for more than 1400 milliseconds but less than 1600 milliseconds, the CMU may or may not cause a LFSA. Field Output Check monitoring shall be disabled when the MAIN CONTACTOR COIL STATUS input is not active. There shall be a programming in the serial memory key to disable Field Output Check monitoring on a field input basis.

6.3.5.16.2 Field Check Status

The CMU shall compare the active states of the field signals with the states reported by the ATC in the Type 81 or Type 67 frame. When a mismatch is detected while a conflict, lack of signal, or multiple fault is timing, Field Check Status shall be reported with the fault to indicate the faulty channel(s) and color(s). Field Output Check monitoring shall be disabled when the MAIN CONTACTOR COIL STATUS input is not active. There

shall be a programming in the serial memory key to disable Field Output Check monitoring on a field input basis.

6.3.5.17 CMU Temperature

The CMU shall measure the temperature at the CMU and report this value in the Type 182 frame. Temperature accuracy shall be at most ± 6 °C over the operating temperature range of the CMU.

6.3.6 Input Signals

There shall be a programming in the serial memory key to disable the Yellow input for each physical channel.

6.3.6.1 2212-HV Field Signal Voltage Inputs

For Lack of Signal monitoring (see 6.3.5.7) A Green or Yellow or Red signal input shall be sensed active when the reported voltage exceeds 70 Volts RMS and shall not be sensed active when it is less than 50 Volts RMS. A Green or Yellow or Red signal between 70 Volts RMS and 50 Volts RMS may or may not be sensed active.

For all other monitoring functions a Green or Yellow or Red signal input shall be sensed active when the reported voltage exceeds 25 Volts RMS and shall not be sensed active when it is less than 15 Volts RMS. A Green or Yellow or Red signal between 15 Volts RMS and 25 Volts RMS may or may not be sensed active.

6.3.6.2 2212-LV Field Signal Voltage Inputs

For Lack of Signal monitoring (see 6.3.5.7) A Green or Yellow or Red signal input shall be sensed active when the reported voltage exceeds 44 Volts RMS and shall not be sensed active when it is less than 40 Volts RMS. A Green or Yellow or Red signal between 44 Volts RMS and 40 Volts RMS may or may not be sensed active.

For all other monitoring functions a Green or Yellow or Red signal input shall be sensed active when the reported voltage exceeds 9 Volts RMS and shall not be sensed active when it is less than 6 Volts RMS. A Green or Yellow or Red signal between 9 Volts RMS and 6 Volts RMS may or may not be sensed active.

6.3.6.3 Field Signal Load Current Outputs

A Green or Yellow or Red signal output shall be sensed active when the reported output load current exceeds the programmed threshold for that output and shall not be sensed active when the reported load current is less than or equal to the programmed threshold.

There shall be a programming in the serial memory key to disable Load Current monitoring on an output basis. There shall be a programming in the serial memory key to specify the Load Current Threshold on an output basis.

6.3.7 Control Signal Inputs

6.3.7.1 Local Flash Status

The LF STATUS input shall be internally connected to the CMU Output Relay common pin. This input shall be considered active when the input voltage exceeds 43 Vdc. This input shall not be considered active when the input voltage is less than 39 Vdc. Signals between 43 Vdc and 39 Vdc may or may not be considered active. Operation of the cabinet in AUTO mode shall place 48 Vdc nominal on this input. Operation of the cabinet in FLASH mode shall be open circuit on this input. The CMU shall report the state of this input in the Type 195, Type 209 and Type 211 frames.

6.3.7.2 Main Contactor Coil Status

The MAIN CONTACTOR COIL STATUS input shall be connected to the Main Contactor signal bus relay coil. An active signal on this input indicates the signal bus should be powering the switch packs. This input shall be considered active when the input voltage exceeds 43 Vdc. This input shall not be considered active when the input voltage is less than 39 Vdc. Signals between 43 Vdc and 39 Vdc may or may not be considered active. The CMU shall report the state of this input in the Type 195, Type 209 and Type 211 frames.

6.3.7.3 Main Contactor Secondary Status

The MAIN CONTACTOR SECONDARY STATUS input shall be connected to the output side of the main contactor signal bus relay. The CMU shall report the state of this input in the Type 195, Type 209 and Type 211 frames. For a 2212-HV CMU, this input shall be considered active when the input voltage exceeds 89 Volts RMS. This input shall not be considered active when the input voltage is less than 70 Volts RMS. For a 2212-LV CMU, this input shall be considered active when the input voltage exceeds 43 Volts RMS. This input shall not be considered active when the input voltage is less than 39 Volts RMS. Signals between 43 Volts RMS and 39 Volts RMS may or may not be considered active.

6.3.7.4 FTR Coil Drive Status

The FTR COIL DRIVE STATUS input shall be connected to the FTR COIL DRIVE signal in the AC SIGNAL POWER BUS. This input shall be considered active when the input voltage exceeds 43 Vdc. This input shall not be considered active when the input voltage is less than 39 Vdc. Signals between 43 Vdc and 39 Vdc may or may not be considered active. The CMU shall report the state of this input in the Type 195, Type 209 and Type 211 frames.

6.3.7.5 Circuit Breaker (CB) Trip Status

The CB TRIP STATUS input shall be connected to the auxiliary switch output of the circuit breaker unit. This input shall be considered active when the input voltage exceeds 43 Vdc. This input shall not be considered active when the input voltage is less

than 39 Vdc. Signals between 43 Vdc and 39 Vdc may or may not be considered active. The CMU shall report the state of this input in the Type 195, Type 209 and Type 211 frames.

6.3.7.6 Front/Rear Door Switch

The CMU shall monitor the DOOR SWITCH FRONT and DOOR SWITCH REAR inputs. These inputs shall be considered active when the input voltage exceeds 14 Vdc (door OPEN condition). These inputs shall not be considered active when the input voltage is less than 10 Vdc. Signals between 14 Vdc and 10 Vdc may or may not be considered active. The CMU shall report the state of these inputs in the Type 195, Type 209 and Type 211 frames.

6.3.7.7 Monitor Interlock

The MONITOR INTERLOCK input shall be connected to VDC GROUND within the CMU.

6.3.7.8 CMU Address Inputs

The address select input pin ADDRESS 0 defines the Serial Bus #1 address of the CMU. The pin is left open for a logical False, and is connected to VDC GROUND for a logical True.

Table 12: CMU Address Inputs

<i>ADDRESS 0</i>	<i>SB#1 Address</i>
False	0x0F
True	0x10

6.3.8 Power and Circuit Requirements

6.3.8.1 MAINS Power

The 2212-HV CMU shall be operational over the voltage range of 80 Volts RMS to 135 Volts RMS. Inrush current on HV MAINS shall be less than 3 Amperes peak.

The 2212-LV CMU shall be operational over the voltage range of 35 Volts RMS to 60 Volts RMS. Inrush current on LV MAINS shall be less than 7 Amperes peak.

6.3.8.2 +24VDC Power Supply

The +24 VDC MONITOR and +12 VDC MONITOR input circuits shall be optically isolated from the MAINS circuitry. The maximum current into the +24 VDC or +12 VDC Monitor inputs over the voltage range of 0 VDC to 30 VDC shall be less than 20 milliamperes.

6.3.9 Failed State Output Circuit

The output relay of the CMU shall have one set of isolated Form A contacts. These relay contacts shall be rated for a minimum of three Amperes at 120 Volts RMS and 100,000 operations. Contact opening/closing time shall be 30 milliseconds or less. The relay coil shall be energized in the no fault state and de-energized in the FSA state.

6.3.10 Front Panel Indicators

All indicators shall be clear LEDs. Clear LEDs shall not depend on a reflector or diffusion as part of its design. Clear LEDs shall only show the die and not appear to be ON when exposed to ambient light. The following indicators shall be provided:

6.3.10.1 Power

A green POWER indicator shall illuminate to indicate MAINS voltage is proper. It shall flash at a 2 Hertz rate when EITHER the NRESET input or POWERDOWN input is low.

For the 2212-HV CMU, it shall remain off when the voltage is less than 80 Volts RMS (± 2 Volts RMS). For the 2212-LV CMU, it shall remain off when the voltage is less than 35 Volts RMS (± 2 Volts RMS).

6.3.10.2 48 VDC Fail

A red 48VDC FAIL indicator shall illuminate when the CMU is in FSA as a result of a 48VDC cabinet power supply fault.

6.3.10.3 24 VDC Fail

A red 24VDC FAIL indicator shall illuminate when the CMU is in FSA as a result of a 24VDC cabinet power supply fault. The 24VDC FAIL indicator shall flash at a 2 Hertz rate when the 24 VDC monitor function is disabled.

6.3.10.4 12 VDC Fail

A red 12VDC FAIL indicator shall illuminate when the CMU is in FSA as a result of a 12VDC cabinet power supply fault.

6.3.10.5 Conflict

A red CONFLICT indicator shall illuminate when the CMU is in FSA as a result of a Conflicting channels fault.

6.3.10.6 Lack of Signal

A red LACK OF SIGNAL indicator shall illuminate when the CMU is in FSA as a result of a Lack of Signal Inputs fault.

6.3.10.7 Multiple

A red MULTIPLE indicator shall illuminate when the CMU is in FSA as a result of a Multiple Inputs fault.

6.3.10.8 CU Flash

A red CU FLASH indicator shall illuminate when the CMU is in FSA as a result of a Type 62 – Send to Local Flash Command from the ATC.

6.3.10.9 Local Flash

A red LOCAL FLASH indicator shall illuminate when the CMU is in FSA as a result of the LOCAL FLASH STATUS input sensed inactive, or CB TRIP STATUS sensed active.

6.3.10.10 Clearance

A red CLEARANCE indicator shall illuminate when the CMU is in FSA as a result of a Yellow Clearance or Yellow Plus Red Clearance fault.

6.3.10.11 Field Check

A red FIELD CHECK indicator shall illuminate when the CMU is in FSA as a result of a field output check fault. The indicator shall flash at a 2 Hertz rate when the CMU is in FSA with field check status as a result of Conflict, Lack of Signal, or Multiple fault.

6.3.10.12 SB#1 Error

A red SB#1 ERROR indicator shall illuminate when the CMU is in FSA as a result of a Serial Bus #1 fault.

6.3.10.13 SB#3 Error

A red SB#3 ERROR indicator shall illuminate when the CMU is in FSA as a result of a Serial Bus #3 fault.

6.3.10.14 Diagnostic

A red DIAGNOSTIC indicator shall illuminate when the CMU is in FSA as a result of a diagnostic fault. The DIAGNOSTIC indicator shall flash at a 4 Hertz rate if the serial memory key is not present and a FSA state does not exist (Front Door open).

6.3.10.15 SB#1 Rx

A yellow indicator shall pulse On each time the CMU correctly receives a frame on Serial Bus #1.

6.3.10.16 SB#3 Rx

A yellow indicator shall pulse On each time the CMU correctly receives a frame on Serial Bus #3.

6.3.11 Mechanical

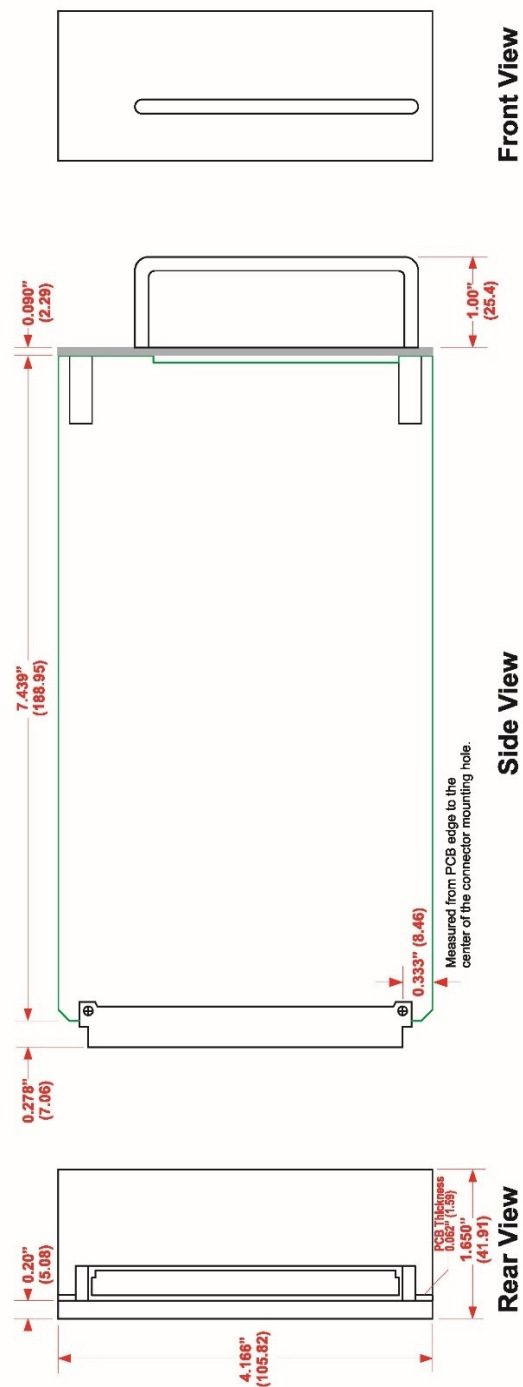


Figure 8: CMU Mechanical Dimensions

6.3.12 Serial Bus #3 Terminations

The Serial Bus #3 RxD+ input shall be terminated on the CMU to the Serial Bus #3 EIA-485 supply voltage through a 560 Ohm resistor. The Serial Bus #3 RxD- input shall be terminated on the CMU to Mains Ground through a 560 Ohm resistor. A 120 Ohm resistor shall be connected on the CMU between RxD+ and RxD-. The CMU Serial Bus #3 TxD drivers shall remain in the mark state with drivers enabled when the CMU is not transmitting a command frame.

Note: A 120 Ohm resistor shall be connected between Serial Bus #3 TxD+ and TxD- in the Auxiliary Display Unit (ADU) or on the Serial Bus #3 cable assembly in the last assembly installed in the Serial Bus #3 daisy chain if an ADU is not installed.

6.3.13 Terminal Port

An Ethernet interface and connector shall be provided for interconnecting the CMU to a personal computer or network. The protocol for this Terminal Port is defined by the manufacturers.

6.3.13.1 Terminal Port Functions

The Ethernet port shall provide support for a static IP and DHCP Client at a minimum. A DHCP single address Server function option shall be provided to facilitate a direct connection to a PC. If a network DHCP Server is discovered on the network then this DHCP Server function shall be automatically disabled. The network settings of the Ethernet port shall be loaded directly from the serial memory key parameters if the download function is enabled.

6.3.13.2 Physical

The connector shall be mounted on the front panel and shall be an 8-position RJ-45 connector.

6.3.14 Serial Memory Key

The CMU shall have a Datakey[™] model KC4210 Keycepticle[™] socket or equivalent serial memory key receptacle, which will intermate with the Datakey[™] model LCK4000-RED, mounted on the front panel. The CMU shall be provided with a Datakey[™] model LCK4000-RED serial memory key or equivalent. The serial memory key shall be rated for -40 to +85 degrees C operation.

Note: Datakey[™] and Keycepticle[™] are registered trademarks of Datakey, Inc.

6.3.14.1 Monitor Unit Serial Memory Key Interface

The CMU shall not provide the capability to program the serial memory key. Writing to the serial memory key may be accomplished through the use of a serial memory key

writer. The serial memory key shall be used by the CMU as a read only device. The 16 bit Frame Check Sequence (FCS) procedure defined in clause 4.5.2 of ISO/IEC 13239-2002(E) shall be used to verify the integrity of the read data. Failure to read the serial memory key correctly shall result in LFSA. Interface circuitry to the device shall utilize the LOFO switch on the serial memory key socket to ensure the device is removed and inserted with no power applied to the interface pins (i.e. dead socket).

6.3.14.2 Serial Memory Key Data

All bytes and bits marked as “reserved” shall be set to “0”.

Table 13 CMU Serial Memory Key Data

Byte #	Contents	Description
1	0x20	Serial Memory Key Version (CMU 2212)
2	Ch 1-9, ..., 1-2	Permissive Programming for channels 1-32: A bit set to “1” programs a channel pair to the permissive state. In the event the CMU has fewer than 32 channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 0. The Contents parameter contains on channel pair per bit position. For example, the contents of byte #5, “Ch 2-3, ..., 1-26”, is as follows: b0 = channel pair 1-26 b1 = channel pair 1-27 b2 = channel pair 1-28 b3 = channel pair 1-29 b4 = channel pair 1-30 b5 = channel pair 1-31 b6 = channel pair 1-32 b7 = channel pair 2-3 If b1 is set, then channel 1 is permissive with channel 27.
3	Ch 1-17, ..., 1-10	
4	Ch 1-25, ..., 1-18	
5	Ch 2-3, ..., 1-26	
6	Ch 2-11, ..., 2-4	
7	Ch 2-19, ..., 2-12	
8	Ch 2-27, ..., 2-20	
9	Ch 3-6, ..., 2-28	
10	Ch 3-14, ..., 3-7	
11	Ch 3-22, ..., 3-15	
12	Ch 3-30, ..., 3-23	
13	Ch 4-10, ..., 3-31	
14	Ch 4-18, ..., 4-11	
15	Ch 4-26, ..., 4-19	
16	Ch 5-7, ..., 4-27	
17	Ch 5-15, ..., 5-8	
18	Ch 5-23, ..., 5-16	
19	Ch 5-31, ..., 5-24	
20	Ch 6-13, ..., 5-32	
21	Ch 6-21, ..., 6-14	
22	Ch 6-29, ..., 6-22	
23	Ch 7-12, ..., 6-30	
24	Ch 7-20, ..., 7-13	
25	Ch 7-28, ..., 7-21	
26	Ch 8-12, ..., 7-29	
27	Ch 8-20, ..., 8-13	

Byte #	Contents	Description
28	Ch 8-28, ..., 8-21	
29	Ch 9-13, ..., 8-29	
30	Ch 9-21, ..., 9-14	
31	Ch 9-29, ..., 9-22	
32	Ch 10-15, ..., 9-30	
33	Ch 10-23, ..., 10-16	
34	Ch 10-31, ..., 10-24	
35	Ch 11-18, ..., 10-32	
36	Ch 11-26, ..., 11-19	
37	Ch 12-14, ..., 11-27	
38	Ch 12-22, ..., 12-15	
39	Ch 12-30, ..., 12-23	
40	Ch 13-19, ..., 12-31	
41	Ch 13-27, ..., 13-20	
42	Ch 14-17, ..., 13-28	
43	Ch 14-25, ..., 14-18	
44	Ch 15-16, ..., 14-26	
45	Ch 15-24, ..., 15-17	
46	Ch 15-32, ..., 15-25	
47	Ch 16-24, ..., 16-17	
48	Ch 16-32, ..., 16-25	
49	Ch 17-25, ..., 17-18	
50	Ch 18-19, ..., 17-26	
51	Ch 18-27, ..., 18-20	
52	Ch 19-22, ..., 18-28	
53	Ch 19-30, ..., 19-23	
54	Ch 20-26, ..., 19-31	
55	Ch 21-23, ..., 20-27	
56	Ch 21-31, ..., 21-24	
57	Ch 22-29, ..., 21-32	
58	Ch 23-28, ..., 22-30	
59	Ch 24-28, ..., 23-29	
60	Ch 25-29, ..., 24-29	
61	Ch 26-31, ..., 25-30	
62	Ch 28-30, ..., 26-32	
63	Ch 31-32, ..., 28-31	
64	Ch 8:1	Lack of Signal Input Enable:

Byte #	Contents	Description
65	Ch 16:9	A bit set to “1” enables the Lack of Signal Input monitoring function for that channel. This bit shall be set to “0” for any channel that has an input mapped to a virtual channel. In the event the CMU has fewer than 32 channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
66	Ch 24:17	
67	Ch 32:25	
68	Ch 8:1	Dark Channel Map #1 A bit set to “1” disables the Lack of Signal Input monitoring function for that channel. For channels that are set to 0, Lack of Signal Input Enable programming shall determine Lack of Signal Input operation. Default programming shall be 0.
69	Ch 16:9	
70	Ch 24:17	
71	Ch 32:25	Dark Channel Map #2 A bit set to “1” disables the Lack of Signal Input monitoring function for that channel. For channels that are set to 0, Lack of Signal Input Enable programming shall determine Lack of Signal Input operation. Default programming shall be 0.
72	Ch 8:1	
73	Ch 16:9	
74	Ch 24:17	Dark Channel Map #3 A bit set to “1” disables the Lack of Signal Input monitoring function for that channel. For channels that are set to 0, Lack of Signal Input Enable programming shall determine Lack of Signal Input operation. Default programming shall be 0.
75	Ch 32:25	
76	Ch 8:1	
77	Ch 16:9	Dark Channel Map #4 A bit set to “1” disables the Lack of Signal Input monitoring function for that channel. For channels that are set to 0, Lack of Signal Input Enable programming shall determine Lack of Signal Input operation. Default programming shall be 0.
78	Ch 24:17	
79	Ch 32:25	
80	Ch 8:1	Dark Channel Map #4 A bit set to “1” disables the Lack of Signal Input monitoring function for that channel. For channels that are set to 0, Lack of Signal Input Enable programming shall determine Lack of Signal Input operation. Default programming shall be 0.
81	Ch 16:9	
82	Ch 24:17	
83	Ch 32:25	

Byte #	Contents	Description
		programming shall determine Lack of Signal Input operation. Default programming shall be 0.
84	Ch 8:1	GY Multiple Channel Enable: A bit set to “1” enables the Green/Yellow Multiple Channel monitoring function for that channel. This bit shall be set to “0” for any channel that has had a Green or Yellow input remapped to a virtual channel. In the event the CMU has fewer than 32 channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
85	Ch 16:9	
86	Ch 24:17	
87	Ch 32:25	
88	Ch 8:1	YR Multiple Channel Enable: A bit set to “1” enables the Yellow/Red Multiple Channel monitoring function for that channel. This bit shall be set to “0” for any channel that has had a Yellow or Red input remapped to a virtual channel. In the event the CMU has fewer than 32 channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
89	Ch 16:9	
90	Ch 24:17	
91	Ch 32:25	
92	Ch 8:1	GR Multiple Channel Enable: A bit set to “1” enables the Green/Red Multiple Channel monitoring function for that channel. This bit shall be set to “0” for any channel that has had a Green or Red input remapped to a virtual channel. In the event the CMU has fewer than 32 channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
93	Ch 16:9	
94	Ch 24:17	
95	Ch 32:25	
96	Ch 8:1	Minimum Yellow Change Enable: A bit set to “1” enables the Minimum Yellow Change monitoring function for that channel. In the event the CMU has
97	Ch 16:9	
98	Ch 24:17	
99	Ch 32:25	

Byte #	Contents	Description
		fewer than 32 channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
100	Ch 8:1	Minimum Yellow Change Plus Red Clearance Enable: A bit set to “1” enables the Minimum Yellow Change Plus Red monitoring function for that channel. In the event the CMU has fewer than 32 channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
101	Ch 16:9	
102	Ch 24:17	
103	Ch 32:25	
104	Ch 8:1	Yellow Input Disable: A bit set to “1” forces the Yellow input to the OFF state for that channel. In the event the CMU has fewer than 32 physical channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 0.
105	Ch 16:9	
106	Ch 24:17	
107	Ch 32:25	
108	Ch 8:1	Output Current Sense Enable Red: A bit set to “1” enables the Lack of Signal current monitoring function for the Red Output of the channel. This bit shall be set to “0” for any Red input that has been mapped to a virtual channel. In the event the CMU has fewer than 32 physical channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
109	Ch 16:9	
110	Ch 24:17	
111	Ch 32:25	
112	Ch 8:1	Output Current Sense Enable Yellow: A bit set to “1” enables the Lack of Signal current monitoring function for the Yellow Output of the channel. This bit shall be set to “0” for any Yellow input that has been mapped to a virtual channel. In the event the CMU has fewer than 32
113	Ch 16:9	
114	Ch 24:17	
115	Ch 32:25	

Byte #	Contents	Description
		physical channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
116	Ch 8:1	Output Current Sense Enable Green: A bit set to “1” enables the Lack of Signal current monitoring function for the Green Output of the channel. This bit shall be set to “0” for any Green input that has been mapped to a virtual channel. In the event the CMU has fewer than 32 physical channels, the bit positions corresponding to the nonexistent channels shall be 0. Default programming shall be 1.
117	Ch 16:9	
118	Ch 24:17	
119	Ch 32:25	
120	Channel 1	Red Output Current Sense Threshold The threshold value for Red current sense shall be programmed in scaled milliamps. IF Param >120 Then Current = (Param *16) – 1800 Else Current = Param If Param > 250 Then Current = “> 2200” mA
121	Channel 2	
122	Channel 3	
123	Channel 4	
124	Channel 5	
125	Channel 6	
126	Channel 7	
127	Channel 8	
128	Channel 9	
129	Channel 10	
130	Channel 11	
131	Channel 12	
132	Channel 13	
133	Channel 14	
134	Channel 15	
135	Channel 16	
136	Channel 17	
137	Channel 18	
138	Channel 19	
139	Channel 20	
140	Channel 21	
141	Channel 22	
142	Channel 23	

Byte #	Contents	Description
143	Channel 24	
144	Channel 25	
145	Channel 26	
146	Channel 27	
147	Channel 28	
148	Channel 29	
149	Channel 30	
150	Channel 31	
151	Channel 32	
152	Channel 1	Yellow Output Current Sense Threshold The threshold value for Yellow current sense shall be programmed in scaled milliamps. IF Param >120 Then Current = (Param *16) – 1800 Else Current = Param If Param > 250 Then Current = "> 2200" mA
153	Channel 2	
154	Channel 3	
155	Channel 4	
156	Channel 5	
157	Channel 6	
158	Channel 7	
159	Channel 8	
160	Channel 9	
161	Channel 10	
162	Channel 11	
163	Channel 12	
164	Channel 13	
165	Channel 14	
166	Channel 15	
167	Channel 16	
168	Channel 17	
169	Channel 18	
170	Channel 19	
171	Channel 20	
172	Channel 21	
173	Channel 22	
174	Channel 23	
175	Channel 24	
176	Channel 25	
177	Channel 26	
178	Channel 27	
179	Channel 28	

Byte #	Contents	Description
180	Channel 29	
181	Channel 30	
182	Channel 31	
183	Channel 32	
184	Channel 1	Green Output Current Sense Threshold The threshold value for Green current sense shall be programmed in scaled milliamps. IF Param >120 Then Current = (Param *16) – 1800 Else Current = Param If Param > 250 Then Current = "> 2200" mA
185	Channel 2	
186	Channel 3	
187	Channel 4	
188	Channel 5	
189	Channel 6	
190	Channel 7	
191	Channel 8	
192	Channel 9	
193	Channel 10	
194	Channel 11	
195	Channel 12	
196	Channel 13	
197	Channel 14	
198	Channel 15	
199	Channel 16	
200	Channel 17	
201	Channel 18	
202	Channel 19	
203	Channel 20	
204	Channel 21	
205	Channel 22	
206	Channel 23	
207	Channel 24	
208	Channel 25	
209	Channel 26	
210	Channel 27	
211	Channel 28	
212	Channel 29	
213	Channel 30	
214	Channel 31	
215	Channel 32	
216	Red Ch 8:1	Field Output Check Enable:

Byte #	Contents	Description
217	Red Ch 16:9	A bit set to “1” enables the Field Output Check monitoring function for that input. This bit shall be set to “0” for any input that has been remapped to a virtual channel. In the event the CMU has fewer than 32 channels, the bit positions corresponding to the nonexistent inputs shall be 0. Default programming shall be 1.
218	Red Ch 24:17	
219	Red Ch 32:25	
220	Yellow Ch 8:1	
221	Yellow Ch 16:9	
222	Yellow Ch 24:17	
223	Yellow Ch 32:25	
224	Green Ch 8:1	
225	Green Ch 16:9	
226	Green Ch 24:17	
227	Green Ch 32:25	
228	Minimum Flash Time	Minimum Flash Time Values of 0 thru 5 shall result in 6 seconds of minimum flash. Maximum value is 16. Default programming shall be 6.
229	Misc Control	Misc Control b0 = Set to 1 shall enable the +12 VDC Power Supply monitor. b1 = Set to 1 shall enable HDFU #2 b2 = Set to 1 shall enable the +48 VDC Power Supply monitor. b3:b7 = Reserved Default programming shall be 0x01.
230	Ch 29 Red	Virtual Channel Assignment Channels that have not been assigned shall be set to 0. Default programming shall be 0. Bits 4:0 specify the physical channel number (1 to 28) Bits 6:5 specify the physical input 01 = Red 10 = Yellow 11 = Green Bit 7 = reserved
231	Ch 29 Yellow	
232	Ch 29 Green	
233	Ch 30 Red	
234	Ch 30 Yellow	
235	Ch 30 Green	
236	Ch 31 Red	
237	Ch 31 Yellow	
238	Ch 31 Green	
239	Ch 32 Red	
240	Ch 32 Yellow	
241	Ch 32 Green	
242	Ch 8:1	HDSP Channel Enable
243	Ch 16:9	A bit set to “1” enables the channel

Byte #	Contents	Description
244	Ch 24:17	monitoring function of the HDSP. This bit shall be set to "0" for any channel that has been disabled or is not installed. Default programming shall be 1.
245	Ch 32:25	
246:28 5	ASCII string	Monitor ID A packed 40 character ID shall be stored in ASCII format. Allowable characters are 020h through 07Eh. If less than 40 characters are used, the unused locations shall be set to 00h. Default programming shall be 00h.

286:32
5 ASCII string

User ID

A packed 40 character ID shall be stored in ASCII format. Allowable characters are 020h through 07Eh. If less than 40 characters are used, the unused locations shall be set to 00h. Default programming shall be 00h.

326	0x00	Ethernet Control b0: 1=DHCP Client Enable b1: 1=DHCP Server Enable (if provided) b2:b6 Reserved b7: 1=Enable download from datakey
327	LS octet	Ethernet IP Address
328	octet	
329	octet	
330	MS octet	
331	LS octet	Ethernet Subnet Mask
332	octet	
333	octet	
334	MS octet	
335	LS octet	Ethernet Gateway Address
336	octet	
337	octet	
338	MS octet	
339	Ascii byte #1	Ethernet Host Name (ascii)
340	Ascii byte #2	

Byte #	Contents	Description
341	Ascii byte #3	alphanumeric characters with no blank spaces. The name must be unique on the network and can contain the following characters: 0-9, A-Z, _, - Unused characters shall be set to 0x00.
342	Ascii byte #4	
343	Ascii byte #5	
344	Ascii byte #6	
345	Ascii byte #7	
346	Ascii byte #8	
347	Ascii byte #9	
348	Ascii byte #10	
349	Ascii byte #11	
350	Ascii byte #12	
351	Ascii byte #13	
352	Ascii byte #14	
353	Ascii byte #15	
354	FYA Pair #1	FYA OLP Parameters b4:b0 = OLP Channel number - 1 b5 = Channel Enable b6 = Flash Rate Detect b7 = Reserved
355	FYA Pair #2	
356	FYA Pair #3	
357	FYA Pair #4	
358	FYA Pair #5	
359	FYA Pair #6	
360	FYA Pair #1	FYA LT Parameters b5:b0 = Green Arrow Channel number b6 = R&Y Input Enable (Reds and Yellows of FYA LT channels shall not be monitored for any faults unless this bit is set) b7 = Reserved
361	FYA Pair #2	
362	FYA Pair #3	
363	FYA Pair #4	
364	FYA Pair #5	
365	FYA Pair #6	
366	FYA Pair #1	FYA OPP Parameters b5:b0 = Opposing Thru Channel number b6 = Reserved b7 = Reserved
367	FYA Pair #2	
368	FYA Pair #3	
369	FYA Pair #4	
370	FYA Pair #5	
371	FYA Pair #6	
372	0, 1:138	App Specific Buffer Size Decimal byte count of the App Specific Buffer including the Mfg ID. This buffer (if used) shall extend from offset 510 down to the Reserved limit of 373. The first (lowest) two bytes of the buffer shall be the NEMA Mfg ID

Byte #	Contents	Description
		number.
373:51 0	0x00	Reserved for future use or App Specific buffer.
511	FCS Isb	16 bit Check Value FCS Polynomial calculation of bytes #1 through #510.
512	FCS Isb	

6.3.15 Serial Memory Key Application Specific Buffer

A section shall be reserved for CMU parameters that are defined by the CMU manufacturer for application specific functionality outside of this standard. This section shall begin at a serial memory key location specified by “511 - App Specific Buffer Size” and end at byte 510. If this section is not used by the CMU, then the App Specific Buffer Size parameter shall be set to zero. The first (lowest) two bytes of this buffer shall be the NEMA Manufacturer Specific ID. The manufacturer specific code is a 16 bit binary number. These codes are maintained and issued by NEMA.

The intention in placing manufacturer specific codes in the serial memory key parameter buffer that are reserved for manufacturers use is to allow equipment to differentiate between serial memory keys that may contain the same data. (Authorized Engineering Information.)

6.3.16 CMU Connector

The CMU connector shall be a DIN 41612 (64 pin) Header Type.

6.3.16.1 CMU Pin Assignments

Signal names are with respect to the CMU.

Table 14: CMU Connector Pin Assignments

Pin #	Function	Pin #	Function
A1	+24VDC Monitor	B1	Reserved
A2	+12VDC Monitor	B2	External Test Reset
A3	24VDC Ground	B3	Serial Bus #1 Disable
A4	Monitor Interlock	B4	Reserved
A5	Address 0	B5	Reserved
A6	Reserved	B6	Reserved
A7	SB#1 TxData +	B7	SB#1 TxData -
A8	SB#1 RxData +	B8	SB#1 RxData -
A9	SB#1 TxClock +	B9	SB#1 TxClock -
A10	SB#1 RxClock +	B10	SB#1 RxClock -

Pin #	Function	Pin #	Function
A11	Reserved	B11	Reserved
A12	Reserved	B12	Reserved
A13	Reserved	B13	Reserved
A14	Reserved	B14	Reserved
A15	Line Sync +	B15	Line Sync -
A16	Nreset +	B16	Nreset -
A17	PowerDown +	B17	PowerDown -
A18	SB#3 TxData +	B18	SB#3 TxData -
A19	SB#3 RxData +	B19	SB#3 RxData -
A20	SB#3 Clock+	B20	SB#3 Clock-
A21	LF Status	B21	LF Status
A22	Output Relay NO	B22	Output Relay NO
A23	CB Trip Status	B23	Reserved
A24	MC Coil Status	B24	Reserved
A25	MC Secondary Status	B25	Reserved
A26	FTR Coil Drive Status	B26	Reserved
A27	Door Switch Front	B27	Reserved
A28	Door Switch Rear	B28	Reserved
A29	Reserved	B29	LV MAINS Power / +48VDC Monitor
A30	Reserved	B30	HV MAINS Power
A31	Equipment Ground	B31	Reserved
A32	Reserved	B32	MAINS Ground (Neutral)

Note #1: Output Relay NO is open during FSA (de-energized).

Note #2: For the 2212-HV CMU, B30 is the Power pin for AC Line. For the 2212-LV CMU, B29 is the Power pin for 48 VDC.

6.4 Cabinet Power Supply Requirements, High Voltage Versions

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	No
Mechanical Dimensions	No

6.4.1 General

Sections 6.4.2 through 6.4.10 provide requirements for the cabinet DC power supplies. Sections 6.4.11, 6.4.12, and 6.4.13 describe the requirements for three basic power supplies with different form factors and ratings that conform to these requirements. Because the interface between the cabinet assemblies and power supply is not controlled by this standard, it is not a conformance requirement that these specific power supplies be used in an ATC cabinet.

6.4.2 AC Line Input

The Power Supply shall be powered from AC Line. The AC input voltage range shall be 90 to 270 Vac, 45 to 65 Hz. Power Factor Correction shall be greater than 0.95 at full load. The TFCS's power supplies shall be designed to operate with an average efficiency of 80% or greater at load.

Additional power from a DC source is optional. DC power source requirements are not controlled by this standard.

6.4.3 Input Inrush Current

The AC Line input Inrush current at power-up shall be less than 50 Amps maximum at rated full load.

6.4.4 Outputs

The following DC outputs shall be provided:

Standard Power Functionality (HV Cabinet)

- 48 Vdc +/- 2 Vdc at 1 Amp minimum
- 24 Vdc +/- 2 Vdc at 4 Amps minimum
- 12 Vdc +/- 1 Vdc at 5 Amps minimum (12 Vdc output is optional)

High Power DC Functionality (LV Cabinet)

- 48 Vdc +/- 2 Vdc at 6 Amps minimum
- 24 Vdc +/- 2 Vdc at 4 Amps minimum

Compact Power Functionality

- 48 Vdc +/- 2 Vdc at 1 Amp minimum
- 24 Vdc +/- 2 Vdc at 3 Amps minimum

Each DC output shall be electrically isolated from AC Mains, Earth Ground, and other DC outputs.

Note: The High Power Functionality is intended for ATCC applications that drive 48Vdc signal heads (LV).

6.4.5 DC Output Ripple

The DC Output Ripple on each output shall be less than 500 mVpp when measured at 20MHz of bandwidth using a 12" twisted pair-wire terminated with a 0.1uf & 47uf capacitor

6.4.6 DC Start-up Time

The DC outputs shall attain regulated output levels within 1500 milliseconds of applied AC Line voltage across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.7 DC Hold-up Time

The DC outputs shall maintain regulated output level for a minimum of 50 milliseconds across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.8 DC Output Connector

The type of DC output connector is not controlled by this standard.

6.4.9 Indicators

Indicators shall be provided to indicate each DC output is active and the fuse is intact.

6.4.10 Mechanical

The mechanical form factor of the enclosure is not controlled by this standard.

6.4.11 Model 2216 Cabinet Power Supply

6.4.11.1 General

The model 2216 Cabinet Power Supply is a modular 19 inch rack mounted power supply device providing the DC voltages necessary for operating the ATC Cabinet in the HV mode. The model 2216-24-HV provides a regulated 24VDC output and a regulated 48 Vdc output. The model 2216-2412-HV model provides a regulated 24 Vdc output, a regulated 48 Vdc output, and a regulated 12 Vdc output option for powering the Input Functionality devices. Power Factor Correction shall be provided. Unless otherwise specified, all model 2216 requirements apply to both model types.

The model 2217 Cabinet Power Supply is a card mounted power supply device providing the DC voltages necessary for operating the ATC Cabinet in the HV mode. The model 2217 provides a regulated 24VDC output and a regulated 48 Vdc output. Power Factor Correction shall be provided.

6.4.11.2 Mechanical

The model 2216 shall be 1U (1.7") in height maximum and designed to mount into a 19 inch EIA rack. The maximum depth of the model 2216 shall be less than 8.0".

6.4.11.3 Inputs

The model 2216 shall be powered from AC Line provided by an AC Line cord with NEMA Type 5-15 plug. The input voltage range shall be 80 to 270 Vac, 45 to 65 Hz. Power Factor Correction shall be greater than 0.95 at full load.

6.4.11.3.1 Input Inrush Current

The AC Line input Inrush current at power-up shall be less than 20 Amps maximum at rated full load.

6.4.11.4 Outputs

6.4.11.4.1 Model 2216-24

The model 2216-24 shall provide:

- 48 Vdc +/- 2 Vdc at 1 Amp maximum
- 24 Vdc +/- 2 Vdc at 5 Amps maximum

Each DC output shall be electrically isolated from AC Mains, Earth Ground, and other DC outputs. Fans shall not be used for cooling.

6.4.11.4.2 Model 2216-2412

The model 2216-2412 shall provide:

- 48 Vdc +/- 2 Vdc at 1 Amp maximum
- 24 Vdc +/- 2 Vdc at 4 Amps maximum
- 12 Vdc +/- 1 Vdc at 5 Amps maximum

Each DC output shall be electrically isolated from AC Mains and Earth Ground. The 24 Vdc and 12 Vdc outputs share a common ground. Fans shall not be used for cooling.

6.4.11.4.3 DC Output Ripple

The DC Output Ripple on each output shall be less than 300 mVpp when measured at 20MHz of bandwidth using a 12" twisted pair-wire terminated with a 0.1uf & 47uf capacitor.

6.4.11.4.4 DC Start-up Time

The DC outputs shall attain regulated output levels within 1000 milliseconds of applied AC Line voltage of 110 Vac nominal across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.11.4.5 DC Hold-up Time

The DC outputs shall maintain regulated output level for a minimum of 50 milliseconds across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.11.5 Front Panel Indicators

All indicators shall be clear LEDs. Clear LEDs shall not depend on a reflector or diffusion as part of its design. Clear LEDs shall not appear to be ON when exposed to ambient light. The following indicators shall be provided:

6.4.11.5.1 AC Line

A green AC Line indicator shall illuminate to indicate Operational input voltage is proper and the AC Line fuse is intact.

6.4.11.5.2 48 VDC

A green indicator shall illuminate to indicate the 48 VDC output is active and the fuse is intact.

6.4.11.5.3 24 VDC

A green indicator shall illuminate to indicate the 24 VDC output is active and the fuse is intact.

6.4.11.5.4 12 VDC (model 2216-2412 Only)

A green indicator shall illuminate to indicate the 12 VDC output is active and the fuse is intact.

6.4.11.6 DC Voltage Test Jacks

Banana style test jacks shall be provided on the front panel for each DC output and DC ground. Mating banana plug spring width shall be 0.175 inches nominal.

6.4.11.7 DC Output Connector

The output connector is a Phoenix Contact #1825161 and mates with a Phoenix Contact #1825352 or equivalent. Pin #1 is the right most pin when viewed from the rear of the supply.

Table 15: Power Supply DC Output Connector

Pin	Function
1	+48VDC
2	48VDC Ground**
3	+24VDC
4	+12VDC (PS-2216-2412 only)
5	24/12 VDC Ground
6	Chassis Ground

**Note that the 48 VDC output shall be electrically isolated from the AC Line input and the 24VDC and 12 VDC outputs. The 48VDC Ground (pin #2) shall be connected within the cabinet to the same MAINS Ground (Neutral) that the Cabinet Monitor Unit (CMU pin B32) is connected to.

6.4.12 Model 2217 Cabinet Power Supply

6.4.12.1 General

The model 2217 Cabinet Power Supply is a card mounted power supply device providing the DC voltages necessary for operating the ATC Cabinet in the HV mode. The model 2217 provides a regulated 24VDC output and a regulated 48 Vdc output. Power Factor Correction shall be provided.

6.4.12.2 Inputs

The model 2217 shall be powered from AC Line. The input voltage range shall be 80 to 270 Vac, 45 to 65 Hz. Power Factor Correction shall be greater than 0.95 at full load.

6.4.12.2.1 Input Inrush Current

The AC Line input Inrush current at power-up shall be less than 20 Amps maximum at rated full load.

6.4.12.3 Outputs

The model 2217 shall provide:

- 48 Vdc +/- 2 Vdc at 1 Amp maximum
- 24 Vdc +/- 2 Vdc at 3 Amps maximum

Each DC output shall be electrically isolated from AC Mains, Earth Ground, and other DC outputs. Fans shall not be used for cooling.

6.4.12.3.1 DC Output Ripple

The DC Output Ripple on each output shall be less than 300 mVpp when measured at 20MHz of bandwidth using a 12" twisted pair-wire terminated with a 0.1uf & 47uf capacitor.

6.4.12.3.2 DC Start-up Time

The DC outputs shall attain regulated output levels within 1000 milliseconds of applied AC Line voltage of 110 Vac nominal across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.12.3.3 DC Hold-up Time

The DC outputs shall maintain regulated output level for a minimum of 50 milliseconds across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.12.4 Front Panel Indicators

All indicators shall be clear LEDs. Clear LEDs shall not depend on a reflector or diffusion as part of its design. Clear LEDs shall not appear to be ON when exposed to ambient light. The following indicators shall be provided:

6.4.12.4.1 AC Line

A green AC Line indicator shall illuminate to indicate Operational input voltage is proper and the AC Line fuse is intact.

6.4.12.4.2 48VDC

A green indicator shall illuminate to indicate the 48 VDC output is active and the fuse is intact.

6.4.12.4.3 24VDC

A green indicator shall illuminate to indicate the 24 VDC output is active and the fuse is intact.

6.4.12.5 Mechanical

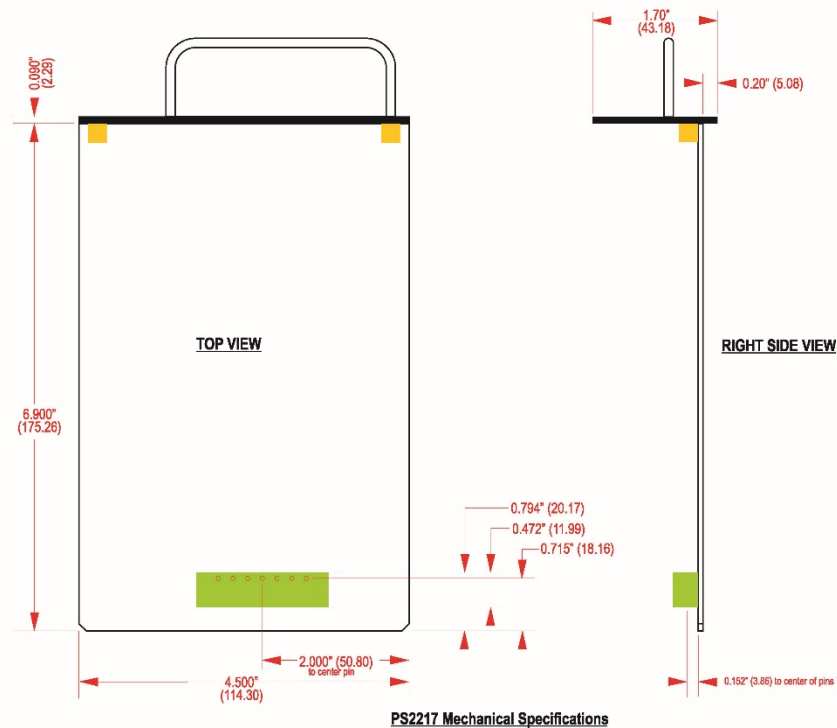


Figure 9: Model 2217 Mechanical Dimensions

6.4.12.5.1 PS2217 Connector

The connector on the model 2217 shall a Phoenix #1776553 or equivalent. The mating connector on the rack is a Phoenix #1785997 or equivalent. Pin #1 is the right most pin when viewed from the rear of the power supply. See Table 16.

Table 16: Model 2217 Power Supply Connector

Pin	Function
1	Chassis Ground
2	AC- Neutral
3	AC+ Power
4	24VDC Ground
5	+24VDC Output
6	48VDC Ground**
7	+48VDC Output

****Note that the 48 VDC output shall be electrically isolated from the AC Line input and the 24VDC output. The 48VDC Ground (pin #6) shall be connected within the cabinet to the same MAINS Ground (Neutral) that the Cabinet Monitor Unit (CMU pin B32) is connected to.**

6.4.13 Model 2248 Cabinet Power Supply

6.4.13.1 General

The model 2248 Cabinet Power Supply is a modular 19 inch rack mounted power supply device providing the DC voltages necessary for operating the ATC Low Voltage (LV) Cabinet. The model 2248 provides a regulated 24Vdc output and a regulated 48 Vdc output, and is intended to provide 48 Vdc power to the Output Functionality components and DC based signal lamps.

6.4.13.2 Mechanical

The model 2248 shall be 1U (1.7") in height maximum and designed to mount into a 19 inch EIA rack. The maximum depth of the model 2248 shall be less than 8.0".

6.4.13.3 Inputs

The model 2248 shall be powered from AC Line provided by an AC Line cord with NEMA Type 5-15 plug. The input voltage range shall be 80 to 270 Vac, 45 to 65 Hz. Power Factor Correction shall be greater than 0.95 at full load.

6.4.13.3.1 Input Inrush Current

The AC Line input Inrush current at power-up shall be less than 50 Amps maximum at rated full load.

6.4.13.4 Outputs

The model 2248 shall provide:

- 48 Vdc +/- 2 Vdc at 6 Amp maximum
- 24 Vdc +/- 2 Vdc at 4 Amps maximum

Each DC output shall be electrically isolated from AC Mains, Earth Ground, and other DC outputs. Fans shall not be used for cooling.

6.4.13.4.1DC Output Ripple

The DC Output Ripple on each output shall be less than 500 mVpp when measured at 20MHz of bandwidth using a 12" twisted pair-wire terminated with a 0.1uf & 47uf capacitor.

6.4.13.4.2DC Start-up Time

The DC outputs shall attain regulated output levels within 1000 milliseconds of applied AC Line voltage of 110 Vac nominal across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.13.4.3DC Hold-up Time

The DC outputs shall maintain regulated output level for a minimum of 50 milliseconds across the operating temperature, input voltage, and input frequency ranges and at rated full load.

6.4.13.5 Front Panel Indicators

All indicators shall be clear LEDs. Clear LEDs shall not depend on a reflector or diffusion as part of its design. Clear LEDs shall not appear to be ON when exposed to ambient light. The following indicators shall be provided:

6.4.13.5.1 AC Line

A green AC Line indicator shall illuminate to indicate Operational input voltage is proper and the AC Line fuse is intact.

6.4.13.5.2 48 VDC

A green indicator shall illuminate to indicate the 48 VDC output is active and the fuse is intact.

6.4.13.5.3 24 VDC

A green indicator shall illuminate to indicate the 24 VDC output is active and the fuse is intact.

6.4.13.6 DC Voltage Test Jacks

Banana style test jacks shall be provided on the front panel for each DC output and DC ground. Mating banana plug spring width shall be 0.175 inches nominal.

6.4.13.7 DC Output Connector

The output connector is a Phoenix Contact #1825161 and mates with a Phoenix Contact #1825352 or equivalent. Pin #1 is the right most pin when viewed from the rear of the supply. See Table 18.

Table 17: Power Supply DC Output Connector

Pin	Function
1	+48VDC
2	48VDC Ground**
3	+24VDC

4	Reserved
5	24VDC Ground
6	Chassis Ground

**Note that the 48 VDC output shall be electrically isolated from the AC Line input and the 24VDC output. The 48VDC Ground (pin #2) shall be connected within the cabinet to the same MAINS Ground (Neutral) that the Cabinet Monitor Unit (CMU pin B32) is connected to.

6.5 Model 2220 Auxiliary Display Unit

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	No
Mechanical Dimensions	No

6.5.1 General

The Auxiliary Display Unit (ADU) is a modular display that provides an enhanced user interface for the ATC Cabinet Monitor Unit system. The ADU shall not be required in the cabinet for the 2212 CMU to operate.

The model 2220 provides a display function in a modular fashion for the installed cabinet monitor unit and utilizes an LED based channel output status display and a menu driven LCD display to view status, configuration, and logs. Five buttons are included to navigate the menu system. The CMU communicates to the model 2220 via Serial Bus #3 (SB#3).

Guidance: The Auxillary Display Unit is an optional assembly for the ATC Cabinet. Mechanical dimensions as well as connector type and pin assignments are not controlled by the standard. The following section, defining the Model 2220 Auxillary Display Unit, is one example of how the user interface for the Cabinet Monitor Unit can be realized.

The display data shall be created and formatted by the installed 2212 CMU.

6.5.2 Mechanical

The model 2220 shall be 1U (1.7") in height maximum and designed to mount into a 19 inch EIA rack. The maximum depth of the model 2220 shall be less than 3.0".

6.5.2.1 Power Connector

The 48VDC power is supplied with a Phoenix Contact 3-pin header 1776511, with a mating harness plug part number 1777992. Pin #1 is the rightmost pin when viewed from the connector end (back side of the ADU).

Table 18: ADU Power Connector Signals

Pin	Function
1	+48VDC
2	48VDC Ground**
3	Earth Ground

**The 48VDC Ground (pin #2) must be connected within the cabinet to the same MAINS GROUND of the Cabinet Monitor Unit (CMU pin B32).

6.5.2.2 SB#3 Connector

This port provides the EIA-485 data path to the CMU Cabinet Monitor Unit (CMU) using SB#3. A temperature rated cable assembly meeting CAT5 performance characteristics at a minimum with modular 8p8c connectors shall be used to make the connection. The cable ends shall be terminated per EIA/TIA-586B to maintain signal integrity. Cable and connectors shall meet the temperature rating of 11.5.4.1.1. The signals shall be MAINS GROUND referenced, which is the same Ground point as the CMU2212. Signal names are with respect to the model 2220.

Table 19: ADU SB#3 Signals

Pin	Function
1	Reserved
2	Reserved
3	Ground (MAINS GROUND / 48VDC GROUND)
4	RxDATA +
5	RxDATA -
6	Ground (MAINS GROUND / 48VDC GROUND)
7	TxDATA +
8	TxDATA -

6.5.3 Power

The model 2220 shall be powered from the cabinet 48 Vdc supply. Input voltage operating range shall be 22.0 Vdc minimum to 60.0 Vdc maximum. The model 2220 input voltage shall be referenced to the cabinet 48 Vdc supply Ground (CMU2212 MAINS GROUND).

6.5.4 Front Panel Indicators

All indicators shall be clear LEDs. Clear LEDs shall not depend on a reflector or diffusion as part of its design. Clear LEDs shall not appear to be ON when exposed to ambient light. The following indicators shall be provided:

6.5.4.1 Channel Status

The model 2220 shall provide 32 columns of LED indicators corresponding to channels 1 through 32. Each column shall be composed of a blue status indicator, a red status indicator, a yellow status indicator, and a green status indicator.

6.5.4.2 CMU Status

A four row by 20 column LCD display shall be provided for displaying data from the CMU2212. The LCD shall provide a backlight function and operate over the full cabinet temperature range. A heater shall be provided to meet the temperature requirement.

6.5.4.3 SB#3 RxD

A yellow indicator shall pulse On each time the model 2220 correctly receives a frame on SB#3 from the CMU2212.

6.5.5 Front Panel Controls

Five buttons shall be provided to navigate the CMU status display menu. The buttons shall be labeled HELP, NEXT, BACK, SEL, and EXIT.

6.6 Sensor Unit

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

6.6.1 General

The Sensor Unit functions as an input device used to sense field status conditions such as:

- NEMA 2-Channel Inductive Loop Detector with (2) Outputs and (2) Status
- NEMA 4-Channel Inductive Loop Detector with (4) Outputs and (4) Status
- Type 222 Inductive Loop Detector with (2) channels without Status
- Type 224 Inductive Loop Detector with (4) channels without Status
- Type 242L DC Isolator with (2) channels and (2) Status
- Type 244L DC Isolator with (4) channels and (4) Status
- Preemptor

6.6.2 Form Factor

Sensor Unit form factor shall conform to NEMA TS2 standard, Section 6.5.2.2.2 (see Figures 6-4, 6-5), for both 2 channel and 4 channel form factors.

6.6.3 Connector and Pin Assignments

Sensor Unit connector and pin assignments shall conform to NEMA TS2 standard Section 6.5.2.28 (see Table 6-2) for NEMA 4-Channel Inductive Loop Detector with four Outputs and four Status without INBUS.

For other Sensor Units listed in 6.6.1 above, the Loop connector pins shall be used for field wire inputs according to the sensor product specifications.

7 SLOTS

Slots include mechanical mounting hardware and electrical connectors necessary to house interchangeable modules.

7.1 Serial Interface Unit Slot

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

7.1.1 General

The Serial Interface Unit Slot functions to mechanically mount and to electrically connect the Serial Interface Unit (SIU2218)

7.1.2 Form Factor

The model 2218 Slot Form Factor shall mechanically and electrically accommodate the model 2218 as shown:

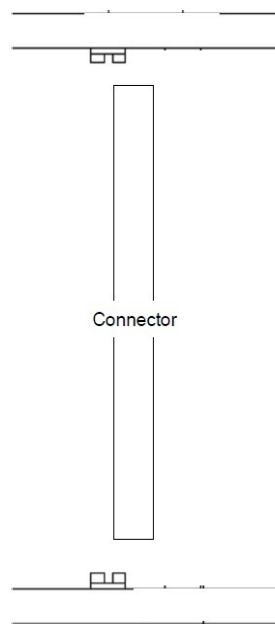


Figure 10: Model 2218 Slot Mechanical Details

The model 2218 Slot shall be capable of being mounted in a EIA 482.6 mm (19 in.) rack (EIA Standard RS-310) and shall not exceed 152.40 mm (6.0 in.) in height and have a

maximum depth of 215.90 mm (8.50 in.). Tolerance is ± 0.51 mm (0.02 in) unless otherwise noted. Connector is centered vertically between card guides. Distance between card guide channel surfaces is $4.52" \pm 0.02"$.

Refer to model 2218 section 6.1.10 of this standard for horizontal dimension from card guide centerline to centerline of model 2218 circuit board

7.1.3 Connector and Pin Assignments

Refer to model 2218 section 6.1.10.6 of this standard for mating connector type, orientation, and pin assignments

7.2 Cabinet Monitor Unit Slot

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

7.2.1 General

Cabinet Monitor Unit Slot functions to mechanically mount and to electrically connect the Cabinet Monitor Unit (CMU2212).

7.2.2 Form Factor

CMU2212 Slot Form Factor shall mechanically and electrically accommodate the CMU2212 as shown:

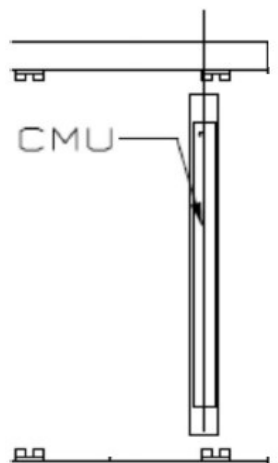


Figure 11: CMU Slot Mechanical Details

1. Refer to CMU2212 section 6.3.11 of this standard for horizontal dimension from card guide centerline to centerline of CMU2212 circuit board.

2. CMU is centered vertically between card guides
3. Distance between card guide channel surfaces is 4.31 ± 0.02 ".
4. Tolerance ± 0.51 mm (0.02 in) unless otherwise noted.

7.2.3 Connector and Pin Assignments

Refer to CMU2212 section 6.3.16 of this standard for connector pin assignments

7.3 IN Slot

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

7.3.1 General

IN Slot functions to mechanically mount and to electrically connect the Sensor Units.

7.3.2 Form Factor

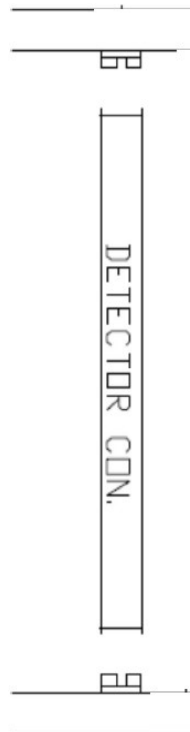


Figure 12: IN Slot Mechanical Details

The IN Slot shall be capable of being mounted in a EIA 482.6 mm (19 in.) rack (EIA Standard RS-310). It shall have a maximum depth of 215.90 mm (8.50 in.) and shall not

exceed 152.40 mm (6.0 in.) in height for shelf mount or 133.35 mm (5.25 in) (3U) for rack mount.

The IN Slot shall provide a 44 terminal, double row, 3.962 mm (0.156 in.) contact spacing, Cinch Jones card edge connector 50-44A-30M, or equivalent centered vertically for each Sensor module.

Detector is centered vertically between card guides. Distance between channel surfaces of card guides is 4.52" $\pm$ 0.02". Tolerance is $\pm$ 0.51 mm (0.02 in) unless otherwise noted.

7.3.3 Connector and Pin Assignments

Refer to Sensor section of this standard for connector orientation and pin assignments.

7.4 OUT Slot

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Type and Pin Assignments	Yes
Mechanical Dimensions	Yes

7.4.1 General

OUT Slot functions to mechanically mount and to electrically connect the 2202 HDSP/FU High Density Switch Pack and Flasher Unit.

7.4.2 Form Factor

OUT Slot Form Factor shall mechanically and electrically accommodate the output devices as shown:

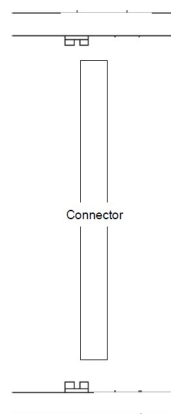


Figure 13: OUT Slot Mechanical Details

The connector shall be centered vertically between card guides. The distance between card guide channel surfaces is 4.52" $\pm$ 0.02". The tolerance is $\pm$ 0.51 mm (0.02 in) unless otherwise noted.

The OUT Slot shall be capable of being shelf mounted or mounted in an EIA 482.6 mm (19 in.) rack (EIA Standard RS-310). It shall have a maximum depth of 215.90 mm (8.50 in.) and shall not exceed 152.40 mm (6.0 in.) in height for shelf mount or 133.35 mm (5.25 in) (3U) for rack mount.

The OUT Slot shall provide a mating connector to DIN 41612 Type E series, 48-pin for each HDSP/FU module. Connector shall be positioned to mate with the HDSP/FU as detailed in that section.

7.4.3 Connector and Pin Assignments

Refer to Model 2202 HDSP/FU section 6.2.7 of this standard for mating connector type, orientation and pin assignments. Support for HDFU pins Channel 1 AUX (In, Sense, Out) and Channel 2 AUX (In, Sense, Out) is optional.

8 INTERFACES

Interfaces interconnect system elements shown in the Block Diagrams.

8.1 SIU IN Interface

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Types and Pin Assignments of SIU Slot and IN Slot	Yes
Mechanical Dimensions of the interface from SIU Slot and IN Slots	No

8.1.1 General

The SIU IN interface functions to electrically connect the IN SLOT connectors to the SIU connectors.

8.1.2 Form Factor

SIU IN mechanical connection from SIU to Sensor Slots is not controlled by this standard.

8.1.3 Input SIU 1 Pin Assignments

Detector functions for channels 25-120 shall be assigned to Input SIU 2 through 5 in the same sequence as shown in

Table 20. Channel 1 and Channel 2 shall be located in Slot #1 and increment sequentially. Signal names are with respect to the Controller Unit.

The Detector Reset outputs shall be connected to pin C of both assigned slots to provide for a two channel or four channel detector reset function.

Table 20: INPUT SIU 1 to IN SLOT Signal Mapping

INPUT SIU 1						
	Row A		Row B		Row C	
Pin#	Pin Name	Function	Pin Name	Function	Pin Name	Function
1	+24V	POWER	+24V	POWER	IO 47	CHANNEL 18 STATUS
2	IO 0	CHANNEL 1-4 RESET	IO 1	CHANNEL 5-8 RESET	IO 48	CHANNEL 19 STATUS
3	IO 2	CHANNEL 9-12 RESET	IO 3	CHANNEL 13-16 RESET	IO 49	CHANNEL 20 STATUS
4	IO 4	CHANNEL 17-20 RESET	IO 5	CHANNEL 21-24 RESET	IO 50	CHANNEL 21 STATUS
5	IO 6	CHANNEL 1 OUT	IO 7	CHANNEL 2 OUT	IO 51	CHANNEL 22 STATUS
6	IO 8	CHANNEL 3 OUT	IO 9	CHANNEL 4 OUT	IO 52	CHANNEL 23 STATUS
7	IO 10	CHANNEL 5 OUT	IO 11	CHANNEL 6 OUT	IO 53	CHANNEL 24 STATUS
8	IO 12	CHANNEL 7 OUT	IO 13	CHANNEL 8 OUT	SB#1 RXD+	
9	IO 14	CHANNEL 9 OUT	IO 15	CHANNEL 10 OUT	SB#1 RXD-	
10	IO 16	CHANNEL 11 OUT	IO 17	CHANNEL 12 OUT	SB#1 TXD+	
11	IO 18	CHANNEL 13 OUT	IO 19	CHANNEL 14 OUT	SB#1 TXD-	
12	IO 20	CHANNEL 15 OUT	IO 21	CHANNEL 16 OUT	SB#1 RXC+	
13	IO 22	CHANNEL 17 OUT	IO 23	CHANNEL 18 OUT	SB#1 RXC-	
14	IO 24	CHANNEL 19 OUT	IO 25	CHANNEL 20 OUT	SB#1 TXC+	
15	IO 26	CHANNEL 21 OUT	IO 27	CHANNEL 22 OUT	SB#1 TXC-	
16	IO 28	CHANNEL 23 OUT	IO 29	CHANNEL 24 OUT	LSYNC+	
17	IO 30	CHANNEL 1 STATUS	IO 31	CHANNEL 2 STATUS	LSYNC-	
18	IO 32	CHANNEL 3 STATUS	IO 33	CHANNEL 4 STATUS	NRESET+	
19	IO 34	CHANNEL 5 STATUS	IO 35	CHANNEL 6 STATUS	NRESET-	
20	IO 36	CHANNEL 7 STATUS	IO 37	CHANNEL 8 STATUS	ASSY ADR	
21	IO 38	CHANNEL 9 STATUS	IO 39	CHANNEL 10 STATUS	INBUS RTS	
22	IO 40	CHANNEL 11 STATUS	IO 41	CHANNEL 12 STATUS	SB#2 RXD+	
23	IO 42	CHANNEL 13 STATUS	IO 43	CHANNEL 14 STATUS	SB#2 RXD-	
24	IO 44	CHANNEL 15 STATUS	IO 45	CHANNEL 16 STATUS	SB#2 TXD+	
25	IO 46	CHANNEL 17 STATUS	OPTO IN 1	PED1	SB#2 TXD-	
26	OPTO IN 2	PED2	OPTO IN 3	PED3	SB#2 RXC+	
27	OPTO IN 4	PED4	OPTO GND	PED GND	SB#2 RXC-	
28	ADR 0		ADR 1		SB#2 TXC+	
29	ADR 2		ADR 3		SB#2 TXC-	
30	INBUS TXD		INBUS RXD		INBUS TXC	
31	EQGND		AC LINE REF		INBUS RXC	
32	DCGND2		DCGND2		DCGND2	

8.2 SIU OUT Interface

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Types and Pin Assignments of SIU Slot and OUT Slot	Yes
Mechanical Dimensions of the interface from SIU Out to OUT Slot	No

8.2.1 General

The SIU OUT interface functions to electrically connect the OUT SLOT connectors to the SIU connectors.

8.2.2 Form Factor

The mechanical form factor of the OUT SLOT interface to the 2202 HDSP/FU is not controlled by this standard.

8.2.3 Connector Pin Assignments

HDSP functions for channels 17-32 shall be assigned to Output SIU 2 in the same sequence as shown in Table 22. Control and SB#1 signal names are with respect to the Controller Unit.

Table 21: First Output SIU Opto Assignments

Pin	First Output Functionality	Housing Type
Opto Input 1	Manual Control Enable (MCE)	All
Opto Input 2	Interval Advance (IA)	All
Opto Input 3	Stop Time (STOPTIME)	All
Opto Input 4	Local Flash (FLASH)	All
Opto Input Common	48VDC	All

Table 22: Output SIU 1 to OUT SLOT Signal Mapping

Switch Pack SP1- SP8						
	Row A		Row B		Row C	
Pin	Pin Name	Switch Pack	Pin Name	Switch Pack	Pin Name	Switch Pack
1	+24V		+24V		IO 47	SP8 CH2 RED

Switch Pack SP1- SP8						
	Row A		Row B		Row C	
Pin	Pin Name	Switch Pack	Pin Name	Switch Pack	Pin Name	Switch Pack
2	IO 0	SP1 CH1 RED	IO 1	SP1 CH1 YEL	IO 48	SP8 CH2 YEL
3	IO 2	SP1 CH1 GRN	IO 3	SP1 CH2 RED	IO 49	SP8 CH2 GRN
4	IO 4	SP1 CH2 YEL	IO 5	SP1 CH2 GRN	IO 50	
5	IO 6	SP2 CH1 RED	IO 7	SP2 CH1 YEL	IO 51	
6	IO 8	SP2 CH1 GRN	IO 9	SP2 CH2 RED	IO 52	
7	IO 10	SP2 CH2 YEL	IO 11	SP2 CH2 GRN	IO 53	
8	IO 12	SP3 CH1 RED	IO 13	SP3 CH1 YEL	SB#1 RXD+	
9	IO 14	SP3 CH1 GRN	IO 15	SP3 CH2 RED	SB#1 RXD-	
10	IO 16	SP3 CH2 YEL	IO 17	SP3 CH2 GRN	SB#1 TXD+	
11	IO 18	SP4 CH1 RED	IO 19	SP4 CH1 YEL	SB#1 TXD-	
12	IO 20	SP4 CH1 GRN	IO 21	SP4 CH2 RED	SB#1 RXC+	
13	IO 22	SP4 CH2 YEL	IO 23	SP4 CH2 GRN	SB#1 RXC-	
14	IO 24	SP5 CH1 RED	IO 25	SP5 CH1 YEL	SB#1 TXC+	
15	IO 26	SP5 CH1 GRN	IO 27	SP5 CH2 RED	SB#1 TXC-	
16	IO 28	SP5 CH2 YEL	IO 29	SP5 CH2 GRN	LSYNC+	
17	IO 30	SP6 CH1 RED	IO 31	SP6 CH1 YEL	LSYNC-	
18	IO 32	SP6 CH1 GRN	IO 33	SP6 CH2 RED	NRESET+	
19	IO 34	SP6 CH2 YEL	IO 35	SP6 CH2 GRN	NRESET-	
20	IO 36	SP7 CH1 RED	IO 37	SP7 CH1 YEL	ASSY ADR	
21	IO 38	SP7 CH1 GRN	IO 39	SP7 CH2 RED	INBUS RTS	
22	IO 40	SP7 CH2	IO 41	SP7 CH2	SB#2	

Switch Pack SP1- SP8						
	Row A		Row B		Row C	
Pin	Pin Name	Switch Pack	Pin Name	Switch Pack	Pin Name	Switch Pack
		YEL		GRN	RXD+	
23	IO 42	SP8 CH1 RED	IO 43	SP8 CH1 YEL	SB#2 RXD-	
24	IO 44	SP8 CH1 GRN	IO 45	--	SB#2 TXD+	
25	IO 46	--	OPTO IN 1	MCE	SB#2 TXD-	
26	OPTO IN 2	IA	OPTO IN 3	STOPTIME	SB#2 RXC+	
27	OPTO IN 4	FLASH	OPTO GND	MAINS_RT 48VDC_RT N	SB#2 RXC-	
28	ADR 0		ADR 1		SB#2 TXC+	
29	ADR 2		ADR 3		SB#2 TXC-	
30	INBUS TXD		INBUS RXD		INBUS TXC	
31	EQGND		AC LINE REF		INBUS RXC	
32	DCGND2		DCGND2		DCGND2	

8.3 SB#1 / SB#2 Interface

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Types and Pin Assignments	Yes
Mechanical Dimensions of the interface from model 2218 to Controller Unit	No

8.3.1 General

8.3.1.1 Serial Bus 1

Serial Bus #1 (SB#1) communicates real-time information required to operate the system. It handles the highest priority, time-sensitive data exchange between the SIUs in the input and output functionalities, controller unit and the CMU. SB#1 is designed for a communications rate of 614,400 bits per second.

8.3.1.2 Serial Bus 2

Serial Bus #2 (SB#2) communicates less time critical information between the Controller Unit and sensors in the input functionalities. An example might be to use NEMA TS2 Port1 protocol on SB#2 to communicate with NEMA TS2 devices. Serial bus 2 is designed for an allowable communications rate of up to 614,400 bits per second.

8.3.1.3 Terminator Unit

An SB#1 and SB#2 terminator unit shall be provided. There are two acceptable methods for implementing termination. Method #1 is accomplished by installing the terminator unit into the DC POWER / COMMUNICATIONS ASSEMBLY DB25S End Connector if that functionality is provided in the cabinet. Method #2 is accomplished by installing the termination unit into the "THROUGH" connector of the last assembly connected to the SB#1 / SB#2 serial chain when a serial "Daisy Chain" method, consisting of "IN" and "THROUGH" on each assembly is used. In either method the terminator unit should be installed at the furthest end of the communications cable from the ATC.

The source of +5VDC and Ground for powering the termination bias resistors shall be derived from Pins 12 and 13, respectively, of the SB#1 / SB#2, 25-pin, D-Sub connector. The terminator unit shall provide a 150 ohm termination resistor between the RxD+ and RxD-, the TxD+ and TxD-, the RxC+ and RxC-, and the TxC+ and TxC- pairs for both serial bus #1 and serial bus #2. The terminator unit shall also provide 1K ohm DC bias resistors from +5VDC ISO to the RxD+, the RxC+, the TxD+, and the TxC+ of both SB#1 and SB#2. The terminator unit shall provide 1K ohm DC bias resistors from ISO GND to the RxD-, the RxC-, the TxD-, and the TxC- of both SB#1 and SB#2. See Figure 15.

8.3.1.4 Form Factor

SB#1 and SB#2 are included in one SB#1 / SB#2 physical interface consisting of a single communications cable terminated by a male 25 pin "D" connector at one end. Connector shall include two 4-40 male screw locks. Cable wiring shall be per Table 23. Cable type and destination connector is not controlled by the standard, but at a minimum must be CAT-5 compliant to carry EIA-485 balanced differential signals at 614,400 bits per second. See Figure 14.

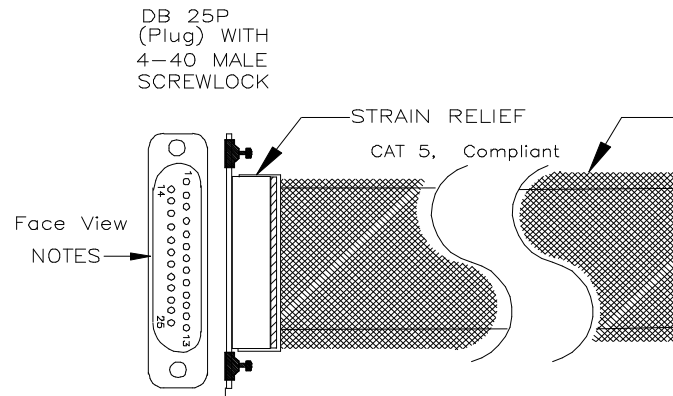


Figure 14: SB#1/SB#2 Communications Interface Form Factor

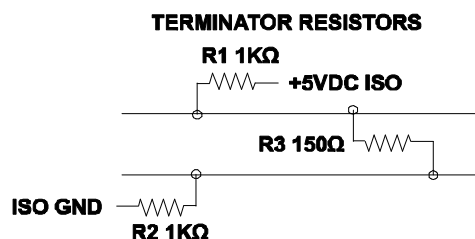
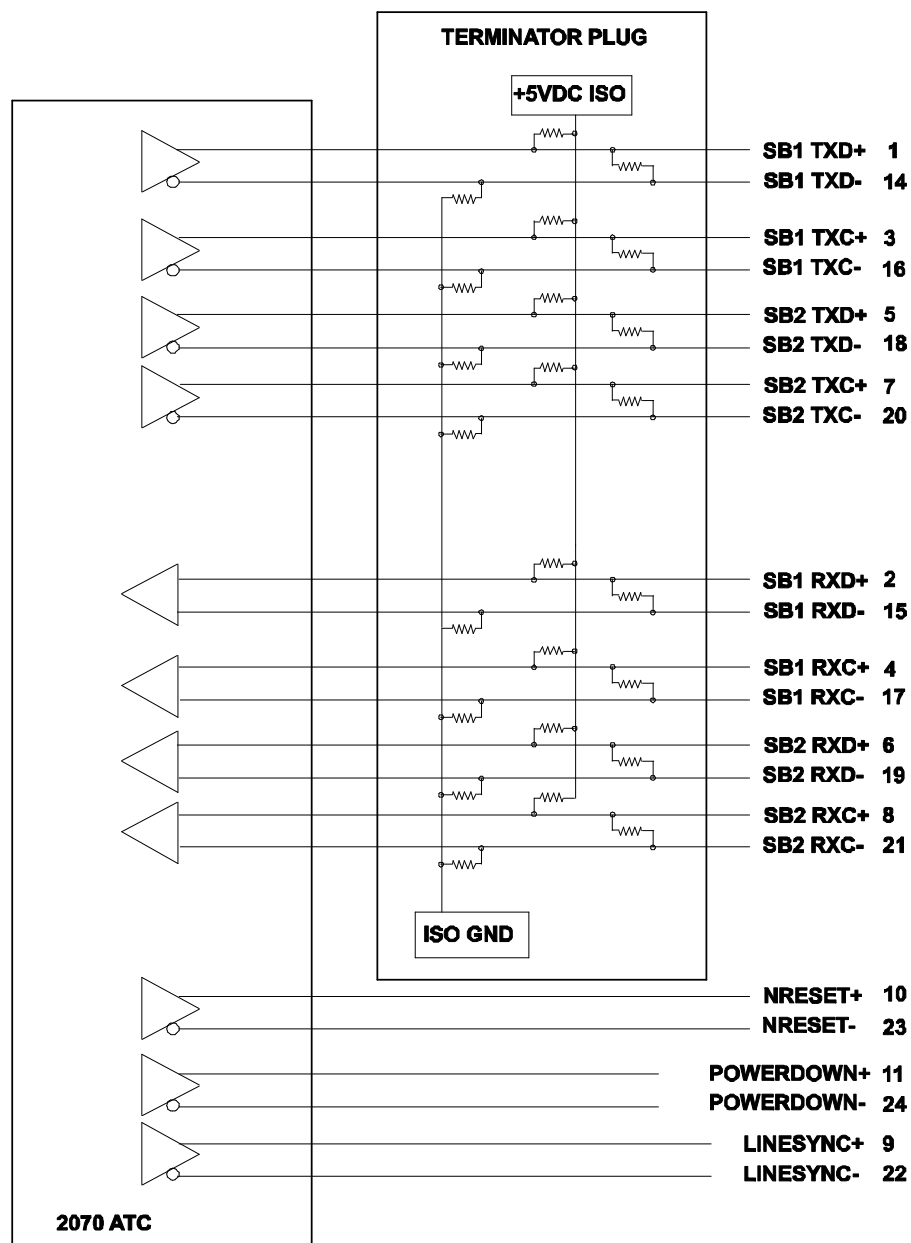


Figure 15: SB#1 / SB#2 Terminator Unit

The R3 value is selected to match the bus characteristic impedance. The value shown for R3 is selected to match characteristic impedance of the transmission line used. Proper selection is verified by the signal waveform measured as being critically-damped without reflections. The R1 = R2 value is selected to bias the receivers to MARK state when not driven.

8.3.2 Connector Pin Assignment

Table 23: SB#1 / SB#2 Connector Pin Assignment

Pin	AT the CU	AT the SIU	AT the CMU	At Port 1 Connector (DA-15S)
1	SB#1 TxD+	SB#1 RxD+	SB#1 RxD+	SB#1 TxD+
2	SB#1 RxD+	SB#1 TxD+	SB#1 TxD+	ISO GND
3	SB#1 TxC+	SB#1 RxC+	SB#1 RxC+	SB#1 TxC+
4	SB#1 RxC+	SB#1 TxC+	SB#1 TxC+	ISO GND
5	SB#2 TxD+	SB#2 RxD+	SB#2 RxD+	SB#1 RxD+
6	SB#2 RxD+	SB#2 TxD+	SB#2 TxD+	ISO GND
7	SB#2 TxC+	SB#2 RxC+	SB#2 RxC+	SB#1 RxC+
8	SB#2 RxC+	SB#2 TxC+	SB#2 TxC+	ISO GND
9	LINE SYNC+	LINE SYNC+	LINE SYNC+	SB#1 TxD-
10	NRESET+	NRESET+	NRESET+	(RESERVED)
11	POWER DOWN+	--	POWER DOWN+	SB#1 TxC-
12	+5 VDC ISO	--	--	EQ GND
13	ISO GND	--	--	SB#1 RxD-
14	SB#1 TxD-	SB#1 RxD-	SB#1 RxD-	(RESERVED)
15	SB#1 RxD-	SB#1 TxD-	SB#1 TxD-	SB#1 RxC-
16	SB#1 TxC-	SB#1 RxC-	SB#1 RxC-	
17	SB#1 RxC-	SB#1 TxC-	SB#1 TxC-	
18	SB#2 TxD-	SB#2 RxD-	SB#2 RxD-	
19	SB#2 RxD-	SB#2 TxD-	SB#2 TxD-	
20	SB#2 TxC-	SB#2 RxC-	SB#2 RxC-	
21	SB#2 RxC-	SB#2 TxC-	SB#2 TxC-	
22	LINE SYNC-	LINE SYNC-	LINE SYNC-	
23	NRESET-	NRESET-	NRESET-	
24	POWER DOWN-	--	POWER DOWN-	
25	EQ GND		--	

8.3.3 Electrical

SB#1 / SB#2 Interface shall consist of the following interface links conforming to the requirements of the Electronic Industries Association EIA-485 Standard for Electrical Characteristics of Generators and Receivers for use in Balanced Digital Multipoint Systems, dated April 1983:

- SB#1 of two interface links (SB#1 TXD $\pm$ and SB#1 RXD $\pm$)
- SB#2 of two interface links (SB#2 TXD $\pm$ and SB#2 RXD $\pm$)
- LINE SYNC of one interface link (LINE SYNC $\pm$)
- NRESET of one interface link (NRESET $\pm$)
- POWER DOWN of one interface link (POWERDOWN $\pm$)

Where differences occur between the EIA-485 standard and this document, this document shall govern

SB#1 / SB#2 Interface shall also include the following power conductors:

- +5 VDC ISO positive bias for Terminator Unit depicted in
- Figure 5
- ISO GND negative bias for Terminator Unit depicted in
- Figure 5
- EQ GND Equipment Ground

8.4 SB#3 Interface

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Types and Pin Assignments	Yes
Mechanical Dimensions of the interface from CMU to HDSP/FU	No

8.4.1 General

The 2202 HDSP/FU and ADU shall respond to serial data frames transmitted by the CMU2212 on the Serial Bus #3 (SB#3).

8.4.2 SB#3 Terminations

No termination loads shall be provided by the 2202 HDSP/FU. Serial Bus #3 termination is provided by the CMU2212 (CMU RXD) and the model 2220 ADU (CMU TXD) if installed. A 120 ohm resistor shall be provided in the model 2220 ADU between RxD+ and RxD-.

8.4.3 SB#3 Electrical

The SB#3 shall consist of two interface links (TXD $\pm$ and RXD $\pm$) conforming to the requirements of the Electronic Industries Association EIA-485, Standard for Electrical Characteristics of Generators and Receivers for use in Balanced Digital Multipoint Systems, dated April 1983. Where differences occur between the EIA-485 standard and this document, this document shall govern.

SB#3 RXD $\pm$ shall be inputs to the 2202 HDSP/FU and ADU from the CMU-2212. SB#3 TXD $\pm$ shall be outputs to the CMU-2212 from the 2202 HDSP/FU and ADU. All voltage

potentials on the SB#3 interface shall be referenced to the same MAINS Ground (Neutral) that the Cabinet Monitor Unit (CMU pin B32) is connected to.

8.4.4 Form Factor

SB#3 interface on cabinet shall consist of (2) RJ-45 8-pin sockets wired in parallel for daisy-chain:

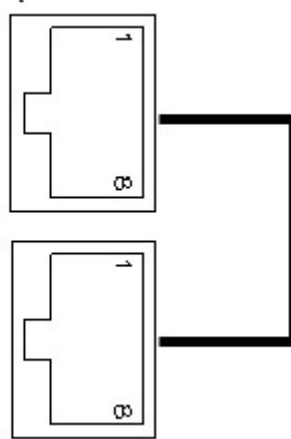


Figure 16: SB#3 Communications Interface Form Factor

SB#3 interface cable shall consist of (2) RJ-45 8-pin plugs on sheathed, color coded cable.

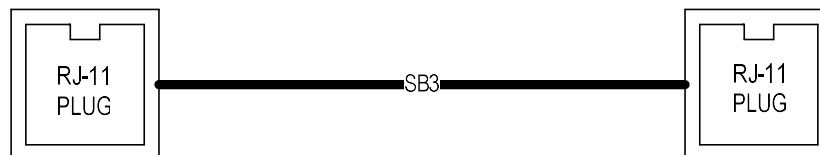


Figure 17: SB#3 Communications Cable Form Factor

8.4.5 Connector Pin Assignments

Table 24: SB#3 Connector Pin Assignment

Pin	Pair Color	AT the CMU	AT the HDSP/FU, ADU
1	White Orange	SB#3 TxC+	SB#3 RxC+
2	Orange	SB#3 TxC-	SB#3 RxC-
3	White Green	Ground (MAINS GROUND / 48VDC GROUND)	Ground (MAINS GROUND / 48VDC GROUND)
4	Blue	SB#3 TxD+	SB#3 RxD+
5	White Blue	SB#3 TxD-	SB#3 RxD-
6	Green	Ground (MAINS GROUND / 48VDC GROUND)	Ground (MAINS GROUND / 48VDC GROUND)

7	White Brown	SB#3 RxD+	SB#3 TxD+
8	Brown	SB#3 RxD-	SB#3 TxD-

8.5 CC Interface

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Types and Pin Assignments	Yes
Mechanical Dimensions of the interface from CMU to Door Switch	No

8.5.1 General

The Cabinet Control (CC) interface functions to electrically connect the various cabinet switches to the functionality containing the CMU.

8.5.2 Form Factor

The CC Interface shall use a Molex 0003091157, with pin number 0002091119, or equivalent intermatable connector on one end and is either hardwired to the appropriate door and Police Panel (PP) switches on the other end or is point-to-point, parallel, wired to an identical connector on the other end, depending on cabinet architecture. A minimum of AWG-22 wire shall be used for all conductors. Free swinging cables shall always be sheathed while fixed cable bundles need only be dressed with zip ties or lacing twine.

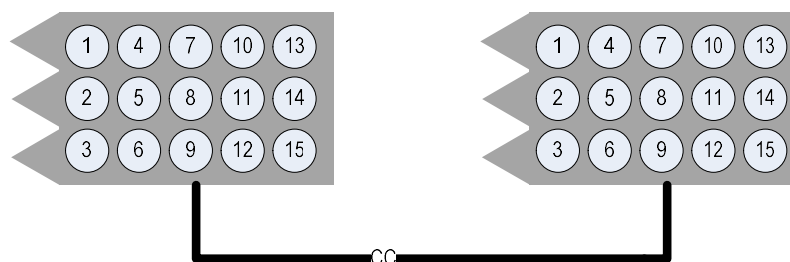


Figure 18: CC Interface Form Factor

8.5.3 Connector Pin Assignments

Table 25: CC Connector Pin Assignments

Pins	Function
1 – 1	MC Coil from PP
2 – 2	48VDC to PP (and Fan if used)
3 - 3	FTR Coil from PP
4 - 4	Reserved (previously Signal

Pins	Function
	Flash)
5 - 5	Front Door Switch
6 - 6	From Auto Flash Switch
7 - 7	Reserved
8 - 8	From Rear Door Switch
9 - 9	CB Trip Status CMU to PP
10 - 10	No Connect
11 - 11	48VDC GROUND
12 - 12	+48VDC
13 - 13	MANUAL CONTROL ENABLE In
14 - 14	INTERVAL ADV Out
15 - 15	STOP TIME Switch

Table 26: Definitions of Terms relating to switches in the Police Panel

Term	Definition
AUTO	The condition whereby both the MC and FTR are energized and the HDSP outputs are connected to the field terminals. The LF status to the CMU is inactive.
FLASH	The condition whereby both the MC and FTR are de-energized and the HDFU outputs are connected to the field terminals. The LF status to the CMU is active.
SIGNALS ON	The state whereby either the AUTO or FLASH condition is present, depending on the position of the AUTO/FLASH switch.
SIGNALS OFF	The condition whereby the MC is deenergized and the FTR is energized and there is no energy is present on the field terminals. The LF status to the CMU is active.

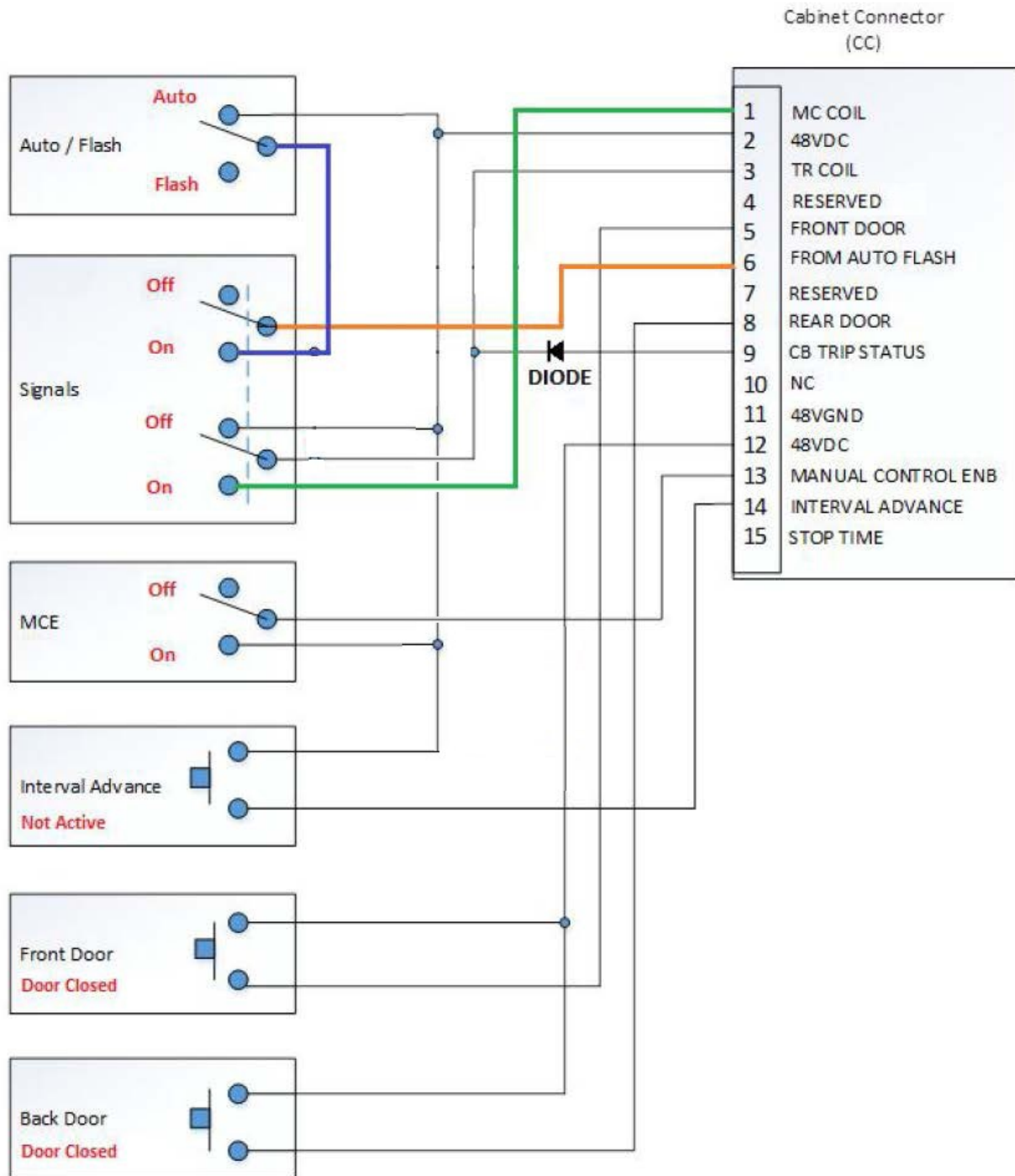


Figure 19: CC Interface Connection Diagram

8.6 CDC Interface

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Types and Pin Assignments	Yes
Mechanical Dimensions of the interface	No

8.6.1 General

The Cabinet D Connector (CDC) interface functions to provide access to the four opto-isolator inputs on each SIU. These inputs may be used to facilitate isolated pedestrian switch signals or other signals requiring isolation.

The CDC interface is optional, but if included, shall conform to this section of the standard.

8.6.1.1 Output SIU #1 Opto Inputs

The Manual Control Enable (MCE), Local Flash (FLASH) switch, Interval Advance (IA) and Stop Time (STOPTIME) switch signals shall be a 48Vdc signal with a series resistor (see 6.1.5.2.2). They shall be low voltage signals referenced to 48VDC GROUND. CDC pin 5 (Opto Common) shall be referenced to 48VDC GROUND.

8.6.1.2 Form Factor

The CDC Interface consists of 9-pin “D” plugs parallel wired (point to point) with AWG-24 minimum wire size. The CDC Interface consists of 9-pin D-subminiature socket connectors (DE9S) with #4-40 mating side screw jacks.

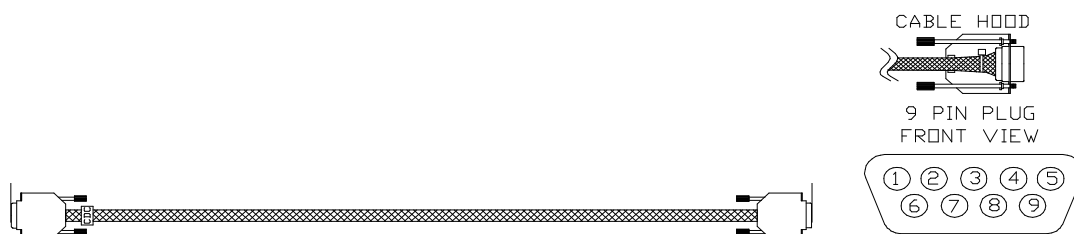


Figure 20: CDC Interface Form Factor

8.6.1.3 CDC Connector Pin Assignments

Table 27: CDC Connector Pin Assignments

DE9S Pin #	SIU Out #1 (or Combo #1)	SIU Out #2	SIU In #1 (or Combo #2)	SIU In #x (x>1)
1	MCE	Opto O2-1	PED1	Opto Ix-1
2	IA	Opto O2-2	PED2	Opto Ix-2
3	STOPTIME	Opto O2-3	PED3	Opto Ix-3
4	FLASH	Opto O2-4	PED4	Opto Ix-4
5	PP-Common	Opto O2-Common	PED GND	Opto Ix-Common
6	(Reserved)	(Reserved)	(Reserved)	(Reserved)
7	(Reserved)	(Reserved)	(Reserved)	(Reserved)
8	CMU Test Reset	Spare 1	Spare3	Spare ...
9	24VDC Ground	Spare 2	Spare4	Spare ...

Note: CMU Test Reset is intended for testing purposes only.

8.7 Output Termination Functionality

Standardization Level	
Description	Controlled by Standard?
Interface Signal Names and Electrical Characteristics	Yes
Connector Types and Pin Assignments	Yes
Mechanical Dimensions of the interface from OUT SLOT to Field Wires	No

8.7.1 General

The Output Termination Functionality interface functions to:

- Provide the electrical connection point between the Signal Head Field Wires and the Output Functionalities' OUT SLOT connectors.
- Physically program the signal flash either as RED or YELLOW
- Under control of the CMU, selects between the HDSP outputs or the Flasher, as the source for feeding the Signal Head Field Wires.

8.7.2 Functional Characteristics

The electrical characteristics and signals of the Output Termination Functionality interface shall be controlled by the standard. The Output Termination Functionality interface shall functionally conform to Figure 21 , which depicts the circuitry required for each High Density Switch Pack, in a cabinet assembly, that provides two channels of RED, YELLOW, and GREEN outputs (six physical outputs).

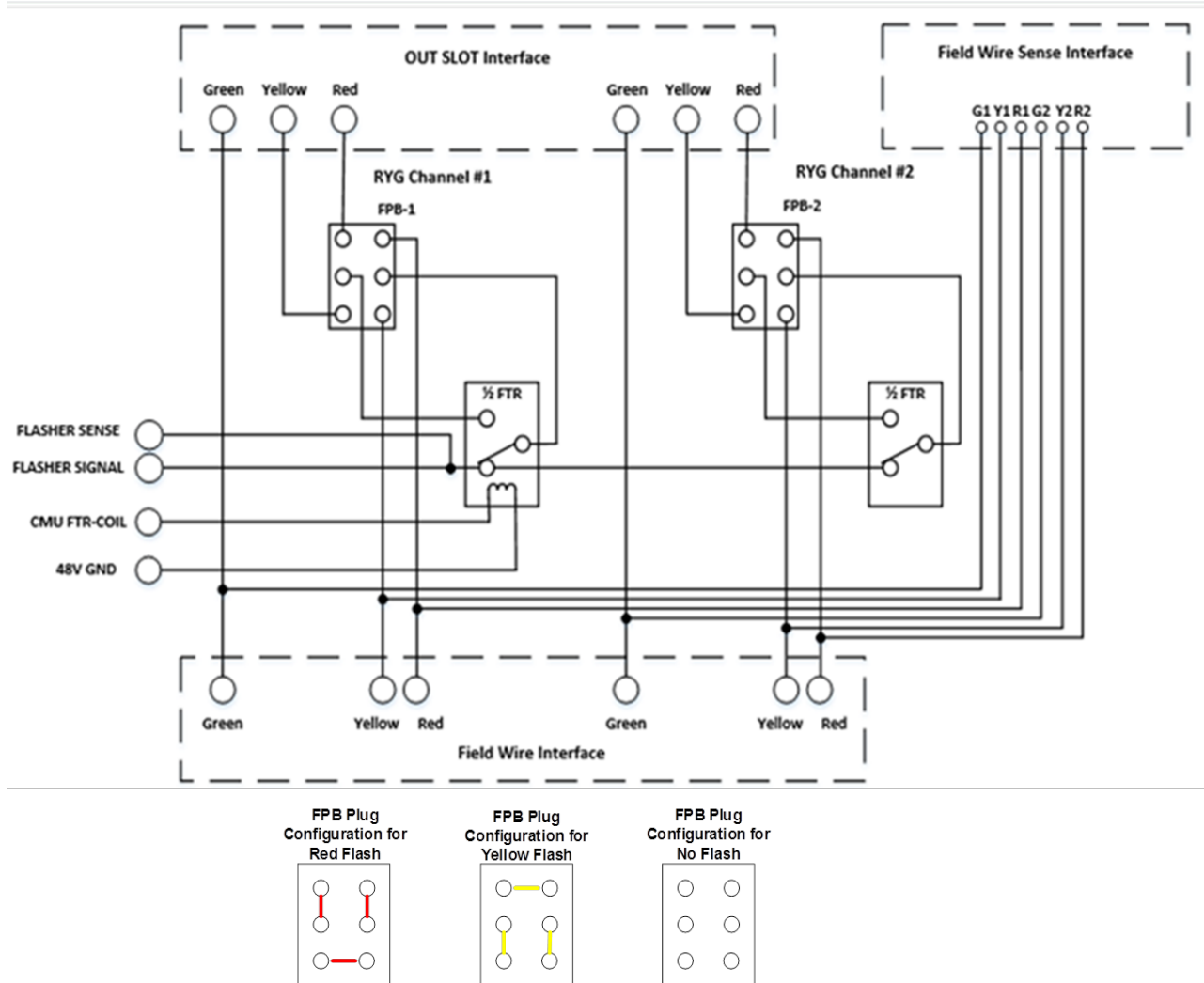


Figure 21: Termination Interface for 2 Switch Pack Channels

8.7.2.1 OUT SLOT Interface

The OUT SLOT interface shall electrically connect to its associated OUT SLOT located on the Output Functionality. The HDSP in that slot controls two sets of RED, YELLOW and GREEN signal indicators.

8.7.2.2 Flash Programming Blocks

Each flash programming block (FPB) shall provide Flash Programming for its HDSP channel. The FPBs shall accept RED plugs to program RED flash and shall accept YELLOW plugs to program YELLOW flash as shown in Figure 21. When a Red plug is installed, the Red signal from its associated HDSP shall be routed through the FTR contacts allowing the FTR to select either the HDSP Red signal or the Flasher signal to be routed to the Field Wire Interface. Yellow FPBs shall operate in a similar manner. FPB1, FPB2 and the programming jumpers shall be electrically sized for the load.

8.7.2.3 Flash Transfer Relay

The Flash Transfer Relay (FTRn) shall provide Flash Transfer that does the following:

- Connects FTR1 field wires and FTR2 field wires to the HDSP outputs when the FTR coil is energized
- Connects FTR1 field wires and FTR2 field wires to the HDFU outputs when the FTR coil is de-energized.
- Connects the FTR coil to FTR DRIVE signal controlled by the CMU. FTR DRIVE is referenced to 48VDC Ground.

The FTR contacts shall be electrically sized for the load. The FTR coil pull-in and drop-out specifications shall provide operation as described above over the service voltage operating range.

8.7.3 Field Wire Sense

Each HDSP includes a sense wire input associated with each of its outputs. The Field Wire Sense Interface shall provide a connection between the HDSP sense inputs and the field wires attached to the Field Wire Interface. The conductors of the Field Wire Sense Interfaces shall be attached as close as possible to the field wire terminals. The sense conductors shall be non-current carrying providing a remote measurement of the field wire voltage that avoids a resistive voltage drop.

8.7.3.1 Flasher Sense Interface

Each HDFU includes a sense wire input associated with each of its outputs. The Flasher Wire Sense Interface shall provide a connection between the HDFU sense inputs and the Flash Signal inputs of the FTRs. The conductors of the Flasher Sense Interfaces shall be attached as close as possible to the FTR terminals. The sense conductors shall be non-current carrying providing a remote measurement of the field wire voltage that avoids a resistive voltage drop.

8.7.4 Form Factor

The mechanical form factor of the Termination interface is not controlled by this standard, including the flash programming blocks, flash transfer relay, and connectors.

8.7.5 Field Wire Interface

The Field Wire Interface connects the field signal loads controlled by the ATCC. The Field Wire Interface conductors and contacts shall be electrically sized for a minimum of 3 Amps.

9 PROTOCOLS

9.1 SB#1 Communications Protocol

9.1.1 Electrical

The Serial Bus #1 shall consist of four interface links (TXD $\pm$ and RXD $\pm$, TXC $\pm$ and RXC $\pm$) conforming to the requirements of the TIA-485-A, *Electrical Characteristics of Generators and Receivers for use in Balanced Digital Multipoint Systems*. Where differences occur between the TIA-485-A standard and this document, this document shall govern.

All voltage potentials on the SB#1 interface shall be referenced to +24VDC Ground.

9.1.2 Data Link Layer

The data link layer protocol is based on a subset of HDLC as defined by ISO/IEC 13239:2002. Each frame shall consist of the following fields:

Flag byte	= 0x7E
Address byte	= 0x01 through 0x20
Control byte	= 0x83
Information field	= defined below in Sections 9.1.5, 9.1.6, and 9.1.7
Frame Check Sequence	= 16 bit FCS procedure defined in clause 2.5.2 of ISO/IEC 13239:2002.
Flag byte	= 0x7E

Transmission shall be in synchronous mode as defined by clause 4.3.1 of ISO/IEC 13239:2002. The format shall be 8 bit data at 614,400 bits per second ($\pm 2\%$). The Primary Station (ATC) shall keep the Tx Clock and Tx Data drivers enabled continuously. Mark Idle shall be added during periods without data transmission from the Primary Station. A secondary transmitting station shall ensure that the last bit of its last byte has been transmitted for the full bit time prior to disabling its transmission clock or transmission drivers.

9.1.3 Procedures

Frames transmitted by the ATC shall be referred to as command frames and frames transmitted by the secondary stations shall be referred to as response frames. Response frames shall only be transmitted as a result of correctly receiving a command frame. The first eight bits in each information field shall contain the Frame Type number.

Reserved bits in any message shall always be set to zero by the transmitting station.

9.1.4 Message Timing

The secondary station shall begin its response to command frames from the ATC within a designated period of time following the correct reception of a complete command frame including the closing flag. This period shall be known as the Service Time and

shall have a maximum value of 500 microseconds. The secondary station link output shall be in the high impedance state outside of the interval defined by the Service Time plus Response Time.

The secondary station shall complete its transmission of the response frame including the closing flag within a designated time known as the Response Time, depending on the number of bytes transmitted in the response frame. The secondary station link output shall transition to a high impedance state a maximum of 25 microseconds following the transmission of the closing flag.

Following the transmission of each command frame, there shall be a Dead Time during which the ATC does not transmit. This Dead Time shall be a minimum of the Service Time plus the Response Time.

9.1.5 SB#1 Frame Types

The frame type shall be determined by the value of the first byte of the message. The command frames type values 112 - 127 and associated response frame type 240 - 255 are allocated to the manufacturer diagnostics. All other frame types not called out are reserved. The command-response frame type values shall be as follows:

Table 28: SB#1 Frame Types

<i>Module Command</i>	<i>I/O Module Response</i>	<i>Description</i>	<i>Destination Device</i>
49	177	Request Module Status	SIU
50	178	MILLISECOND CTR. Mgmt.	SIU
51	179	Configure Inputs	SIU
52	180	Poll Raw Input Data	SIU
53	181	Poll Filtered Input Data	SIU
54	182	Poll Input Transition Buffer	SIU
55	183	Set Outputs	SIU
56	184	Configure Input Tracking Functions	SIU
57	185	Configure Complex Output Functions	SIU
58	186	Reserved	--
59	187	Reserved	--
60	188	Module Identification	SIU, CMU
61	189	Reserved (deprecated)	--
62	190	Send to Local Flash	CMU
63	191	Reserved	--
64	192	Reserved	--
65	193	Reserved (deprecated)	--
66	---	Time / Date	ALL
67	195	Switch Pack Drivers (short)	CMU

<i>Module Command</i>	<i>I/O Module Response</i>	<i>Description</i>	<i>Destination Device</i>
80	208	Module Description	SIU, CMU
81	209	Switch Pack Drivers (long)	CMU
82	210	CMU Configuration	CMU
83	211	Switch Pack Drivers (high Res)	CMU

Messages 62/190, 67/195, 81/209, 82/210 and 83/211 shall be for the CMU. See CMU2212 SB#1 Frame Types section 9.1.7 for command and response frames. Message 66 is a broadcast message to Address 255 containing the current time. Any device may receive and process this message if it has the software capacity.

9.1.5.1 Module Identification

The Module Identification command frame shall be used to request the identification value for the SIU and CMU. Reply message shall use the following addresses: ATC field I/Os shall respond with address 20. SIUs respond with their own address ranging from 0-14. CMUs respond with their own addresses, ranging from 15-18.

The command and response frames shall be shown as follows:

Table 29: Module Identification Command

<i>Description</i>	<i>msb</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number= 60)	0	0	1	1	1	1	0	0		Byte 1

Table 30: Module Identification Response

<i>Description</i>	<i>msb</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number= 188)	1	0	1	1	1	1	0	0		Byte 1
model 2218 ID byte	x	x	x	x	x	x	x	x		Byte 2

9.1.5.2 Type 66 Command _Time and Date Broadcast

Table 31: Type 66 Command

<i>Byte #</i>	<i>Contents</i>	<i>Description</i>
1	66	Frame Type
2	0x01:0x0C	Month

<i>Byte #</i>	<i>Contents</i>	<i>Description</i>
3	0x01:0x1F	Day
4	0x00:0x63	Year
5	0x00:0x17	Hour
6	0x00:0x3B	Minutes
7	0x00:0x3B	Seconds
8	0x00:0x09	Tenth Seconds

This frame shall be transmitted from the ATC once every second. The destination address shall be the all-station address of 255 (0xFF). No response from secondary devices is allowed to the all station address. The transmission of this frame shall begin within ± 100 milliseconds of the ATC real time contained within the frame.

9.1.5.3 Module Description

The Module Description command frame shall be used to request the device type for the model 2218 and CMU. A text string is provided for device properties.

Table 32: Module Description Command

<i>Byte #</i>	<i>Contents</i>	<i>Description</i>
1	80	Frame Type

Table 33: Module Description Response

<i>Byte #</i>	<i>Contents</i>	<i>Description</i>
1	208	Frame Type
2	Device	Enumerated Device Type 1: CMU-212 2: CMU-2212-LV 3: CMU-2212-HV 4: CMU-2212-VHV (Reserved) 5: SIU-218 6: SIU-2218 7:255 Reserved
3:42	ASCII Text	A packed 40 character field describing the device properties. Allowable ASCII characters are 020h through 07Eh. If less than 40 characters are used, the unused locations shall be set to 00h. Default

Byte #	Contents	Description
		programming shall be 00h.

9.1.6 SIU2218 SB#1 Frame Types

9.1.6.1 Request Module Status

The command shall be used to request model 2218 status information response. Command/response frames shall be as follows:

Table 34: Request Module Status Command

Description	ms b								lsb	Byte Number
(Type Number = 49)	0	0	1	1	0	0	0	1		Byte 1
Reset Status Bits	P	E	K	R	T	M	L	W		Byte 2

Table 35: Request Module Status Response

Description	ms b								lsb	Byte Number
(Type Number = 177)	1	0	1	1	0	0	0	1		Byte 1
System Status	P	E	K	R	T	M	L	W		Byte 2
SCC Receive Error Count	Receive Error Count									Byte 3
SCC Transmit Error Count	Transmit Error Count									Byte 4
MC Timestamp MSB	MC Timestamp MSB									Byte 5
MC Timestamp NMSB	MC Timestamp NMSB									Byte 6
MC Timestamp NLSB	MC Timestamp NLSB									Byte 7
MC Timestamp LSB	MC Timestamp LSB									Byte 8

The response status bits are defined as follows:

- P - Indicates model 2218 hardware reset
- E - Indicates a communications loss of greater than two seconds
- M - Indicates an error with the Millisecond Counter interrupt
- L - Indicates an error in the LINESYNC
- W - Indicates that the model 2218 has been reset by the Watchdog
- R - Indicates that the EIA-485 receive error count byte has rolled over
- T - Indicates that the EIA-485 transmit error count byte has rolled over
- K - Not Used

Each of these bits shall be individually reset by a '1' in the corresponding bit of any subsequent Request Module Status frame, and the response frame shall report the current status bits. The SCC error count bytes shall not be reset. When a count rolls over (255 - 0), its corresponding roll-over flag shall be set.

9.1.6.2 Millisecond Counter Management

The millisecond counter management frame shall be used to set the value of the millisecond counter. The 'S' bit shall return status '0' on completion or '1' on error. The 32-bit value shall be loaded into the millisecond counter at the next 0-1 transition of the LINESYNC signal. The frames shall be as follows:

Table 36: Millisecond Counter Management Command

<i>Description</i>	<i>ms b</i>								<i>Isb</i>	<i>Byte Number</i>
(Type Number = 50)	0	0	1	1	0	0	1	0		Byte 1
New MC Timestamp MSB	x	x	x	x	x	x	x	x		Byte 2
New MC Timestamp NMSB	x	x	x	x	x	x	x	x		Byte 3
New MC Timestamp NLSB	x	x	x	x	x	x	x	x		Byte 4
New MC Timestamp LSB	x	x	x	x	x	x	x	x		Byte 5

Table 37: Millisecond Counter Management Response

<i>Description</i>	<i>ms b</i>								<i>Isb</i>	<i>Byte Number</i>
(Type Number = 178)	1	0	1	1	0	0	1	0		Byte 1
Status	0	0	0	0	0	0	0	S		Byte 2

9.1.6.3 Configure Inputs

The configure inputs command frame shall be used to change input configurations. The command-response frames shall be as follows:

Table 38: Configure Inputs Command

<i>Description</i>	<i>ms b</i>								<i>Isb</i>	<i>Byte Number</i>
(Type Number = 51)	0	0	1	1	0	0	1	1		Byte 1
Number of Items (n)	n	n	n	n	n	n	n	n		Byte 2

Item # - Byte 1	E	Input Number (I0 – I59)	Byte 3(I-1)+3
Item # - Byte 2	Leading edge filter (e)		Byte 3(I-1)+4
Item # - Byte 3	Trailing edge filter (r)		Byte 3(I-1)+5

Table 39: Configure Inputs Response

<i>Description</i>	<i>ms b</i>	<i>lsb</i>	<i>Byte Number</i>
(Type Number = 179)	1	0 1 1 0 0 1 1	Byte 1
Status	0	0 0 0 0 0 0 0 S	Byte 2

Block field definitions shall be as follows:

- E - Ignore Input Flag. "1" = do not record transition entries for this input, "0" = record transition entries for this input
- e - A one-byte leading edge filter specifying the number of consecutive input samples which must be "0" before the input is considered to have entered to "0" state from "1" state (range 1 to 255, 0 = filtering disabled)
- r - A one-byte trailing edge filter specifying the number of consecutive input samples which must be "1" before the input is considered to have entered to "1" state from "0" state (range 1 to 255, 0 = filtering disabled)
- S - return status S = '0' on completion or '1' on error

9.1.6.4 Poll Raw Inputs Data

The poll raw input data frame shall be used to poll the model 2218 for the current unfiltered status of all inputs. The response frame shall contain 8 bytes (Inputs 0-63) of information indicating the current input status. The frames shall be as follows:

Table 40: Poll Raw Input Data

<i>Description</i>	<i>ms b</i>	<i>lsb</i>	<i>Byte Number</i>
(Type Number = 52)	0	0 1 1 0 1 0 0	Byte 1

Table 41: Poll Raw Data Response

<i>Description</i>	<i>ms b</i>	<i>lsb</i>	<i>Byte Number</i>
(Type Number = 180)	1	0 1 1 0 1 0 0	Byte 1

Inputs I0 (lsb) to I7 (msb)	x	x	x	x	x	x	x	x	Byte 2
Inputs I8 to I53	x	x	x	x	x	x	x	x	Bytes 3 to 8
Inputs I56 to I59, A0 to A3	A3	A2	A1	A0	I59	I58	I57	I56	Byte 9
MC Timestamp MSB	x	x	x	x	x	x	x	x	Byte 10
MC Timestamp NMSB	x	x	x	x	x	x	x	x	Byte 11
MC Timestamp NLSB	x	x	x	x	x	x	x	x	Byte 12
MC Timestamp LSB	x	x	x	x	x	x	x	x	Byte 13

9.1.6.5 Poll Filtered Input Data

The poll filtered input data frame shall be used to poll the model 2218 for the current filtered status of all inputs. The response frame shall contain 8 bytes (Inputs 0-63) of information indicating the current filtered status of the inputs. Raw input data shall be provided in the response for inputs that are not configured for filtering. The frames shall be as follows:

Table 42: Poll Filtered Data Command

<i>Description</i>	<i>msb</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 53)	0	0	1	1	0	1	0	1		Byte 1

Table 43: Poll Filtered Data Response

<i>Description</i>	<i>msb</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 181)	1	0	1	1	0	1	0	1		Byte 1
Inputs I0 (lsb) to I7 (msb)	x	x	x	x	x	x	x	x		Byte 2
Inputs I8 to I53, I56 to I59	x	x	x	x	x	x	x	x		Bytes 3 to 9
MC Timestamp MSB	x	x	x	x	x	x	x	x		Byte 10
MC Timestamp NMSB	x	x	x	x	x	x	x	x		Byte 11
MC Timestamp NLSB	x	x	x	x	x	x	x	x		Byte 12
MC Timestamp LSB	x	x	x	x	x	x	x	x		Byte 13

9.1.6.6 Poll Input Transition Buffer

The poll input transition buffer frame shall poll the model 2218 for the contents of the input transition buffer. The response frame shall include a three-byte information field for each of the input changes that have occurred since the last interrogation. The frames are as follows:

Table 44: Poll Input Transition Buffer Command

<i>Description</i>	<i>ms b</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 54)	0	0	1	1	0	1	1	0		Byte 1
Block Number	x	x	x	x	x	x	x	x		Byte 2

Table 45: Poll Input Transition Buffer Response

<i>Description</i>	<i>ms b</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 182)	1	0	1	1	0	1	1	0		Byte 1
Block Number	x	x	x	x	x	x	x	x		Byte 2
Number of Entries = N	x	x	x	x	x	x	x	x		Byte 3
Item #	S	Input Number (I0 – I59)								Byte 3(I-1)+4
Item # MC Timestamp NLSB	x	x	x	x	x	x	x	x		Byte 3(I-1)+5
Item # MC Timestamp LSB	x	x	x	x	x	x	x	x		Byte 3(I-1)+6
Status	0	0	0	0	C	F	E	G		Byte 3(I-1)+7
MC Timestamp MSB	x	x	x	x	x	x	x	x		Byte 3(N-1)+8
MC Timestamp NMSB	x	x	x	x	x	x	x	x		Byte 3(N-1)+9
MC Timestamp NLSB	x	x	x	x	x	x	x	x		Byte 3(N-1)+10
MC Timestamp LSB	x	x	x	x	x	x	x	x		Byte 3(N-1)+11

Each detected state transition for each active input (see 6.1.7.1 and 9.1.6.3) is placed in the queue as it occurs. Bit definitions are as follows:

- S - Indicates the state of the input after the transition, bit is 1 if the Input is ON after the transition, bit is 0 if the Input is OFF after the transition
- C - Indicates the 255 entry buffer limit has been exceeded
- F - Indicates the 1024 buffer limit has been exceeded
- G - Indicates the requested block number is out of monotonic increment sequence
- E - Same block number requested, E is set in response

The entries provided within the transition buffer poll response shall be ordered from the start of the reply as the oldest to newest. The very first access provides the oldest entry. The model 2218 device shall initialize, upon Power Up or Reset, its last block number received value to 0xFF in order to facilitate suppression of the G Bit response when the ATC program starts and uses 0x00 as the first block number. Subsequent responses are subject to the old-buffer purge mechanism stated below.

The ATC program monotonically increases the block number after each command issued to purge the old buffer. When the model 2218 module receives this command, it shall compare the associated block number with the block number of the previously received command. If it is the same, the previous buffer shall be re-sent to the ATC and

the 'E' flag set in the status response frame. If it is not equal to the previous block number, the old buffer shall be purged and the next block of data sent. If the block number is not incremented by one, the status G bit shall be set. The block number received becomes the current number (even if out of sequence). The block number byte sent in the response block shall be the same as that received in the command block. The block number counter rollover (0xFF becomes 0x00) shall be considered as a normal increment.

9.1.6.7 Set Outputs

The set outputs frame shall be used to command the model 2218 to set the outputs according to the data in the frame. If there is any error configuring the outputs, the 'E' flag in the response frame shall be set to '1'. If the LINESYNC reference has been lost, the 'L' bit in the response frame shall be set. Loss of LINESYNC reference shall also be indicated in system status information. The ACTIVE indicator shall be assigned as O54, and O55 is not used. These command and response frames are as follows:

Table 46: Set Outputs Command

<i>Description</i>	<i>ms b</i>								<i>ls b</i>	<i>Byte Number</i>
(Type Number = 55)	0	0	1	1	0	1	1	1	1	Byte 1
Outputs O0 (lsb) to O7 (msb) Data	x	x	x	x	x	x	x	x	x	Byte 2
Outputs O8 to O47 Data	x	x	x	x	x	x	x	x	x	Bytes 3 to 7
Outputs O48 to O55 Data	x	x	x	x	x	x	x	x	0	Byte 8
Outputs O56 to O63 Data (reserved)	0	0	0	0	0	0	0	0	0	Byte 9
Outputs O0 (lsb) to O7 (msb) Control	x	x	x	x	x	x	x	x	x	Byte 10
Outputs O8 to O47 Control	x	x	x	x	x	x	x	x	x	Bytes 11 to 15
Outputs O48 to O55 Control	x	x	x	x	x	x	x	x	0	Byte 16
Outputs O56 to O63 Control (reserved)	0	0	0	0	0	0	0	0	0	Byte 17

Table 47: Set Outputs Response

<i>Description</i>	<i>ms b</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 183)	1	0	1	1	0	1	1	1	1	Byte 1
Status	0	0	0	0	0	0	0	L	E	Byte 2

9.1.6.8 Configure Input Tracking Functions

The Configure Input Tracking Functions frame shall be used to configure the definition for an output that responds to transitions on a particular input. The maximum number of active definitions is 8.

Please note that Configure Input Tracking Functions is not intended for use with Traffic Signal Control Applications. (Authorized Engineering Information.)

9.1.6.8.1 Command and Response Frames

The command and response frames for Input Tracking Functions shall be as follows:

Table 48: Configure Input Tracking Function Command

<i>Description</i>	<i>ms</i> <i>b</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 56)	0	0	1	1	1	0	0	0	Byte 1	
Number of Items	Number of Items									Byte 2
Item # - Byte 1	E	Output Number (O0 – O54)								Byte 2(l-1)+3
Item # - Byte 2	I	Input Number (I0 – I59)								Byte 2(l-1)+4

Number of Items: 0-16 Tracking Definitions are contained in this message.

Field Definitions:

E	'1'	-	Enable Input Tracking function for this Output
	'0'	-	Remove Input Tracking function for this Output
I	'1'	-	Output is OFF when Input is ON, ON when Input OFF
	'0'	-	Output is ON when Input is ON, OFF when Input is OFF

Output Number: 0 - Maximum Output Number for the model 2218 device type.

Input Number: 0 - Maximum Input Number for the model 2218 device type.

Table 49: Configure Input Tracking Function Response

<i>Description</i>	<i>ms</i> <i>b</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 184)	1	0	1	1	1	0	0	0	Byte 1	
Status	0	0	0	0	0	0	0	V	Byte 2	
MC Timestamp MSB	x	x	x	x	x	x	x	x	Byte 3	
MC Timestamp NMSB	x	x	x	x	x	x	x	x	Byte 4	
MC Timestamp NLSB	x	x	x	x	x	x	x	x	Byte 5	
MC Timestamp LSB	x	x	x	x	x	x	x	x	Byte 6	

Field Definitions:

V '1' - Maximum number of configurable outputs will be exceeded.
 '0' - No error

9.1.6.8.2 Timestamp

The timestamp value shall be sampled prior to the response frame.

9.1.6.8.3 Output Updates

Outputs, which track inputs, shall be updated no less than once per millisecond. Input to output signal propagation delay shall not exceed 2 milliseconds.

9.1.6.8.4 Tracking Functions Overview

A maximum of eight different output numbers may be activated by specifying eight definitions.

One complete definition for an output that tracks an Input consists of two bytes containing four parameters: 1) the instruction to install or to remove the definition, 2) the output number, 3) the relationship of the state of the output to the input and 4) the input number.

Each definition specifies the controlling input number for that unique output number. More than one output definition may specify the same input controlling source. [That is, the same input may be used as the control source for more than one tracking output.]

A complete definition is called an item in the command message frame. The 'Number of Items' byte specifies the quantity of complete definitions contained in the command frame. If the value is 0, all existing active input tracking definitions shall be removed.

The transmission of a definition may do the following:

- a) Install a new active tracking definition.
- b) Remove an existing active tracking definition. When an input tracking definition is removed, the output is set according to the most recently received set outputs command.
- c) Convert an active output definition from complex or square wave definition to tracking conversion removes the existing definition and assigns the tracking definition without a transition through the 'output is set according to the most recently received set outputs command' state. The most recent state of the output remains until the new function changes it.
- d) Redefine an existing tracking definition.

If a command frames to be processed by the model 2218 would result in having more than the maximum number (8) of definitions activated, the entire command frame shall be rejected. The response V bit shall be set to 1.

The V bit response is based on counting the current active quantity plus the projected enable definitions after accounting for remove definitions and invalid output numbers.

The V bit response evaluation takes the currently active definition quantity, adds the projected enable definitions, subtracts the remove definitions, ignores invalid input and output numbers and compares the result to the maximum number of active tracking definitions allowed. If the quantity of active definitions would become greater than the maximum number of active tracking definitions, or if there are more remove definitions than existing active definitions, the V bit shall be set in the response.

While processing an enable request, an out of range input number shall preclude processing for that definition.

The out of range output and input numbers shall not affect the active definition count. No error response is returned.

The rest of the message shall be processed.

The “Number of Items” field is valid from 0 to 16 because the longest message may contain eight Enable and eight Remove definitions.

The input state always comes from the filtered input data source.

Valid Input and Output Number Ranges:
model 2218 device types: Inputs 0 - 53 & 56 - 59, Outputs 0 - 54

9.1.6.9 Configure Complex Output Functions

The Configure Complex Output Functions frame shall be used to configure the definition for an output that provides a complex operation. The maximum number of active definitions is eight.

Note that Configure Complex Output Functions is not intended for use with Traffic Signal Control Applications. (Authorized Engineering Information.)

9.1.6.9.1 Command and Response Frames

The command and response frames shall be as follows:

Table 50: Configure Complex Outputs

<i>Description</i>	<i>ms</i> <i>b</i>	<i>lsb</i>	<i>Byte Number</i>
(Type Number = 57)	0	0 1 1 1 0 0 1	Byte 1
Number of Items	Number of Items		Byte 2

Item # - Byte 1	0	Output Number (O0 – O54)	Byte 7(l-1)+3
Item # - Byte 2	Primary Duration (MSB)		Byte 7(l-1)+4
Item # - Byte 3	Primary Duration (LSB)		Byte 7(l-1)+5
Item # - Byte 4	Secondary Duration (MSB)		Byte 7(l-1)+6
Item # - Byte 5	Secondary Duration (LSB)		Byte 7(l-1)+7
Item # - Byte 6	0	Input Number (I0 – I59)	Byte 7(l-1)+8
Item # - Byte 7	P	W G E J F R L	Byte 7(l-1)+9

Number of Items: 0-16 Complex Output Definitions are contained in this message.

Output Number: 0 - Maximum Output Number for the model 2218 device type.

Primary Duration: MSB & LSB form a 16 bit Hex numerical value 0x0000 - 0xffff.

Secondary Duration: MSB & LSB form a 16 bit Hex numerical value 0x0000 - 0xffff.

Input Number: 0 - Maximum Input Number for the model 2218 device type.

Field Definitions:

- P '1' - The output is configured for single-pulse operation.
Once complete, the complex output function shall be disabled.
- '0' - The output is configured for continuous oscillation.
- W '1' - It is triggered by the specified input.
Triggered complex output shall commence within 2 milliseconds of the associated trigger recognition.
- '0' - Operation shall begin within two milliseconds of the command receipt.
- G '1' - Operation shall be gated active by the specified input.
- '0' - Gating is inactive.
- E '1' Enable complex output function for this output
- '0' Remove complex output function for this output
- J '1' During primary duration, the output shall be written as a logic '1'.
During secondary duration, the output shall be written as a logic '0'.
- '0' During primary duration, the output shall be written as a logic '0'.
During secondary duration, the output shall be written as a logic '1'
- F '1' - The trigger or gate shall be acquired subsequent to filtering the specified input.
The raw input signal shall be used if filtering is not enabled for the specified input.
- '0' - The trigger or gate shall be derived from the raw input.
- R '1' - For triggered output, the output shall be triggered by an ON-to-OFF transition of thespecified input and shall be triggered immediately upon command receipt if the input isOFF.
For gated output, the output shall be active while the input is OFF.
- '0' - For triggered output, the output shall be triggered by an OFF-to-ON transition of the specified input and shall be triggered immediately upon command receipt if the input is ON. For gated output, the output shall be active while the input is ON.
- L '1' - The LINESYNC based clock shall be used for the time ticks.

'0' - The millisecond counter shall be used for the time ticks.

Table 51: Configure Complex Output Functions Response

<i>Description</i>	<i>ms b</i>								<i>lsb</i>	<i>Byte Number</i>
(Type Number = 185)	1	0	1	1	1	0	0	1		Byte 1
Status	0	0	0	0	0	0	0	V		Byte 2
MC Timestamp MSB	x	x	x	x	x	x	x	x		Byte 3
MC Timestamp NMSB	x	x	x	x	x	x	x	x		Byte 4
MC Timestamp NLSB	x	x	x	x	x	x	x	x		Byte 5
MC Timestamp LSB	0	x	x	x	x	x	x	x		Byte 6

Field Definitions:

V '1' - Maximum number of configurable outputs will be exceeded.
'0' - No error.

9.1.6.9.2 Sampling Rate

Controlling input signals shall be sampled at least once per millisecond.

9.1.6.9.3 Complex Output Functions Overview

A maximum of eight different output numbers may be activated by specifying eight definitions. One complete definition for a complex output consists of seven bytes containing fourteen parameters:

- 1) The Output Number
- 2, 3) Primary Duration: MSB & LSB form a 16 bit Hex numerical value
- 4, 5) Secondary Duration: MSB & LSB form a 16 bit Hex numerical value
- 6) The Input Number
- 7) Bit P: One Pulse or Continuous Oscillation
- 8) Bit W: Output Operation is Edge Triggered by Input or Not Triggered by Input
- 9) Bit G: Output Operation is Gated by Input or is Continuous Oscillation
- 10) Bit E: Enable Definition or Remove Definition
- 11) Bit J: Defines Primary/Secondary Duration relationship: ON/OFF or OFF/ON
- 12) Bit F: Input from Filtered or Raw Data
- 13) Bit R: Selects Edge for Triggered by Input ON to OFF or OFF to ON. Bit R: Selects State for Gated to be active by Input OFF or by Input ON
- 14) Linesync edges or millisecond counter provides tick timing

Each definition specifies the controlling input number for that unique output number. The input is a functional control only when the operation is specified as Triggered (W=1) or Gated (G=1). Otherwise, the input number is ignored. More than one output definition may specify the same Input controlling source. [That is, the same input may be used as

the control source for more than one complex output.] If both W=1 and G=1 are set in the definition, the G=1 shall be used as if W=0.

The primary duration is the first timed interval of a pulse or the first portion of a continuous oscillation. The first portion follows acquisition of a trigger or gated input. If not triggered or gated, the first portion follows the activation of the definition.

The secondary duration follows the primary duration.

A complete definition is called an item in the command message frame.

The 'Number of Items' byte specifies the quantity of complete definitions contained in the command frame. If the value is 0, all existing active complex output definitions shall be removed.

The transmission of a definition may do the following:

- a) Install a new active complex output definition.
- b) Remove an existing active complex output definition. When a complex output definition is removed, the output is set according to the most recently received set outputs command.
- c) Convert an active output definition from tracking or square wave definition to complex output. Conversion removes the existing definition and assigns the complex output definition without a transition through the 'output is set according to the most recently received Set Outputs Command' state. The most recent state of the output remains until the new function changes it.
- d) Redefine an existing complex output definition.

If a command frame to be processed by the model 2218 would result in having more than the maximum number (8) definitions activated, the entire command frame shall be rejected. The response V bit shall be set to 1.

The V bit response evaluation takes the currently active definition quantity, adds the projected enable definitions, subtracts the remove definitions, ignores invalid input and output number,s and compares the result to the maximum number of active complex definitions allowed. If the quantity of active definitions would become greater than the maximum number of active complex output definitions, or if there are more Remove definitions than existing active complex output definitions, the V bit shall be set in the response.

While processing an Enable request that requires triggered or gated operation, an out of range input number shall preclude processing for that definition.

The out of range output and input numbers shall not affect the active definition count. No error response is returned. The rest of the message shall be processed.

The “Number of Items” field is valid from 0 to 16 because the longest message may contain eight enable and eight remove definitions.

The input state comes from the filtered or raw input data source as specified by the bit F value.

Valid Input and Output Number Ranges:

Model 2218 device types: Inputs 0 - 53 & 56 - 59, Outputs 0 - 54

The LINESYNC based clock shall used both the rising and falling edges providing a nominal 8.33 millisecond time tick.

9.1.7 CMU2212 SB#1 Frame Types

9.1.7.1 Type 62 – Send to Local Flash Command – SET FSA

Table 52 Type 62 Command

<i>Byte #</i>	<i>Contents</i>	<i>Description</i>
1	62	Frame Type
2	FSA mode	Set Failed State Action b0 = Set LFSA (L) b1 = Set NFSA (N) b2:7 = reserved

9.1.7.2 Type 190 Response – Send to Local Flash Response – Set FSA

Table 53: Type 190 Response

<i>Byte #</i>	<i>Contents</i>	<i>Description</i>
1	190	Frame Type

9.1.7.3 Type 67 Command – Switch Pack Drivers

The format of this command frame shall be identical to Message 81 and 83 but requests a short status response from the CMU. This frame may be used as an alternate to Message 81 or 83. The destination for this frame shall be the CMU. The channel numbers in the description column below refer to the channel numbers of the CMU. The ATC shall include a definition, via program entry, of the CMU channel to ATC signal driver group utilization.

The Dark Channel Map Select bits shall select a preprogrammed mask in the CMU serial memory key that disables Lack of Signal Input monitoring for the selected channels.

Table 54: CMU Type 67 Command

Byte #	Contents	Description
1	67	Frame Type
2	Channel 8:1 Red	Switch Pack Status A bit set to 1 indicates the HDSP output is set ON.
3	Channel 16:9 Red	
4	Channel 24:17 Red	
5	Channel 32:25 Red	
6	Channel 8:1 Yellow	
7	Channel 16:9 Yellow	
8	Channel 24:17 Yellow	
9	Channel 32:25 Yellow	
10	Channel 8:1 Green	
11	Channel 16:9 Green	
12	Channel 24:17 Green	
13	Channel 32:25 Green	
14	Map Select	Dark Channel Map Select Bit 1 and bit 0 shall select one of four Dark Channel Maps programmed in the serial memory key that disables Lack of Signal Input monitoring for a selected channel. Bits 2 thru 7 are reserved. 00 = Mask #1 01 = Mask #2 10 = Mask #3 11 = Mask #4

9.1.7.4 Type 195 Response – CMU Short Status

If the CMU is in FSA (byte #2, Fault Type not equal to 0), then the channel color status bytes shall contain an exact image of the signals that were applied to the CMU at the point in time of the detection of the failure. Control Status 1, Control Status 2, and Output Assembly Flasher Status shall always reflect current status.

Table 55: CMU Type 195 Response

<i>Byte #</i>	<i>Contents</i>	<i>Description</i>
1	195	Frame Type
2	Fault Type	Enumerated fault code 00 = No Fault 01 = CMU/HDSP +24 VDC 02 = CMU +12 VDC 03 = Conflict 04 = Serial Bus #1 05 = Serial Bus #3 06 = ATC LFSA Flash (Type 62) 07 = ATC NFSA Flash (Type 62) 08 = Diagnostic 09 = Multiple 10 = Lack of Signal Input 11 = Short Yellow Clearance 12 = Skipped Yellow Clearance 13 = Yellow + Red Clearance 14 = Field Output Check 15 = Serial Memory Key absent 16 = Serial Memory Key FCS error 17 = Serial Memory Key Data error 18 = Local Flash 19 = CB Trip 20 = CMU/HDSP Mains Fail 21 = NRESET Active 22 = HDSP Diagnostic 23 = FYA Flash Rate Fault 24 = CMU +48 VDC 25:127 = Reserved 128:255 = Spare (Reserved for Mfg Use)
3	Channel Fault Status 8:1	Channel Fault Status

Byte #	Contents	Description
4	Channel Fault Status 16:9	Channel Fault Status bits shall be set to 1 for channels that were detected in fault for fault types 03, 09, 10, 11, 12, 13, and 14. For fault type 01, 05, 20, and 22 a bit shall be set in Channel Fault Status 32:1 for each enabled HDSP channel that failed. For type 01, if the CMU 24VDC input failed then Channel Fault Status shall be set to 0. For type 20, if the CMU MAINS input failed then Channel Fault Status shall be set to 0. For all other fault types the Channel Fault Status bits shall be set to 0.
5	Channel Fault Status 24:17	
6	Channel Fault Status 32:25	
7	Channel Red Status 8:1	Channel Color Status Channel Color Status bits shall be set to 1 for channels that are sensed active. For channel inputs that have been remapped to a virtual channel (29-32), the Channel Color Status bits shall be set to 0. For virtual channel (29-32) inputs that have not been assigned to a physical output, the Channel Color Status bits shall be set to 0.
8	Channel Red Status 16:9	
9	Channel Red Status 24:17	
10	Channel Red Status 32:25	
11	Channel Yellow Status 8:1	
12	Channel Yellow Status 16:9	
13	Channel Yellow Status 24:17	
14	Channel Yellow Status 32:25	
15	Channel Green Status 8:1	
16	Channel Green Status 16:9	
17	Channel Green Status 24:17	
18	Channel Green Status 32:25	
19	Control Status #1	Control Status #1 b7 = Start-up Call (1=Exit from Flash) b6 = Flasher Output Fail (1 = Fail) b5 = Rear Door (1=Open) b4 = Front Door (1=Open) b3 = MAIN CONTACTOR Coil (1=Active) b2 = MAIN CONTACTOR Secondary (1=Active) b1 = FTR Coil Drive (1= Active) b0 = Output Relay Transfer (1=Fault)
20	Control Status #2	Control Status #2 b7 = Reserved

Byte #	Contents	Description
		b6 = Reserved b5 = Reserved b4 = Reserved b3 = Reserved b2 = HDFU #2 Comm Error b1 = HDFU #1 Comm Error b0 = Configuration Change This bit set to 1 indicates the configuration programming has changed since the last poll of frame Type 82. It shall also be set to 1 when the CMU is exiting flash (Start-up Call bit changes from 0 to 1) and at CMU power-up.
21	Flasher Status	Flasher Status (1=Fail) b0 = HDFU #1 FL1-1 b1 = HDFU #1 FL1-2 b2 = HDFU #1 FL2-1 b3 = HDFU #1 FL2-2 b4 = HDFU #2 FL1-1 b5 = HDFU #2 FL1-2 b6 = HDFU #2 FL2-1 b7 = HDFU #2 FL2-2
22	0x00	Reserved
23	0x00	Reserved
24	0x00	Reserved

9.1.7.5 Type 80 Command – Module Description

Table 56: Serial Bus #1 Type 80 Command

Byte #	Contents	Description
1	80	Frame Type

9.1.7.6 Type 208 Response – Module Description

Table 57: Serial Bus #1 Type 208 Response

Byte #	Contents	Description
1	208	Frame Type
2	Device	Enumerated Device Type 1: CMU-212 2: CMU-2212-LV 3: CMU-2212-HV 4: CMU-2212-VHV (Reserved) 5: SIU-218 6: model 2218 7:255 Reserved
3:42	ASCII Text	A packed 40 character field describing the device properties. Allowable ASCII characters are 020h through 07Eh. If less than 40 characters are used, the unused locations shall be set to 00h. Default programming shall be 00h.

9.1.7.7 Type 81 Command – Switch Pack Drivers (Long)

The destination for this frame shall be the CMU. The channel numbers in the description column below refer to the channel numbers of the CMU. The ATC shall include a definition, via program entry, of the CMU channel to ATC signal driver group utilization. The Dark Channel Map Select bits shall select a preprogrammed mask in the CMU serial memory key that disables Lack of Signal Input monitoring for the selected channels.

Table 58: Serial Bus #1 Type 81 Command

Byte #	Contents	Description
1	81	Frame Type
2	Channel 8:1 Red	HDSP Status A bit set to 1 indicates the HDSP output is set ON.
3	Channel 16:9 Red	
4	Channel 24:17 Red	
5	Channel 32:25 Red	
6	Channel 8:1 Yellow	
7	Channel 16:9 Yellow	
8	Channel 24:17 Yellow	

Byte #	Contents	Description
9	Channel 32:25 Yellow	
10	Channel 8:1 Green	
11	Channel 16:9 Green	
12	Channel 24:17 Green	
13	Channel 32:25 Green	
14	Map Select	Dark Channel Map Select Bit 1 and bit 0 shall select one of four Dark Channel Maps programmed in the serial memory key that disables Lack of Signal Input monitoring for a selected channel. 00 = Mask #1 01 = Mask #2 10 = Mask #3 11 = Mask #4 Bits 2 thru 7 are Reserved

9.1.7.8 Type 209 Response – CMU Status

If the CMU is in FSA (byte #2, Fault Type not equal to 0), then all bytes of the information field of this frame except Control Status 1, Control Status 2 and Output Assembly Flasher Status shall contain an exact image of the signals that were applied to the CMU at the point in time of the detection of the failure. Control Status 1, Control Status 2 and Output Assembly Flasher Status shall always reflect current status.

Table 59: Serial Bus #1 Type 209 Response

Byte #	Contents	Description
1	209	Frame Type
2	Fault Type	Enumerated fault code 00 = No Fault 01 = CMU/HDSP +24 VDC 02 = CMU +12 VDC 03 = Conflict 04 = Serial Bus #1 05 = Serial Bus #3 06 = ATC LFSA Flash (Type 62) 07 = ATC NFSA Flash (Type 62) 08 = Diagnostic

Byte #	Contents	Description
		09 = Multiple 10 = Lack of Signal Input 11 = Short Yellow Clearance 12 = Skipped Yellow Clearance 13 = Yellow + Red Clearance 14 = Field Output Check 15 = Serial Memory Key absent 16 = Serial Memory Key FCS error 17 = Serial Memory Key Data error 18 = Local Flash 19 = CB Trip 20 = CMU/HDSP Mains Fail 21 = NRESET Active 22 = HDSP Diagnostic 23 = FYA Flash Rate Fault 24 = CMU +48 VDC 25:127 = Reserved 128:255 = Spare (Reserved for Mfg Use)
3	Channel Fault Status 8:1	Channel Fault Status Channel Fault Status bits shall be set to 1 for channels that were detected in fault for fault types 03, 09, 10, 11, 12, 13, and 14. For fault type 01, 05, 20, and 22 a bit shall be set in Channel Fault Status 32:1 for each enabled HDSP channel that failed. For type 01, if the CMU 24VDC input failed then Channel Fault Status shall be set to 0. For type 20, if the CMU MAINS input failed then Channel Fault Status shall be set to 0. For all other fault types the Channel Fault Status bits shall be set to 0.
4	Channel Fault Status 16:9	
5	Channel Fault Status 24:17	
6	Channel Fault Status 32:25	
7	Channel Red Status 8:1	Channel Color Status Channel Color Status bits shall be set to 1 for channels that are sensed active. For channel inputs that have been remapped to a virtual channel (29-32), the Channel
8	Channel Red Status 16:9	
9	Channel Red Status 24:17	
10	Channel Red Status 32:25	
11	Channel Yellow Status 8:1	

Byte #	Contents	Description
12	Channel Yellow Status 16:9	Color Status bits shall be set to 0. For virtual channel (29-32) inputs that have not been assigned to a physical output, the Channel Color Status bits shall be set to 0.
13	Channel Yellow Status 24:17	
14	Channel Yellow Status 32:25	
15	Channel Green Status 8:1	
16	Channel Green Status 16:9	
17	Channel Green Status 24:17	
18	Channel Green Status 32:25	
19	Channel Red Status 8:1	Field Check Status Field Check Status bits shall be set to 1 for channels that are sensed with field check status. For channel inputs that have been remapped to a virtual channel (29-32), the Field Check Status bits shall be set to 0. For virtual channel (29-32) inputs that have not been assigned to a physical output, the Field Check Status bits shall be set to 0.
20	Channel Red Status 16:9	
21	Channel Red Status 24:17	
22	Channel Red Status 32:25	
23	Channel Yellow Status 8:1	
24	Channel Yellow Status 16:9	
25	Channel Yellow Status 24:17	
26	Channel Yellow Status 32:25	
27	Channel Green Status 8:1	
28	Channel Green Status 16:9	
29	Channel Green Status 24:17	
30	Channel Green Status 32:25	
31	Control Status #1	Control Status #1 b7 = Start-up Call (1=Exit from Flash) b6 = Flasher Output Fail (1 = Fail) b5 = Rear Door (1=Open) b4 = Front Door (1=Open) b3 = MAIN CONTACTOR Coil (1=Active) b2 = MAIN CONTACTOR Secondary (1=Active) b1 = FTR Coil Drive (1= Active) b0 = Output Relay Transfer (1=Fault)
32	Control Status #2	Control Status #2

Byte #	Contents	Description
		b7 = Reserved b6 = Reserved b5 = Reserved b4 = Reserved b3 = Reserved b2 = HDFU #2 Comm Error b1 = HDFU #1 Comm Error b0 = Configuration Change This bit set to 1 indicates the configuration programming has changed since the last poll of frame Type 82. It shall also be set to 1 when the CMU is exiting flash (Start-up Call bit changes from 0 to 1) and at CMU power-up.
33	Mains Raw Voltage	CMU Mains Voltage
34	Assembly #1 Signal Power Voltage	HDSP Signal Voltage Signal Voltage for any HDSP in assembly #1 (channels 1:16) Signal Voltage for any HDSP in assembly #2 (channels 17:32)
35	Assembly #2 Signal Power Voltage	
36	Channel 1 Red Voltage	Channel Input Voltages The Channel input voltages shall be the most recent values reported from the HDSPs. For channel inputs that have been remapped to a virtual channel (29-32), the Channel Voltage value shall be set to 0.
37	Channel 2 Red Voltage	
38	Channel 3 Red Voltage	
39	Channel 4 Red Voltage	
40	Channel 5 Red Voltage	
41	Channel 6 Red Voltage	
42	Channel 7 Red Voltage	
43	Channel 8 Red Voltage	
44	Channel 9 Red Voltage	
45	Channel 10 Red Voltage	
46	Channel 11 Red Voltage	
47	Channel 12 Red Voltage	
48	Channel 13 Red Voltage	
49	Channel 14 Red Voltage	
50	Channel 15 Red Voltage	
51	Channel 16 Red Voltage	
52	Channel 17 Red Voltage	

Byte #	Contents	Description
53	Channel 18 Red Voltage	
54	Channel 19 Red Voltage	
55	Channel 20 Red Voltage	
56	Channel 21 Red Voltage	
57	Channel 22 Red Voltage	
58	Channel 23 Red Voltage	
59	Channel 24 Red Voltage	
60	Channel 25 Red Voltage	
61	Channel 26 Red Voltage	
62	Channel 27 Red Voltage	
63	Channel 28 Red Voltage	
64	Channel 29 Red Voltage	
65	Channel 30 Red Voltage	
66	Channel 31 Red Voltage	
67	Channel 32 Red Voltage	
68	Channel 1 Yellow Voltage	
69	Channel 2 Yellow Voltage	
70	Channel 3 Yellow Voltage	
71	Channel 4 Yellow Voltage	
72	Channel 5 Yellow Voltage	
73	Channel 6 Yellow Voltage	
74	Channel 7 Yellow Voltage	
75	Channel 8 Yellow Voltage	
76	Channel 9 Yellow Voltage	
77	Channel 10 Yellow Voltage	
78	Channel 11 Yellow Voltage	
79	Channel 12 Yellow Voltage	
80	Channel 13 Yellow Voltage	
81	Channel 14 Yellow Voltage	
82	Channel 15 Yellow Voltage	
83	Channel 16 Yellow Voltage	
84	Channel 17 Yellow Voltage	
85	Channel 18 Yellow Voltage	
86	Channel 19 Yellow Voltage	
87	Channel 20 Yellow Voltage	
88	Channel 21 Yellow Voltage	
89	Channel 22 Yellow Voltage	
90	Channel 23 Yellow Voltage	

Byte #	Contents	Description
91	Channel 24 Yellow Voltage	
92	Channel 25 Yellow Voltage	
93	Channel 26 Yellow Voltage	
94	Channel 27 Yellow Voltage	
95	Channel 28 Yellow Voltage	
96	Channel 29 Yellow Voltage	
97	Channel 30 Yellow Voltage	
98	Channel 31 Yellow Voltage	
99	Channel 32 Yellow Voltage	
100	Channel 1 Green Voltage	
101	Channel 2 Green Voltage	
102	Channel 3 Green Voltage	
103	Channel 4 Green Voltage	
104	Channel 5 Green Voltage	
105	Channel 6 Green Voltage	
106	Channel 7 Green Voltage	
107	Channel 8 Green Voltage	
108	Channel 9 Green Voltage	
109	Channel 10 Green Voltage	
110	Channel 11 Green Voltage	
111	Channel 12 Green Voltage	
112	Channel 13 Green Voltage	
113	Channel 14 Green Voltage	
114	Channel 15 Green Voltage	
115	Channel 16 Green Voltage	
116	Channel 17 Green Voltage	
117	Channel 18 Green Voltage	
118	Channel 19 Green Voltage	
119	Channel 20 Green Voltage	
120	Channel 21 Green Voltage	
121	Channel 22 Green Voltage	
122	Channel 23 Green Voltage	
123	Channel 24 Green Voltage	
124	Channel 25 Green Voltage	
125	Channel 26 Green Voltage	
126	Channel 27 Green Voltage	
127	Channel 28 Green Voltage	
128	Channel 29 Green Voltage	

Byte #	Contents	Description
129	Channel 30 Green Voltage	
130	Channel 31 Green Voltage	
131	Channel 32 Green Voltage	
132	Channel 1 Red Current	Scaled Load Current The output load current values for current sense shall be programmed in scaled milliamps. IF Param >120 Then Current = (Param *16) – 1800 Else Current = Param If Param > 250 Then Current = "> 2200" mA For channel inputs that have been remapped to a virtual channel (29-32), the Channel Current value shall be set to 0.
133	Channel 2 Red Current	
134	Channel 3 Red Current	
135	Channel 4 Red Current	
136	Channel 5 Red Current	
137	Channel 6 Red Current	
138	Channel 7 Red Current	
139	Channel 8 Red Current	
140	Channel 9 Red Current	
141	Channel 10 Red Current	
142	Channel 11 Red Current	
143	Channel 12 Red Current	
144	Channel 13 Red Current	
145	Channel 14 Red Current	
146	Channel 15 Red Current	
147	Channel 16 Red Current	
148	Channel 17 Red Current	
149	Channel 18 Red Current	
150	Channel 19 Red Current	
151	Channel 20 Red Current	
152	Channel 21 Red Current	
153	Channel 22 Red Current	
154	Channel 23 Red Current	
155	Channel 24 Red Current	
156	Channel 25 Red Current	
157	Channel 26 Red Current	
158	Channel 27 Red Current	
159	Channel 28 Red Current	
160	Channel 29 Red Current	
161	Channel 30 Red Current	
162	Channel 31 Red Current	
163	Channel 32 Red Current	
164	Channel 1 Yellow Current	
165	Channel 2 Yellow Current	
166	Channel 3 Yellow Current	

Byte #	Contents	Description
167	Channel 4 Yellow Current	
168	Channel 5 Yellow Current	
169	Channel 6 Yellow Current	
170	Channel 7 Yellow Current	
171	Channel 8 Yellow Current	
172	Channel 9 Yellow Current	
173	Channel 10 Yellow Current	
174	Channel 11 Yellow Current	
175	Channel 12 Yellow Current	
176	Channel 13 Yellow Current	
177	Channel 14 Yellow Current	
178	Channel 15 Yellow Current	
179	Channel 16 Yellow Current	
180	Channel 17 Yellow Current	
181	Channel 18 Yellow Current	
182	Channel 19 Yellow Current	
183	Channel 20 Yellow Current	
184	Channel 21 Yellow Current	
185	Channel 22 Yellow Current	
186	Channel 23 Yellow Current	
187	Channel 24 Yellow Current	
188	Channel 25 Yellow Current	
189	Channel 26 Yellow Current	
190	Channel 27 Yellow Current	
191	Channel 28 Yellow Current	
192	Channel 29 Yellow Current	
193	Channel 30 Yellow Current	
194	Channel 31 Yellow Current	
195	Channel 32 Yellow Current	
196	Channel 1 Green Current	
197	Channel 2 Green Current	
198	Channel 3 Green Current	
199	Channel 4 Green Current	
200	Channel 5 Green Current	
201	Channel 6 Green Current	
202	Channel 7 Green Current	
203	Channel 8 Green Current	
204	Channel 9 Green Current	

Byte #	Contents	Description
205	Channel 10 Green Current	
206	Channel 11 Green Current	
207	Channel 12 Green Current	
208	Channel 13 Green Current	
209	Channel 14 Green Current	
210	Channel 15 Green Current	
211	Channel 16 Green Current	
212	Channel 17 Green Current	
213	Channel 18 Green Current	
214	Channel 19 Green Current	
215	Channel 20 Green Current	
216	Channel 21 Green Current	
217	Channel 22 Green Current	
218	Channel 23 Green Current	
219	Channel 24 Green Current	
220	Channel 25 Green Current	
221	Channel 26 Green Current	
222	Channel 27 Green Current	
223	Channel 28 Green Current	
224	Channel 29 Green Current	
225	Channel 30 Green Current	
226	Channel 31 Green Current	
227	Channel 32 Green Current	
228	Binary Seconds	CMU Time and Date
229	Binary Minutes	
230	Binary Hours (0:23)	
231	Binary Date	
232	Binary Month	
233	Binary Year	
234	Voltage * 4	CMU 24VDC Supply Voltage
235	Voltage * 4	CMU 48VDC Supply Voltage
236	degrees F + 40	CMU Temperature (Fahrenheit)
237	Red Status 8:1	Channel Current Sense Status Status bits shall be set to 1 for channels that are sensed active. Status bits shall be set to 0 for channels that have the Current Sense monitor function disabled.
238	Red Status 16:9	
239	Red Status 24:17	
240	Red Status 32:25	
241	Yellow Status 8:1	
242	Yellow Status 16:9	

Byte #	Contents	Description
243	Yellow Status 24:17	
244	Yellow Status 32:25	
245	Green Status 8:1	
246	Green Status 16:9	
247	Green Status 24:17	
248	Green Status 32:25	
249	Flasher Status	Flasher Status (1=Fail) b0 = HDFU #1 FL1-1 b1 = HDFU #1 FL1-2 b2 = HDFU #1 FL2-1 b3 = HDFU #1 FL2-2 b4 = HDFU #2 FL1-1 b5 = HDFU #2 FL1-2 b6 = HDFU #2 FL2-1 b7 = HDFU #2 FL2-2

9.1.7.9 Type 82 Command – Get CMU Configuration

The ATC shall request the serial memory key programming using this message and validate that the CMU permissive program settings are equal or less permissive than the ATC controller unit programming, and that the HDSP configuration programming is consistent with Output Functionality model 2218 assignment. If these checks are not validated, then the ATC shall issue a Type 62 frame with the L bit set causing a LFSA in the CMU. This check shall be performed at initialization and when the Type 195, Type 209 or Type 211 Control Status 2, Configuration Change bit is set to 1.

Table 60: CMU Type 82 Command

Byte #	Contents	Description
1	82	Frame Type

9.1.7.10 Type 210 Response – CMU Configuration

Table 61: CMU Type 210 Response

Byte #	Contents	Description
1	210	Frame Type
2:513	Bytes #1 thru #512	Serial Memory Key Contents

9.1.7.11 Type 83 Command – Switch Pack Drivers (High Resolution)

The destination for this frame shall be the CMU. The channel numbers in the description column below refer to the channel numbers of the CMU. The ATC shall include a definition, via program entry, of the CMU channel to ATC signal driver group utilization. The Dark Channel Map Select bits shall select a preprogrammed mask in the CMU serial memory key that disables Lack of Signal Input monitoring for the selected channels.

Table 62: Serial Bus #1 Type 83 Command

Byte #	Contents	Description
1	83	Frame Type
2	Channel 8:1 Red	HDSP Status A bit set to 1 indicates the HDSP output is set ON.
3	Channel 16:9 Red	
4	Channel 24:17 Red	
5	Channel 32:25 Red	
6	Channel 8:1 Yellow	
7	Channel 16:9 Yellow	
8	Channel 24:17 Yellow	
9	Channel 32:25 Yellow	
10	Channel 8:1 Green	
11	Channel 16:9 Green	
12	Channel 24:17 Green	
13	Channel 32:25 Green	
14	Map Select	Dark Channel Map Select Bit 1 and bit 0 shall select one of four Dark Channel Maps programmed in the serial memory key that disables Lack of Signal Input monitoring for a selected channel. 00 = Mask #1 01 = Mask #2 10 = Mask #3 11 = Mask #4 Bits 2 thru 7 are Reserved

9.1.7.12 Type 211 Response – CMU Status (High Resolution)

If the CMU is in FSA (byte #2, Fault Type not equal to 0), then all bytes of the information field of this frame except Control Status 1, Control Status 2, HDFU Load Current, and Output Assembly Flasher Status shall contain an exact image of the signals that were applied to the CMU at the point in time of the detection of the failure. Control Status 1, Control Status 2, HDFU Load Current, and Output Assembly Flasher Status shall always reflect current status.

Table 63: Serial Bus #1 Type 211 Response

Byte #	Contents	Description
1	211	Frame Type
2	Fault Type	Enumerated fault code 00 = No Fault 01 = CMU/HDSP +24 VDC 02 = CMU +12 VDC 03 = Conflict 04 = Serial Bus #1 05 = Serial Bus #3 06 = ATC LFSA Flash (Type 62) 07 = ATC NFSA Flash (Type 62) 08 = Diagnostic 09 = Multiple 10 = Lack of Signal Input 11 = Short Yellow Clearance 12 = Skipped Yellow Clearance 13 = Yellow + Red Clearance 14 = Field Output Check 15 = Serial Memory Key absent 16 = Serial Memory Key FCS error 17 = Serial Memory Key Data error 18 = Local Flash 19 = CB Trip 20 = CMU/HDSP Mains Fail 21 = NRESET Active 22 = HDSP Diagnostic 23 = FYA Flash Rate Fault 24 = CMU +48 VDC 25:127 = Reserved 128:255 = Spare (Reserved for Mfg Use)
3	Channel Fault Status 8:1	Channel Fault Status Channel Fault Status bits shall be set to 1 for channels that were detected
4	Channel Fault Status 16:9	
5	Channel Fault Status 24:17	

Byte #	Contents	Description
6	Channel Fault Status 32:25	in fault for fault types 03, 09, 10, 11, 12, 13, and 14. For fault type 01, 05, 20, and 22 a bit shall be set in Channel Fault Status 32:1 for each enabled HDSP channel that failed. For type 01, if the CMU 24VDC input failed then Channel Fault Status shall be set to 0. For type 20, if the CMU MAINS input failed then Channel Fault Status shall be set to 0. For all other fault types the Channel Fault Status bits shall be set to 0.
7	Channel Red Status 8:1	Channel Color Status Channel Color Status bits shall be set to 1 for channels that are sensed active. For channel inputs that have been remapped to a virtual channel (29-32), the Channel Color Status bits shall be set to 0. For virtual channel (29-32) inputs that have not been assigned to a physical output, the Channel Color Status bits shall be set to 0.
8	Channel Red Status 16:9	
9	Channel Red Status 24:17	
10	Channel Red Status 32:25	
11	Channel Yellow Status 8:1	
12	Channel Yellow Status 16:9	
13	Channel Yellow Status 24:17	
14	Channel Yellow Status 32:25	
15	Channel Green Status 8:1	
16	Channel Green Status 16:9	
17	Channel Green Status 24:17	Field Check Status Field Check Status bits shall be set to 1 for channels that are sensed with field check status. For channel inputs that have been remapped to a virtual channel (29-32), the Field Check Status bits shall be set to 0. For virtual channel (29-32) inputs that have not been assigned to a physical output, the Field Check Status bits shall be set to 0.
18	Channel Green Status 32:25	
19	Channel Red Status 8:1	
20	Channel Red Status 16:9	
21	Channel Red Status 24:17	
22	Channel Red Status 32:25	
23	Channel Yellow Status 8:1	
24	Channel Yellow Status 16:9	
25	Channel Yellow Status 24:17	
26	Channel Yellow Status 32:25	
27	Channel Green Status 8:1	
28	Channel Green Status 16:9	
29	Channel Green Status 24:17	
30	Channel Green Status 32:25	
31	Control Status #1	Control Status #1 b7 = Start-up Call (1=Exit from Flash) b6 = Flasher Output Fail (1 = Fail) b5 = Rear Door (1=Open) b4 = Front Door (1=Open)

Byte #	Contents	Description
		b3 = MAIN CONTACTOR Coil (1=Active) b2 = MAIN CONTACTOR Secondary (1=Active) b1 = FTR Coil Drive (1= Active) b0 = Output Relay Transfer (1=Fault)
32	Control Status #2	Control Status #2 b7 = Reserved b6 = Reserved b5 = Reserved b4 = Reserved b3 = Reserved b2 = HDFU #2 Comm error b1 = HDFU #1 Comm error b0 = Configuration Change This bit set to 1 indicates the configuration programming has changed since the last poll of message 82. It shall also be set to 1 when the CMU is exiting flash (Start-up Call bit changes from 0 to 1) and at CMU power-up.
33	Mains Raw Voltage	CMU Mains Voltage
34	Assembly #1 Signal Power Voltage	HDSP Signal Voltage Signal Voltage for any HDSP in assembly #1 (channels 1:16) Signal Voltage for any HDSP in assembly #2 (channels 17:32)
35	Assembly #2 Signal Power Voltage	
36	Channel 1 Red Voltage	Channel Input Voltages The Channel input voltages shall be the most recent values reported from the HDSPs. For channel inputs that have been remapped to a virtual channel (29-32), the Channel Voltage value shall be set to 0.
37	Channel 2 Red Voltage	
38	Channel 3 Red Voltage	
39	Channel 4 Red Voltage	
40	Channel 5 Red Voltage	
41	Channel 6 Red Voltage	
42	Channel 7 Red Voltage	
43	Channel 8 Red Voltage	
44	Channel 9 Red Voltage	
45	Channel 10 Red Voltage	
46	Channel 11 Red Voltage	
47	Channel 12 Red Voltage	
48	Channel 13 Red Voltage	
49	Channel 14 Red Voltage	
50	Channel 15 Red Voltage	

Byte #	Contents	Description
51	Channel 16 Red Voltage	
52	Channel 17 Red Voltage	
53	Channel 18 Red Voltage	
54	Channel 19 Red Voltage	
55	Channel 20 Red Voltage	
56	Channel 21 Red Voltage	
57	Channel 22 Red Voltage	
58	Channel 23 Red Voltage	
59	Channel 24 Red Voltage	
60	Channel 25 Red Voltage	
61	Channel 26 Red Voltage	
62	Channel 27 Red Voltage	
63	Channel 28 Red Voltage	
64	Channel 29 Red Voltage	
65	Channel 30 Red Voltage	
66	Channel 31 Red Voltage	
67	Channel 32 Red Voltage	
68	Channel 1 Yellow Voltage	
69	Channel 2 Yellow Voltage	
70	Channel 3 Yellow Voltage	
71	Channel 4 Yellow Voltage	
72	Channel 5 Yellow Voltage	
73	Channel 6 Yellow Voltage	
74	Channel 7 Yellow Voltage	
75	Channel 8 Yellow Voltage	
76	Channel 9 Yellow Voltage	
77	Channel 10 Yellow Voltage	
78	Channel 11 Yellow Voltage	
79	Channel 12 Yellow Voltage	
80	Channel 13 Yellow Voltage	
81	Channel 14 Yellow Voltage	
82	Channel 15 Yellow Voltage	
83	Channel 16 Yellow Voltage	
84	Channel 17 Yellow Voltage	
85	Channel 18 Yellow Voltage	
86	Channel 19 Yellow Voltage	
87	Channel 20 Yellow Voltage	
88	Channel 21 Yellow Voltage	
89	Channel 22 Yellow Voltage	
90	Channel 23 Yellow Voltage	
91	Channel 24 Yellow Voltage	
92	Channel 25 Yellow Voltage	
93	Channel 26 Yellow Voltage	

Byte #	Contents	Description
94	Channel 27 Yellow Voltage	
95	Channel 28 Yellow Voltage	
96	Channel 29 Yellow Voltage	
97	Channel 30 Yellow Voltage	
98	Channel 31 Yellow Voltage	
99	Channel 32 Yellow Voltage	
100	Channel 1 Green Voltage	
101	Channel 2 Green Voltage	
102	Channel 3 Green Voltage	
103	Channel 4 Green Voltage	
104	Channel 5 Green Voltage	
105	Channel 6 Green Voltage	
106	Channel 7 Green Voltage	
107	Channel 8 Green Voltage	
108	Channel 9 Green Voltage	
109	Channel 10 Green Voltage	
110	Channel 11 Green Voltage	
111	Channel 12 Green Voltage	
112	Channel 13 Green Voltage	
113	Channel 14 Green Voltage	
114	Channel 15 Green Voltage	
115	Channel 16 Green Voltage	
116	Channel 17 Green Voltage	
117	Channel 18 Green Voltage	
118	Channel 19 Green Voltage	
119	Channel 20 Green Voltage	
120	Channel 21 Green Voltage	
121	Channel 22 Green Voltage	
122	Channel 23 Green Voltage	
123	Channel 24 Green Voltage	
124	Channel 25 Green Voltage	
125	Channel 26 Green Voltage	
126	Channel 27 Green Voltage	
127	Channel 28 Green Voltage	
128	Channel 29 Green Voltage	
129	Channel 30 Green Voltage	
130	Channel 31 Green Voltage	
131	Channel 32 Green Voltage	
132	Channel 1 Red Current (LS)	HDSP Load Current scaled to 1 mA per bit
133	Channel 1 Red Current (MS)	
134	Channel 2 Red Current (LS)	
135	Channel 2 Red Current (MS)	
136	Channel 3 Red Current (LS)	

Byte #	Contents	Description
137	Channel 3 Red Current (MS)	
138	Channel 4 Red Current (LS)	
139	Channel 4 Red Current (MS)	
140	Channel 5 Red Current (LS)	
141	Channel 5 Red Current (MS)	
142	Channel 6 Red Current (LS)	
143	Channel 6 Red Current (MS)	
144	Channel 7 Red Current (LS)	
145	Channel 7 Red Current (MS)	
146	Channel 8 Red Current (LS)	
147	Channel 8 Red Current (MS)	
148	Channel 9 Red Current (LS)	
149	Channel 9 Red Current (MS)	
150	Channel 10 Red Current (LS)	
151	Channel 10 Red Current (MS)	
152	Channel 11 Red Current (LS)	
153	Channel 11 Red Current (MS)	
154	Channel 12 Red Current (LS)	
155	Channel 12 Red Current (MS)	
156	Channel 13 Red Current (LS)	
157	Channel 13 Red Current (MS)	
158	Channel 14 Red Current (LS)	
159	Channel 14 Red Current (MS)	
160	Channel 15 Red Current (LS)	
161	Channel 15 Red Current (MS)	
162	Channel 16 Red Current (LS)	
163	Channel 16 Red Current (MS)	
164	Channel 17 Red Current (LS)	
165	Channel 17 Red Current (MS)	
166	Channel 18 Red Current (LS)	
167	Channel 18 Red Current (MS)	
168	Channel 19 Red Current (LS)	
169	Channel 19 Red Current (MS)	
170	Channel 20 Red Current (LS)	
171	Channel 20 Red Current (MS)	
172	Channel 21 Red Current (LS)	
173	Channel 21 Red Current (MS)	
174	Channel 22 Red Current (LS)	
175	Channel 22 Red Current (MS)	
176	Channel 23 Red Current (LS)	
177	Channel 23 Red Current (MS)	
178	Channel 24 Red Current (LS)	
179	Channel 24 Red Current (MS)	

Byte #	Contents	Description
180	Channel 25 Red Current (LS)	
181	Channel 25 Red Current (MS)	
182	Channel 26 Red Current (LS)	
183	Channel 26 Red Current (MS)	
184	Channel 27 Red Current (LS)	
185	Channel 27 Red Current (MS)	
186	Channel 28 Red Current (LS)	
187	Channel 28 Red Current (MS)	
188	Channel 29 Red Current (LS)	
189	Channel 29 Red Current (MS)	
190	Channel 30 Red Current (LS)	
191	Channel 30 Red Current (MS)	
192	Channel 31 Red Current (LS)	
193	Channel 31 Red Current (MS)	
194	Channel 32 Red Current (LS)	
195	Channel 32 Red Current (MS)	
196	Channel 1 Yellow Current (LS)	
197	Channel 1 Yellow Current (MS)	
198	Channel 2 Yellow Current (LS)	
199	Channel 2 Yellow Current (MS)	
200	Channel 3 Yellow Current (LS)	
201	Channel 3 Yellow Current (MS)	
202	Channel 4 Yellow Current (LS)	
203	Channel 4 Yellow Current (MS)	
204	Channel 5 Yellow Current (LS)	
205	Channel 5 Yellow Current (MS)	
206	Channel 6 Yellow Current (LS)	
207	Channel 6 Yellow Current (MS)	
208	Channel 7 Yellow Current (LS)	
209	Channel 7 Yellow Current (MS)	
210	Channel 8 Yellow Current (LS)	
211	Channel 8 Yellow Current (MS)	
212	Channel 9 Yellow Current (LS)	
213	Channel 9 Yellow Current (MS)	
214	Channel 10 Yellow Current (LS)	
215	Channel 10 Yellow Current (MS)	
216	Channel 11 Yellow Current (LS)	
217	Channel 11 Yellow Current (MS)	
218	Channel 12 Yellow Current (LS)	
219	Channel 12 Yellow Current (MS)	

Byte #	Contents	Description
220	Channel 13 Yellow Current (LS)	
221	Channel 13 Yellow Current (MS)	
222	Channel 14 Yellow Current (LS)	
223	Channel 14 Yellow Current (MS)	
224	Channel 15 Yellow Current (LS)	
225	Channel 15 Yellow Current (MS)	
226	Channel 16 Yellow Current (LS)	
227	Channel 16 Yellow Current (MS)	
228	Channel 17 Yellow Current (LS)	
229	Channel 17 Yellow Current (MS)	
230	Channel 18 Yellow Current (LS)	
231	Channel 18 Yellow Current (MS)	
232	Channel 19 Yellow Current (LS)	
233	Channel 19 Yellow Current (MS)	
234	Channel 20 Yellow Current (LS)	
235	Channel 20 Yellow Current (MS)	
236	Channel 21 Yellow Current (LS)	
237	Channel 21 Yellow Current (MS)	
238	Channel 22 Yellow Current (LS)	
239	Channel 22 Yellow Current (MS)	
240	Channel 23 Yellow Current (LS)	
241	Channel 23 Yellow Current (MS)	
242	Channel 24 Yellow Current (LS)	
243	Channel 24 Yellow Current (MS)	
244	Channel 25 Yellow Current (LS)	
245	Channel 25 Yellow Current (MS)	
246	Channel 26 Yellow Current (LS)	
247	Channel 26 Yellow Current (MS)	
248	Channel 27 Yellow Current (LS)	

Byte #	Contents	Description
249	Channel 27 Yellow Current (MS)	
250	Channel 28 Yellow Current (LS)	
251	Channel 28 Yellow Current (MS)	
252	Channel 29 Yellow Current (LS)	
253	Channel 29 Yellow Current (MS)	
254	Channel 30 Yellow Current (LS)	
255	Channel 30 Yellow Current (MS)	
256	Channel 31 Yellow Current (LS)	
257	Channel 31 Yellow Current (MS)	
258	Channel 32 Yellow Current (LS)	
259	Channel 32 Yellow Current (MS)	
260	Channel 1 Green Current (LS)	
261	Channel 1 Green Current (MS)	
262	Channel 2 Green Current (LS)	
263	Channel 2 Green Current (MS)	
264	Channel 3 Green Current (LS)	
265	Channel 3 Green Current (MS)	
266	Channel 4 Green Current (LS)	
267	Channel 4 Green Current (MS)	
268	Channel 5 Green Current (LS)	
269	Channel 5 Green Current (MS)	
270	Channel 6 Green Current (LS)	
271	Channel 6 Green Current (MS)	
272	Channel 7 Green Current (LS)	
273	Channel 7 Green Current (MS)	
274	Channel 8 Green Current (LS)	
275	Channel 8 Green Current (MS)	
276	Channel 9 Green Current (LS)	
277	Channel 9 Green Current (MS)	
278	Channel 10 Green Current (LS)	
279	Channel 10 Green Current (MS)	
280	Channel 11 Green Current (LS)	
281	Channel 11 Green Current (MS)	
282	Channel 12 Green Current (LS)	
283	Channel 12 Green Current (MS)	

Byte #	Contents	Description
284	Channel 13 Green Current (LS)	
285	Channel 13 Green Current (MS)	
286	Channel 14 Green Current (LS)	
287	Channel 14 Green Current (MS)	
288	Channel 15 Green Current (LS)	
289	Channel 15 Green Current (MS)	
290	Channel 16 Green Current (LS)	
291	Channel 16 Green Current (MS)	
292	Channel 17 Green Current (LS)	
293	Channel 17 Green Current (MS)	
294	Channel 18 Green Current (LS)	
295	Channel 18 Green Current (MS)	
296	Channel 19 Green Current (LS)	
297	Channel 19 Green Current (MS)	
298	Channel 20 Green Current (LS)	
299	Channel 20 Green Current (MS)	
300	Channel 21 Green Current (LS)	
301	Channel 21 Green Current (MS)	
302	Channel 22 Green Current (LS)	
303	Channel 22 Green Current (MS)	
304	Channel 23 Green Current (LS)	
305	Channel 23 Green Current (MS)	
306	Channel 24 Green Current (LS)	
307	Channel 24 Green Current (MS)	
308	Channel 25 Green Current (LS)	
309	Channel 25 Green Current (MS)	
310	Channel 26 Green Current (LS)	
311	Channel 26 Green Current (MS)	
312	Channel 27 Green Current (LS)	

Byte #	Contents	Description
313	Channel 27 Green Current (MS)	
314	Channel 28 Green Current (LS)	
315	Channel 28 Green Current (MS)	
316	Channel 29 Green Current (LS)	
317	Channel 29 Green Current (MS)	
318	Channel 30 Green Current (LS)	
319	Channel 30 Green Current (MS)	
320	Channel 31 Green Current (LS)	
321	Channel 31 Green Current (MS)	
322	Channel 32 Green Current (LS)	
323	Channel 32 Green Current (MS)	
324	HDFU #1, 1-1 Current (LS)	HDFU Load Current scaled to 1 mA per bit
325	HDFU #1, 1-1 Current (MS)	
326	HDFU #1, 1-2 Current (LS)	
327	HDFU #1, 1-2 Current (MS)	
328	HDFU #1, 2-1 Current (LS)	
329	HDFU #1, 2-1 Current (MS)	
330	HDFU #1, 2-2 Current (LS)	
331	HDFU #1, 2-2 Current (MS)	
332	HDFU #1, Aux 1Current (LS)	
333	HDFU #1, Aux 1Current (MS)	
334	HDFU #1, Aux 2Current (LS)	
335	HDFU #1, Aux 2Current (MS)	
336	HDFU #2, 1-1 Current (LS)	
337	HDFU #2, 1-1 Current (MS)	
338	HDFU #2, 1-2 Current (LS)	
339	HDFU #2, 1-2 Current (MS)	
340	HDFU #2, 2-1 Current (LS)	
341	HDFU #2, 2-1 Current (MS)	
342	HDFU #2, 2-2 Current (LS)	
343	HDFU #2, 2-2 Current (MS)	
344	HDFU #2, Aux 1Current (LS)	
345	HDFU #2, Aux 1Current (MS)	
346	HDFU #2, Aux 2Current (LS)	
347	HDFU #2, Aux 2Current (MS)	
348	Binary Seconds	CMU Time and Date
349	Binary Minutes	
350	Binary Hours (0:23)	

Byte #	Contents	Description
351	Binary Date	
352	Binary Month	
353	Binary Year	
354	Voltage * 4	CMU 24VDC Supply Voltage
355	Voltage * 4	CMU 48VDC Supply Voltage
356	degrees F + 40	CMU Temperature (Fahrenheit)
357	Red Status 8:1	Channel Current Sense Status Status bits shall be set to 1 for channels that are sensed active. Status bits shall be set to 0 for channels that have the Current Sense monitor function disabled.
358	Red Status 16:9	
359	Red Status 24:17	
360	Red Status 32:25	
361	Yellow Status 8:1	
362	Yellow Status 16:9	
363	Yellow Status 24:17	
364	Yellow Status 32:25	
365	Green Status 8:1	
366	Green Status 16:9	
367	Green Status 24:17	
368	Green Status 32:25	
369	Flasher Status	Flasher Status (1=Fail) b0 = HDFU #1 FL1-1 b1 = HDFU #1 FL1-2 b2 = HDFU #1 FL2-1 b3 = HDFU #1 FL2-2 b4 = HDFU #2 FL1-1 b5 = HDFU #2 FL1-2 b6 = HDFU #2 FL2-1 b7 = HDFU #2 FL2-2

9.2 SB#2 Communications Protocol

9.2.1 General

Serial Bus #2 (SB#2) shall be dedicated to gathering preprocessed data from the Cabinet Smart Input Devices resident in the input functionalities. SB#2 connects in a multi-drop topology with C12S of the ATC 5201 Controller Unit and one or more SIUs. Each model 2218 SIU routes SB#2 to one or more sensor devices using the INBUS function. This SB#2 can be for ATC data collection such as sensor operational status, detection speed reports, occupancy reports, vehicle counts, etc. Alternatively, NEMA TS-2 Port 1 can operate on SB#2 as an option, using TxD, RxD, TxC and RxC signal pairs of Table 7 in Section 6.1.10.6.. In this case, communication functions per the BIU Port 1 section of the NEMA TS-2 standard.

9.2.2 Data Link Layer

For INBUS SB#2 shall be designed to operate Asynchronous

- (1) START BIT
- Programmable number of DATA BITS
- (1) or (2) STOP BIT(s)
- Optional PARITY BIT
- Selectable data rates up to 19,200 bits per second.

For other applications defined by the ATC, the SB#2 is designed to operate up to a 614.4K bps synchronous data rate.

9.2.3 Communication Frames

The protocol for SB#2 is not controlled by this standard.

9.3 SB#3 Communications Protocol

9.3.1 General

Serial Bus #3 is used by the CMU2212 to communicate with the 2202 HDSP/FU and the model 2220 ADU.

9.3.2 Serial Bus Terminations

See section 8.4.2.

9.3.3 Electrical

The Serial Bus #3 shall consist of two interface links (TXD $\pm$ and RXD $\pm$) conforming to the TIA-485-A, *Electrical Characteristics of Generators and Receivers for use in Balanced Digital Multipoint Systems*. Where differences occur between the TIA-485-A standard and this document, this document shall govern.

All voltage potentials on the 2202 HDSP/FU SB#3 interface shall be referenced to CMU MAINS GROUND.

9.3.4 Data Link Layer

The data link layer protocol is based on a subset of HDLC as defined by ISO/IEC 13239:2002. Each frame shall consist of the following fields:

Flag byte	= 0x7E
Address byte ADU2220	= 0x01 through 0x20 for HDFSP/FU, 0xFA for ADU2220
Control byte	= 0x13 (U Format)
Information field	= defined below in Section 9.3.7, 9.3.8, and 9.3.9
Frame Check Sequence	= 16 bit FCS procedure defined in clause 2.5.2 of ISO/IEC 13239:2002.
Flag byte	= 0x7E

Transmission shall be in start/stop mode with basic transparency defined by clause 4.3.2.2 of ISO/IEC 13239:2002 applied. The format shall be 8 bit data, 1 stop bit, no parity, and 307,200 bits per second ($\pm 2\%$). Only asynchronous half duplex operation shall be permitted.

9.3.5 Procedures

Frames transmitted by the CMU-2212 shall be referred to as command frames and frames transmitted by the 2202 HDSP/FU and ADU shall be referred to as response frames. Response frames shall only be transmitted as a result of correctly receiving a command frame. The first eight bits in each information field shall contain the Frame Type number.

Reserved bits in any message shall always be set to zero by the transmitting station.

9.3.6 Message Timing

The 2202 HDSP/FU and ADU shall begin its response to command frames from the CMU-2212 within a designated period of time following the correct reception of a complete command frame including the closing flag. This period shall be known as the Service Time and shall have a maximum value of 500 microseconds. The 2202 HDSP/FU and ADU TXD link output shall be in the high impedance state outside of the interval defined by the Service Time plus Response Time.

The 2202 HDSP/FU and ADU shall complete its transmission of the response frame including the closing flag within a designated time known as the Response Time, depending on the number of bytes transmitted in the response frame. The 2202 HDSP/FU and ADU TXD link output shall transition to a high impedance state a maximum of 25 microseconds following the transmission of the closing flag. The Response Time period shall have a maximum value of:

$$(1.1) * (\# \text{ of bytes in information field} + 6) * (10/307200)$$

Note that due to the transparency mechanism, any occurrence of the flag byte (0x7E) or control escape byte (0x7D) in the information field adds a second byte to the count. Thus, the number of bytes in the information field could be higher if data characters are 0x7E or 0x7D.

Following the transmission of each command frame, there shall be a Dead Time during which the CMU-2212 does not transmit. This Dead Time shall be a minimum of the Service Time plus the Response Time.

9.3.7 SB#3 Information Field Formats

The first eight bits in each information field shall contain the frame Type Number. There shall be a maximum of 64 different command frame types defined by this standard and 64 different response frame types defined by this standard. Additionally, there shall be 64 different command frame types reserved for manufacturers use and 64 different response frame types reserved for manufacturer's use, as outlined below.

Table 64: SB#3 Frame Types

Frame Types	Function
0-63	Command Frame
64-127	Command Frame, Manufacturers Use
128-191	Response Frame
192-255	Response Frame, Manufacturers Use

The information fields of frame types 64 through 127 and 192 through 255 shall contain a manufacturer specific code (MFG Code). The manufacturer specific code is a 16 bit

binary number which shall be placed in bits 9–24 of the information field. These codes are maintained and issued by NEMA.

Reserved bits shall always be set to zero by the transmitting station.

The SB#3 frames to and from the model 2220 shall use only Command frame types 64-127 and corresponding Response frame types 192-255.

The intention in placing manufacturer specific codes in frame types that are reserved for manufacturers use is to allow equipment to differentiate between frames that may contain the same frame type. (Authorized Engineering Information.)

9.3.8 2202 HDSP/FU SB#3 Frame Types

9.3.8.1 Voltage / Current Encoding

RMS voltage and current data shall be reported as a 14 bit positive integer, and shall be encoded in the frames as follows:

- The LSB byte in the frame is composed of the least significant 7 bits (b6:b0) of the reported value with bit 7 of the frame byte set to “1”.
- The MSB byte in the frame is composed of the next most significant 7 bits (b13:b7) of the reported value with bit 7 of the frame byte set to “1”.

Bit #	7	6	5	4	3	2	1	0
LSB	1	b6	b5	b4	b3	b2	b1	b0
MSB	1	b13	b12	b11	b10	b9	b8	b7

9.3.8.2 2202 HDSP/FU Status

9.3.8.2.1 Type 01 Command – HDSP Status

Table 65: SB#3 Type 01 Command

Byte #	Contents	Description
1	01	Frame Type
2	Channel Fault Indicator	Channel ID Indicators b0 = Channel 1a indicator active b1 = Channel 2a indicator active b2 = Channel 1b indicator active b3 = Channel 2b indicator active b4 = Display Sleep mode b5 = Disable RxTx Indicators b6 = reserved b7 = 1

Byte #	Contents	Description
3	Channel Output Disable	Channel 1, 2 Output Disable (set bit disables output) b0 = Channel 1 Red Disable b1 = Channel 1 Yellow Disable b2 = Channel 1 Green Disable b3 = Channel 2 Red Disable b4 = Channel 2 Yellow Disable b5 = Channel 2 Green Disable b6 = Reserved b7 = 1

9.3.8.2.2 Type 129 Response – HDSP Status

Table 66: SB#3 Type 129 Response

Byte #	Contents	Description
1	129	Frame Type
2	Status	HDSP Status b0 = set to 1 if +24 VDC MONITOR input is Low b1 = set if 2202 HDSP/FU has reset since last poll b2 = set if last RMS period data was not transmitted b3 = diagnostic failure b4:6 = Reserved b7 = 1
3	Signal Power LSB	HV+SIGNAL or LV+SIGNAL RMS Voltage (Units are 0.1 Vrms) Field Output RMS Voltage (Units are 0.1 Vrms)
4	Signal Power MSB	
5	Channel 1 Red LSB	
6	Channel 1 Red MSB	
7	Channel 1 Yellow LSB	
8	Channel 1 Yellow MSB	
9	Channel 1 Green LSB	
10	Channel 1 Green MSB	
11	Channel 2 Red LSB	
12	Channel 2 Red MSB	
13	Channel 2 Yellow LSB	
14	Channel 2 Yellow MSB	
15	Channel 2 Green LSB	

Byte #	Contents	Description
16	Channel 2 Green MSB	Field Output RMS Current (Units are 1 Milliampere)
17	Channel 1 Red LSB	
18	Channel 1 Red MSB	
19	Channel 1 Yellow LSB	
20	Channel 1 Yellow MSB	
21	Channel 1 Green LSB	
22	Channel 1 Green MSB	
23	Channel 2 Red LSB	
24	Channel 2 Red MSB	
25	Channel 2 Yellow LSB	
26	Channel 2 Yellow MSB	
27	Channel 2 Green LSB	
28	Channel 2 Green MSB	
29	Channel Input Control	Channel 1, 2 Input State (1=Input Active) b0 = Channel 1 Red On b1 = Channel 1 Yellow On b2 = Channel 1 Green On b3 = Channel 2 Red On b4 = Channel 2 Yellow On b5 = Channel 2 Green On b6 = reserved b7 = 1
30	Channel Output Disabled	Channel 1, 2 (1=Output Disabled) b0 = Channel 1 Red Disabled b1 = Channel 1 Yellow Disabled b2 = Channel 1 Green Disabled b3 = Channel 2 Red Disabled b4 = Channel 2 Yellow Disabled b5 = Channel 2 Green Disabled b6 = reserved b7 = 1

9.3.8.2.3 Type 02 Command – HDFU Status

Table 67: SB#3 Type 02 Command

Byte #	Contents	Description
1	02	Frame Type
2	Channel Output Disable	Channel 1, 2 Output Disable (1=Output Disabled) b0 = Channel 1-1 Disable b1 = Channel 1-2 Disable b2 = Channel 1 Aux Disable (Ch1 Green) b3 = Channel 2 Aux Disable (Ch 2 Red) b4 = Channel 2-1 Disable b5 = Channel 2-2 Disable b6 = Disable RxTx Indicators b7 = 1
3	Channel Fault Indicator	Channel ID Indicators b0 = Channel 1a indicator active b1 = Channel 2a indicator active b2 = Channel 1b indicator active b3 = Channel 2b indicator active b4 = Display Sleep mode b5 = Use display bits b0:3 b6 = reserved b7 = 1 If b5 is set, then the HDFU will use b3:b0 to drive the Channel ID LEDs.

9.3.8.2.4 Type 130 Response – HDFU Status

Table 68: SB#3 Type 130 Response

Byte #	Contents	Description
1	130	Frame Type
2	HDFU Status	HDFU Status b0 = Reserved b1 = set if HDFU has reset since last poll b2 = set if last RMS period data was not transmitted b3 = diagnostic failure b4:6 = Reserved b7 = 1
3	Signal Power LSB	HV+SIGNAL or LV+SIGNAL RMS Voltage (Units are 0.1 Vrms)
4	Signal Power MSB	
5	Channel 1-1 LSB	Field Output RMS Voltage (Units are 0.1 Vrms)

Byte #	Contents	Description
6	Channel 1-1 MSB	
7	Channel 1-2 LSB	
8	Channel 1-2 MSB	
9	Channel 1 Aux LSB	
10	Channel 1 Aux MSB	
11	Channel 2 Aux LSB	
12	Channel 2 Aux MSB	
13	Channel 2-1 LSB	
14	Channel 2-1 MSB	
15	Channel 2-2 LSB	
16	Channel 2-2 MSB	
17	Channel 1-1 LSB	Field Output RMS Current (Units are 1 Milliampere)
18	Channel 1-1 MSB	
19	Channel 1-2 LSB	
20	Channel 1-2 MSB	
21	Channel 1 Aux LSB	
22	Channel 1 Aux MSB	
23	Channel 2 Aux LSB	
24	Channel 2 Aux MSB	
25	Channel 2-1 LSB	
26	Channel 2-1 MSB	
27	Channel 2-2 LSB	
28	Channel 2-2 MSB	
29	Channel Input Control	Channel 1, 2 Input State b0 = 1-1 Control On b1 = 1-2 Control On b2 = Channel X On b3 = Channel Y On b4 = 2-1 Control On b5 = 2-2 Control On b6 = Reserved b7 = 1
30	Channel Output Disabled	Channel 1, 2 (1=Output Disabled) b0 = Channel 1-1 Disabled b1 = Channel 1-2 Disabled b2 = Channel X Disabled b3 = Channel Y Disabled b4 = Channel 2-1 Disabled b5 = Channel 2-2 Disabled b6 = Reserved b7 = 1

9.3.8.2.5 Type 60 Command – HDFU Module ID

Table 69: SB#3 Type 60 Command

Byte #	Contents	Description
1	60	Frame Type

9.3.8.2.6 Type 188 Response – HDFU Module ID

Table 70: SB#3 Type 188 Response

Byte #	Contents	Description
1	188	Frame Type
2	Device Type	2202 HDSP/FU Type 1 = 2202 HDSP/FU-HV 2 = 2202 HDSP/FU-LV 3 = 2202 HDSP/FU-VHV (reserved) 4 = 2200 HDSP-HV 5 = 2200 HDSP-LV 6 = 2200 HDSP-VHV (reserved) 7 = 2204 HDFU-HV 8 = 2204 HDFU-LV 9 = 2204 HDFU-VHV (reserved) 10:255 = Reserved
3:62	ID String	Manufacturer-specified Identification String A packed 60 character ID shall be stored in ASCII format. Allowable characters are 020h through 07Eh. If less than 60 characters are used, the unused locations shall be set to 20h. Default programming shall be 20h.

9.3.8.2.7 Type 01, 02 Response – Negative Acknowledge

This frame shall be transmitted from the 2202 HDSP/FU to the CMU as a negative acknowledge response frame if the 2202 HDSP/FU correctly receives a Type 01 command frame or a Type 02 command frame with an invalid parameter, including the Frame Type.

Table 71: SB#3 Type 128 Response

Byte #	Contents	Description
--------	----------	-------------

1	128	Frame Type
2	Status	2202 HDSP/FU SB#3 Error Type b0 = set to 1 if invalid frame type received b1 = set to 1 if invalid parameter is received b2:6 = reserved b7 = 1
3	Parameter Number	Invalid Parameter Byte Number The byte number of the Type 01 or Type 02 frame that contained the invalid parameter. If more than one parameter is invalid then this offset shall be the first byte in the frame in error.

9.3.9 Model 2220 SB#3 Frame Types

The model 2220 shall respond to serial data frames transmitted by the CMU2212 on the Serial Bus #3. The SB#3 address of the model 2220 shall be 0xFA.

9.3.9.1 Type 70 Command – Set LEDs

Table 72: ADU Type 70 Command

Byte #	Contents	Description
1	70	Frame Type
2	0-255	MFG Code LS
3	0-255	MFG Code MS
4	LED Control	b0 = LED Enable b1 = Disable RxTx Indicators b7:2 = reserved
5	Ch 8:1	Blue LEDs
6	Ch 16:9	
7	Ch 24:17	
8	Ch 32:25	
9	Ch 8:1	Red LEDs
10	Ch 16:9	
11	Ch 24:17	
12	Ch 32:25	
13	Ch 8:1	Yellow LEDs
14	Ch 16:9	
15	Ch 24:17	
16	Ch 32:25	
17	Ch 8:1	Green LEDs
18	Ch 16:9	
19	Ch 24:17	
20	Ch 32:25	

9.3.9.2 Type 198 Response – Set LEDs

Table 73: ADU Type 198 Response

Byte #	Contents	Description
1	198	Frame Type
2	0-255	MFG Code LS
3	0-255	MFG Code MS
4	Button Status	(A value of 1 shall be set when the button is in the pressed state) b0 = Help b1 = Next b2 = Back b3 = Select b4 = Exit b7:5 = reserved
5	0	Reserved for future use

9.3.9.3 Type 71 Command – Set LCD

Table 74: ADU Type 71 Command

Byte #	Contents	Description
1	71	Frame Type
2	0-255	MFG Code LS
3	0-255	MFG Code MS
4	LCD Control	b0 = Back Light Enable b1 = Heater Enable b7:2 = reserved
5	Line Number	LCD Line Number
6	Ascii Char	ASCII Text for the 20 Character Line Character #1 is the leftmost character
7	Ascii Char	
8	Ascii Char	
9	Ascii Char	
10	Ascii Char	
11	Ascii Char	
12	Ascii Char	
13	Ascii Char	
14	Ascii Char	
15	Ascii Char	
16	Ascii Char	
17	Ascii Char	

Byte #	Contents	Description
18	Ascii Char	
19	Ascii Char	
20	Ascii Char	
21	Ascii Char	
22	Ascii Char	
23	Ascii Char	
24	Ascii Char	
25	Ascii Char	

9.3.9.4 Type 199 Response – Set LCD

Table 75: ADU Type 199 Response

Byte #	Contents	Description
1	199	Frame Type
2	0-255	MFG Code LS
3	0-255	MFG Code MS
4	Button Status	(A value of 1 shall be set when the button is in the pressed state) b0 = Help b1 = Next b2 = Back b3 = Select b4 = Exit b7:5 = reserved
5	0	Reserved for future use

10 PRODUCT SAFETY AND RELIABILITY

10.1 Mercury

Mercury shall not be used in components that will create a disposal issue.

10.2 Shock Hazard

For the ATCC Low Voltage Version, voltages shall be constrained to eliminate shock hazard as defined in NFPA 70B and 70E.

10.3 Arc Flash

Electromotive force of electrical signals shall be constrained to eliminate arc flash hazard as defined in NFPA 70B and 70E.

10.4 Flashing Operation During Service and Replacement

The cabinet shall be capable of remaining in Flashing Mode when any element of the cabinet is removed or replaced except the combined Power & Service Functionality and Field Termination functionality.

10.5 Mean Time Between Failure (MTBF)

If published, MTBF shall be calculated using the latest revision of MIL standard 217 and published with the calculation criteria.

10.6 Derating

The parts used to create the elements of the TFCS shall not be operated at greater than 80 percent of their rated values unless otherwise indicated within this standard.

11 ENVIRONMENTAL AND TESTING REQUIREMENTS

NOTE: All references in this section to NEMA TS2 explicitly refer to the NEMA Standards Publication TS2-2016 v3.07, *Traffic Controller Assemblies with NTCIP Requirements*.

11.1 General

This section establishes the environmental and operational conditions for a First Article ATCC, and defines the minimum test procedures which may be used to demonstrate conformance of an ATCC cabinet with the provisions of the Standard.

Testing shall be performed either within an environmental chamber or on a bench.

These test procedures do not verify equipment performance under every possible combination of environmental requirements covered by the Standard. However, nothing in this testing profile shall be construed as to relieve the requirement that the equipment provided must fully conform with the Standard under all environmental conditions stated herein.

11.2 Inspection

A visual and physical inspection of the ATCC unit shall verify mechanical, dimensional, and assembly conformance to all parts of this standard.

11.3 Testing Certification

A complete quality control and final test report shall be supplied with each item.

11.4 Definitions of Design Acceptance Testing (DAT) and Production Testing.

Design Acceptance Testing (DAT) is performed on the first article ATCC unit and is a part of the pre-production process.

Production testing is performed on all units prior to their shipment to an agency.

11.5 Environmental and Operating Requirements

The requirements (voltage, temperature, etc.) of this section shall apply in any combination.

11.5.1 Voltage and Frequency

11.5.1.1.1 Operating Voltage

The nominal operating voltage shall be 120 VAC, unless otherwise noted for the ATCC High Voltage version (HV) and at 48 VDC for the ATCC Low Voltage version (LV).

11.5.1.1.2 Operating Frequency

The nominal AC line operating frequency range shall be 60 Hz (± 3.0 Hz), unless otherwise noted for the ATCC High Voltage version (HV) and at DC for the ATCC Low Voltage version (LV).

11.5.2 Transients, Power Service (DAT)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.1.6, Transients, Power Service.

11.5.3 Nondestructive Transient Immunity (DAT)

11.5.3.1 Nondestructive Transient Immunity (DAT)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.1.8, Nondestruct Transient Immunity, with the following variations: Test voltage amplitude shall be 2000 ± 100 V, both positive and negative polarity.

11.5.3.2 Nondestructive Transient protection – Output

11.5.3.2.1 Output AC transient protection

The HV cabinet shall provide protection for AC outputs from voltage surges introduced through output wiring. Surge devices shall have a peak surge current rating between 26kA and 39kA @ 240 Joules of energy, clamp voltage of 340V @ 1KA.

11.5.3.2.2 Output DC transient protection

The LV Cabinet shall provide protection for all DC outputs from voltage surges introduced through peripheral load devices (i.e. loop, camera, LED Lamp etc.). The device must be rated to clamp to DC 60 V at a surge current of 250A. An example of such a device would have the same form factor as an HE103C-9.

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.1.8, Nondestruct Transient Immunity, with the following variations:

- Test voltage amplitude shall be 2000 ± 100 V, both positive and negative polarity.

11.5.4 Temperature and Humidity

The ATC unit under test shall maintain all programmed functions when the temperature and humidity ambients are within the specified limits defined herein.

11.5.4.1.1 Ambient Temperature

The operating ambient temperature range shall be from -37°C to +74°C. The storage temperature range shall be from -45°C to +85°C.

The rate of change in ambient temperature shall not exceed 18°C per hour, during which the relative humidity shall not exceed 95 percent.

11.5.4.1.2 Humidity

The relative humidity shall not exceed 95 percent non-condensing over the temperature range of -37°C to +74°C.

Above +46°C, constant absolute humidity shall be maintained. This will result in the relative humidities shown in Table 76 for dynamic testing.

Table 76: Ambient Temperature versus Relative Humidity At Barometric Pressures
(29.92 In. Hg.) (Non-Condensing)

Ambient Temperature/ Dry Bulb (°C)	Relative Humidity (in%)	Ambient Temperature/ Wet Bulb (°C)
-37.0 to 1.1	10	-17.2 to 42.7
1.1 to 46.0	95	42.7
48.8	70	42.7
54.4	50	42.7
60.0	38	42.7
65.4	28	42.7
71.2	21	42.7
74.0	18	42.7

11.6 Test Facilities

All instrumentation required in the test procedures, such as voltmeters, ammeters, thermocouples, pulse timers, etc. shall be selected in accordance with good engineering practice. Calibration records for all test equipment shall be provided with test documentation. In all cases where time limit tests are required, the allowance for any instrumentation errors shall be included in the limit test.

1. Variable Voltage Source: A variable source capable of supplying 20 A from 0 VAC to 135 VAC for the ATCC High Voltage version and 0- 48 VDC for the ATCC Low Voltage version.
2. Environmental Chamber: An environmental chamber capable of attaining temperatures of -37°C to +74°C and relative humidities given in Table 76.

3. Transient Generator(s): Transient generator(s) capable of supplying the transients outlined in 11.5.2 and 11.5.3.

11.7 Test Procedures: Transients, Temperature, Voltage, and Humidity

11.7.1 Test A: Placement in Environmental Chamber and Check-Out of Hook-Up (DAT and Production Testing)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.1, Test A: Placement in Environmental Chamber and Check-Out of Hook-Up, with the following variations:

- For DAT purposes only, the transient generator shall be connected to the AC input circuit at a point at least 25 feet from the AC power source and not over 10 feet from the input to the test unit

Upon the satisfactory completion and verification of the test hook-up, proceed with Test B (DAT) or Test C (DAT and Production Testing). Figure 22 describes the test profile used for Tests C through G to demonstrate the ability of the test unit to function reliably under stated conditions of temperature, voltage and humidity. Where the characteristics of Figure 22 differ from those specified in NEMA TS2, Figure 2-1, those in Figure 22 will apply unless specified otherwise.

11.7.2 Test B: Temperature Cycling and Applied Transient Tests (Power Service) (DAT)

Note: In this section the term “dwell” refers to an application in its resting state awaiting inputs, and the actual resting state depends on the application. The goal is to determine if the transients introduced affect the operation of the application by triggering false inputs and/or outputs causing inappropriate application program activity (e.g. improper timing, false calls, error alarms, etc).

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.2, Test B: Transient Tests (Power Service), with the following variations:

- Test voltage amplitude in Step 13 shall be 2000 ± 100 V, both positive and negative polarity.

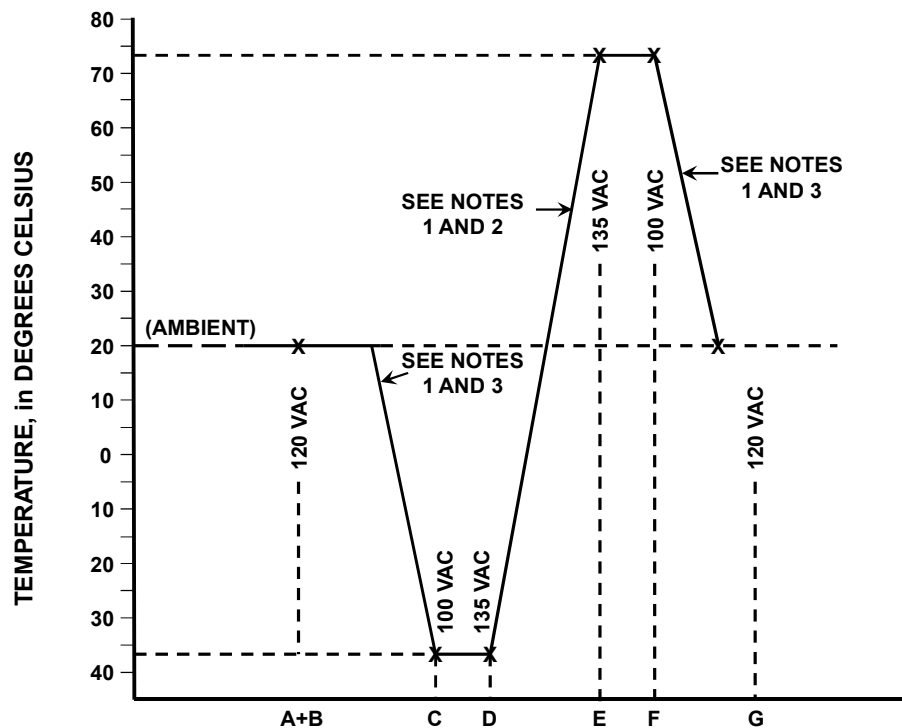


Figure 22: Test Profile

NOTES:

1. The rate of change in temperature shall not exceed 18° C per hour.
2. Humidity controls shall be set in conformance with the humidities given in Table 8-1 during the temperature change between Test D and Test E.
3. If a change in both voltage and temperature are required for the next test, the voltage shall be selected prior to the temperature change.

11.7.3 Test C: Low-Temperature Low-Voltage Tests (DAT and Production Testing)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.3, Test C: Low-Temperature Low-Voltage Tests, with the following variations:

- Utilize Figure 22 for the test profile (temperature: -37°C, low voltage).
- Skip tests 2.c.2) 2.2.11 Timing Accuracy Tests and 2.c.3) 2.2.11.2 Repeatability.

11.7.4 Test D: Low-Temperature High-Voltage Tests (DAT and Production Testing)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.4, Test D: Low-Temperature High-Voltage Tests, with the following variations:

- Utilize Figure 22 for the test profile (temperature: -37°C).
- Skip tests 2.c.2) 2.2.11 Timing Accuracy Tests and 2.c.3) 2.2.11.2 Repeatability.

11.7.5 Test E: High-Temperature High-Voltage Tests (DAT and Production Testing)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.5, Test E: High-Temperature High-Voltage Tests, with the following variations:

- Utilize Table Table 76 for the humidity controls.
- In Step 3, utilize the temperature range of +1.1° C to +46° C and +42.7° C wet bulb.
- Skip tests 2. 2.2.11 Timing Accuracy Tests and 2. 2.2.11.2 Repeatability.

11.7.6 Test F: High-Temperature Low-Voltage Tests (DAT and Production Testing)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.6, Test F: High-Temperature Low-Voltage Tests, with the following variations:

- Utilize Table Table 76 for the humidity controls.
- In Step 3, utilize the temperature range of +1.1° C to +46° C and +42.7° C wet bulb.
- Skip tests 2. 2.2.11 Timing Accuracy Tests and 2. 2.2.11.2 Repeatability.

11.7.7 Test G: Test Termination (All tests)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.7, Test G: Test Termination, with the following variations:

- Utilize Figure 22 for the test profile.

11.7.8 Test H: Appraisal of Equipment Under Test

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.7.8, Test H: Appraisal of Equipment Under Test.

11.8 Vibration Test (DAT)

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.8, Vibration Test.

Shock and vibration tests shall be performed prior to environmental tests.

11.9 Shock (Impact) Test (DAT)

11.9.1 Purpose of Test

The ATC unit under test shall meet all requirements as defined in NEMA TS2, Section 2.2.9, Shock (Impact) Test, with the following modifications:

- a) Program the table to subject the test unit to a 10G force having a duration of 11 ms.
- b) The test shall use a waveform suitable to simulate a drop test, such as saw tooth. The waveform shall be a saw tooth with 11 ms rise time and 0 ms fall time.
- c) Subject the test unit to the above test setup three times in both the positive and negative direction.

Shock and vibration tests shall be performed prior to environmental tests.

11.10 Electronic Emissions

The TFCS shall adhere to the code of Federal Regulations, Title 47 – Telecommunications, Chapter 1 – Federal Communications Commission, Part 15, Class A, tested with the TFCS functionality mounted in a grounded electrical enclosure.

11.11 Electrostatic Discharge

The TFCS shall adhere to applicable sections of International Electrotechnical Commission (IEC) Standard 61000-4-2:2008 “Electromagnetic compatibility (EMC) – Part 4-2: Testing and Measurement Techniques – Electrostatic Discharge Immunity Test”, tested with the TFCS functionality mounted in a grounded electrical enclosure and using the Human Body model at 2KV, 4KV, 6KV and 8KV. USB testing shall be limited to 4KV.

12 WIRE REQUIREMENTS

12.1 Capacity ratings

All wire used in shall be copper. All #14 to 30 AWG shall be stranded. Wires shall be sized per the ampacity ratings of Table 77.

Table 77: Ampacity

AWG Wire Size	Ampacity Rating
30-24	0.25
22	2
20	4
18	5
16	10
14	15
12	20
10	30
8	50
6	70

12.2 Conductors

Conductors shall conform to military specification MIL-W-16878D, Electrical Insulated High Temperature Wire, type B. Conductors #14 AWG or larger shall be permitted to be UL type THHN.

12.3 Splicing

Conductors shall not be spliced between terminations.

12.4 Cables

Cable may be used in lieu of individual conductors to interconnect equipment or panels within the controller cabinet. The cable shall be constructed of #28 AWG or larger conductors. Insulation shall have a voltage rating of 300 volts minimum and shall be rated for use at 105° C. Cables shall be provided with strain relief.

13 ACRONYMS

Term	Definition
AASHTO	American Association of State Highway and Transportation Officials
AC	alternating current
ADU	CMU Auxilliary Display Unit
ATC	Advanced Transportation Controller
ATCC	ATC Cabinet
BIU	Bus Interface Unit
CB	Circuit Breaker
CBD	Central Business District
CMU	Cabinet Monitor Unit
ConOps	Concept of Operations
DC	direct current
EMI	electromagnetic interference
FHWA	Federal Highway Administration
ESD	Electrostatic Discharge
GFI	Ground Fault Interrupt
GUI	Graphical User Interface
HDFU	High Density Flasher Unit
HDSP	High Density Switch Pack
IEEE	Institute of Electrical and Electronics Engineers
I/F	interface
I/O	input/output
IEC	International Electrotechnical Commission
IMSA	International Municipal Signal Association
IPC	Formerly, the Institute for Printed Circuits. This same institution was later called the Institute Interconnecting and Packaging Electronic Circuits. It is now referred to as IPC-Association Connecting Electronics Industries.
ITE	Institute of Transportation Engineers
ITS	Intelligent Transportation System
JPO	Joint Program Office
LED	light emitting diode
MTBF	mean time between failures
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association
NRTL	Nationally Recognized Testing Lab
NTCIP	National Transportation Communications for ITS Protocol
NFPA	National Fire Protection Association
PF	power factor
RFI	radio frequency interference
SDO	Standard Development Organization
SEMP	System Engineering Management Plan

Term	Definition
SEP	Systems Engineering Process
SIU	Serial Interface Unit
SSC	Small-Sized Cabinet
StdRS	Standards Requirements Specification
StdHLD	Standards High Level Design
TFCS	transportation field cabinet system
UPS	Uninterruptible Power Source
USA	United States of America
USDOT	United States Department of Transportation
VAC	voltage alternating current
VDC	voltage direct current
WG	Working Group

14 NEEDS TO REQUIREMENTS TO DESIGN TRACEABILITY

Needs to Requirements of this section were posted on the ITE Community Website in April, 2016 minus Related Design Elements field that was added in May 2018 for traceability to sections of this document.

Verify the Requirement					
Requirement ID		Requirement Title			
5.1.1.1		Patents and Copyrights			
Requirement Text					
Interfaces that are patented or copyrighted shall not be included as part of the TFCS					
Justification for the Requirement:					
1. The user needs the TFCS to be specified as an open architecture system					
Source for Justification:					
1. UN ID 4.3.1.1					
Requirement Text (Comments/Changes)					
Guidance:					
Related Design Elements					
1. Section 9 Describes the standardized interfaces					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.

6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement	
Requirement ID	Requirement Title
5.1.1.2	Commonly-Available Components
Requirement Text Components that make up the TFCS shall be available for purchase from distribution or shall be able to be fabricated from commonly available materials.	
Justification for the Requirement: 1. The user needs the TFCS to be specified as an open architecture system.	
Source for Justification: 1. UN ID 4.3.1.1	
Requirement Text (Comments/Changes) Guidance: After discussion, includes purchased parts available from distributors and fabricated parts available from the manufacturer, such as custom sheet metal or electronic subassemblies.	
Related Design Elements Fabricated from commonly available materials: 1. 6.1.10Model 2218 Mechanical Details 2. 6.2.82202 HDSP/FU Mechanical 3. 6.3.11Mechanical 4. 6.5.2Mechanical 5. 6.6.2Form Factor 6. 7.1.2Form Factor 7. 7.2.2Form Factor 8. 7.3.2Form Factor 9. 7.4.2Form Factor 10. 8.1.2Form Factor 11. 8.2.2Form Factor 12. 8.3.1.4Form Factor	

13. 8.4.4Form Factor				
14. Form FactorForm Factor				
15. 8.6.1.2Form Factor				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement	
Requirement ID	Requirement Title
5.1.2	Rack Mountable
Requirement Text	
The TFCS subsystems and assemblies shall be suitable for mounting in a 19 inch rack	
Justification for the Requirement:	
1. The user needs the TFCS to be specified with a modular internal structure.	
Source for Justification:	
1. UN ID 4.3.1.2	
Requirement Text (Comments/Changes)	
Guidance: It is the intent of the design team, to focus on the two door cabinet design but the solution should not preclude the use of a single door cabinet with a removable rack.	
Related Design Elements	
1. Figure 2: High Level Functional Block Diagram, High Voltage Version	

2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version

Requirement Criteria		Yes	No
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	
3	Is the requirement unambiguous?	X	
4	Is the requirement feasible?	X	
5	Is the requirement verifiable?	X	
		Insp.	Anal. Test Demo.
6	If verifiable, by which method?	X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.			

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.5		Utilizes a Controller Subsystem		
Requirement Text				
The TFCS shall utilize a Controller Subsystem (CS) to perform the functions of an application computer and provide external communication interfaces for the TFCS				
Justification for the Requirement:				
1. The user needs the TFCS to be specified with a modular internal structure.				
2. The user needs the TFCS to be scalable.				
3. The user needs the TFCS to be expandable.				
4. The user needs the TFCS to provide a means to quickly transfer cabinet and application data from an external computer to a TFCS or from one TFCS to another.				
5. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard.				
6. The user needs the TFCS to provide uniform internal interfaces and protocols.				
7. The user needs the TFCS to have a non-volatile method of maintaining the system reports and logs.				
Source for Justification:				
1. UN ID 4.3.1.2				
2. UN ID 4.3.1.3				
3. UN ID 4.3.1.4				
4. UN ID 4.3.1.13				
5. UN ID 4.3.3.1				
6. UN ID 4.3.11.2				
7. UN ID 4.3.8.3				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements				
1. 8.3.1.1Serial Bus 1				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
				Demo.

6	If verifiable, by which method?			X	
Note: An answer of no requires a comment or change in the Comments/Change field.					

Verify the Requirement			
Requirement ID		Requirement Title	
5.1.6		Utilizes an Input Assembly	
Requirement Text The TFCS shall utilize an input assembly (IA) to accept devices used for field inputs.			
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.			
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4			
Requirement Text (Comments/Changes) Guidance: “Utilize” means that an input assembly is required when input devices are used. Does not mean that an input assembly is always required. Scalable means deployable for a range of applications. Expandable means can add capacity later.			
Related Design Elements 1. 6.6Sensor Unit 2. 7.3IN Slot			
Requirement Criteria		Yes	No
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	
3	Is the requirement unambiguous?	X	
4	Is the requirement feasible?	X	
5	Is the requirement verifiable?	X	

		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement			
Requirement ID	Requirement Title		
5.1.7	Utilizes an Output Assembly		
Requirement Text The TFCS shall utilize an output assembly (OA) to accept devices used for field outputs.			
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.			
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4			
Requirement Text (Comments/Changes) Guidance: "Utilize" means that an output assembly is required when output devices are used. Does not mean that an output assembly is always required. Scalable means deployable for a range of applications. Expandable means can add capacity later.			
Related Design Elements 1. 6.2Model 2202 High Density Switch Pack / Flasher Unit (HDSP/FU) 2. 7.4OUT Slot			
Requirement Criteria			
	Yes	No	
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	
3	Is the requirement unambiguous?	X	

4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.8		Utilizes an Input/Output Assembly		
Requirement Text The TFCS shall utilize an input/output assembly (IOA) to accept devices used for both field inputs and field outputs.				
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.				
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4				
Requirement Text (Comments/Changes) Guidance: "Utilize" means that an input/output assembly is required when input and output devices are desired in the same rack to save space. Does not mean that an input/output assembly is always required. Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements 1. Figure 2 "Typical Input Functionality" (Note 1) 2. Figure 3 "Typical Input Functionality" (Note 1) Mechanical packaging of Input/Output is not controlled by the Standard. Standard provides common signal names, electrical characteristics and protocols to support Input/Output function.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the				

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement				
Requirement ID		Requirement Title		
5.1.9		Utilizes a Field Terminal Assembly		
Requirement Text The TFCS shall utilize a field termination assembly (FTA) to secure field wiring within the cabinet.				
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.				
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4				
Requirement Text (Comments/Changes) Guidance: "Utilize" means that a field terminal assembly is required to secure field wires inside the cabinet, not a direct connection of field wires to assemblies. Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements 1. Figure 2 Typical Output Termination Functionality (Note 1) 2. Figure 3 Typical Output Termination Functionality (Note 1) 3. 8.7 Output Termination Functionality				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.10		Utilizes a Monitor Assembly		
Requirement Text The TFCS shall utilize a monitor assembly (MA) to monitor the operation of the TFCS and provide diagnostic information.				
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.				
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4				
Requirement Text (Comments/Changes) Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements 1. Figure 2: High Level Functional Block Diagram, High Voltage Version 2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version 3. 6.3Model 2212 Cabinet Monitor Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.11		Utilizes a Service Assembly		
Requirement Text				
The TFCS shall utilize a service assembly (SA) to provide raw power and clean power.				
Justification for the Requirement:				
1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.				
Source for Justification:				
1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later. Raw Power is power that is protected against surges and spikes. Clean Power is protected against surges and spikes and filtered to regulate electrical noise.				
Related Design Elements				
1. Figure 2 Service Functionality (Note 1) 2. Figure 3 Service Functionality (Note 1) 3. 10.4Flashing Operation During Service and Replacement				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.12		Utilizes a Serial Bus Assembly		
Requirement Text The TFCS shall utilize a serial bus assembly (SBA) to facilitate access to the serial bus by the elements of the TFCS.				
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.				
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4				
Requirement Text (Comments/Changes) Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements 1. Figure 2 SB1 / SB2 2. Figure 3 SB1 / SB2 3. 8.3SB#1 / SB#2 Interface				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID	Requirement Title			
5.1.13	Utilizes a Clean Power Bus Assembly			
Requirement Text The TFCS shall utilize a power assembly (CPBA) to distribute clean power for the elements of the TFCS.				
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.				
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4				
Requirement Text (Comments/Changes) Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements 1. Figure 2 Power Functionality (Note 1) 2. Figure 3 Power Functionality (Note 1) 3. 6.4Cabinet Power Supply Requirements, High Voltage Versions				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the				

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement				
Requirement ID		Requirement Title		
5.1.14		Utilizes a Cabinet Power Supply Assembly		
Requirement Text				
The TFCS shall utilize a cabinet power supply to provide DC power to the elements of the TFCS.				
Justification for the Requirement:				
1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable.				
Source for Justification:				
1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements				
1. Figure 2 Power Functionality (Note 1) 2. Figure 3 Power Functionality (Note 1) 3. 6.4Cabinet Power Supply Requirements, High Voltage Versions				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.7		Hot-Swappable Input Devices		
Requirement Text				
The TFCS input modules shall be hot-swappable within the input assembly.				
Justification for the Requirement:				
1. The user needs the TFCS to be specified with a modular internal structure.				
Source for Justification:				
1. UN ID 4.3.1.2				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.6Sensor Unit (refer to specifications for each sensor type listed)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.8		Multiple Input Assemblies		
Requirement Text				
The TFCS shall support the use of multiple Input Assemblies in a single TFCS.				
Justification for the Requirement:				
1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable. 4. The user needs the TFCS to support input devices that have higher channel density than commonly deployed field input devices.				
Source for Justification:				
1. UN ID 4.3.1.2 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4 4. UN ID 4.3.2.2				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements				
1. Figure 2 Typical Input Functionality (Note 1) 2. Figure 3 Typical Input Functionality (Note 1) 3. 6.1.5Input Functions 4. 6.1.5.5SB#1 Address Select Inputs				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.1.3		Open Standard		
Requirement Text The TFCS's physical, electrical and communication interfaces shall be based on existing open standards or defined herein.				
Justification for the Requirement: 1. The user needs the TFCS to be specified with a modular internal structure. 2. The user needs the TFCS to be extensible. 3. The user needs the TFCS to provide standardized external interfaces and protocols.				
Source for Justification: 1. UN ID 4.3.1.2 2. UN ID 4.3.1.5 3. UN ID 4.3.9.2				
Requirement Text (Comments/Changes) Guidance: Constrain requirement. Communications conform to a referenced standard. If not, the communications interface must be defined in this standard. Extensible relates to "future proofing," meaning the interfaces can accommodate future local needs that are unknown at design.				
Related Design Elements 1. Section 8INTERFACES (normative references to EIA standards or defined in this standard)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.5		48 Input Channels		
Requirement Text The TFCS shall provide an input assembly that provides for up to 48 input channels.				
Justification for the Requirement: 2. The user needs the TFCS to be scalable. 3. The user needs the TFCS to be expandable. 4. The user needs the TFCS to support the use of commonly deployed field sensors 5. The user needs the TFCS to support input devices that have higher channel density than commonly deployed field input devices.				
Source for Justification: 2. UN ID 4.3.1.3 3. UN ID 4.3.1.4 4. UN ID 4.3.2.1 5. UN ID 4.3.2.2				
Requirement Text (Comments/Changes) Guidance: Field sensors external to the cabinet				
Related Design Elements 1. Figure 2 Typical Input Functionality (Note 1) 2. Figure 3 Typical Input Functionality (Note 1)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.8		Multiple Input Assemblies		
Requirement Text				
The TFCS shall support the use of multiple Input Assemblies in a single TFCS.				
Justification for the Requirement:				
1. The user needs the TFCS to be scalable. 2. The user needs the TFCS to be expandable. 3. The user needs the TFCS to be specified with a modular internal structure. 4. The user needs the TFCS to support input devices that have higher channel density than commonly deployed field input devices.				
Source for Justification:				
1. UN ID 4.3.1.3 2. UN ID 4.3.1.4 3. UN ID 4.3.1.2 4. UN ID 4.3.2.2				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements				
1. Figure 2 Typical Input Functionality (Note 1) 2. Figure 3 Typical Input Functionality (Note 1) 3. 6.1.5Input Functions 4. 6.1.5.5SB#1 Address Select Inputs				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.4.2		At least seven high density switch pack modules		
Requirement Text				
The TFCS' output assembly shall accept at least 7 high density switch pack (HDSP) modules.				
Justification for the Requirement:				
1. The user needs the TFCS to be scalable.				
2. The user needs the TFCS to be expandable.				
3. The user needs the TFCS to be of a design that reduces the time required for maintenance personnel to perform maintenance actions in the field.				
4. The user needs the TFCS to support input devices that have higher channel density than commonly deployed field input devices.				
5. The user needs the TFCS to support the use of output devices that have higher channel density than the commonly deployed field output devices				
Source for Justification:				
1. UN ID 4.3.1.3				
2. UN ID 4.3.1.4				
3. UN ID 4.3.1.20				
4. UN ID 4.3.2.2				
5. UN ID 4.3.4.1				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements				
1. Figure 2 Typical Output Functionality (Note 1)				
2. Figure 3 Typical Output Functionality (Note 1)				
3. 6.1.6Output Functions				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
		Demo.		
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the				

Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID	Requirement Title			
5.1.42	Subsystems not interdependent			
Requirement Text				
Replacement of any individual TFCS subsystems in any combination shall not affect the operation of other TFCS subsystems.				
Justification for the Requirement:				
1. The user needs the TFCS to be scalable.				
Source for Justification:				
1. UN ID 4.3.1.3				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements				
1. 10.4Flashing Operation During Service and Replacement				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.43		Subsystem Quantities		
Requirement Text				
The Subsystem quantities shall match those addresses assigned in Section 4.7.14.7.1 of the cabinet V1 standard				
Justification for the Requirement:				
1. The user needs the TFCS to be scalable. 2. The user needs the TFCS to be expandable.				
Source for Justification:				
1. UN ID 4.3.1.3 2. UN ID 4.3.1.4				
Requirement Text (Comments/Changes)				
Guidance: Scalable means deployable for a range of applications. Expandable means can add capacity later.				
Related Design Elements				
1. 9.1.6SIU2218 SB#1 Frame Types includes the addresses of the V1 standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.4.8		Multiple Output Assemblies		
Requirement Text				
The TFCS shall support the use of multiple Output Assemblies in a single TFCS.				
Justification for the Requirement:				
1. The user needs the TFCS to be expandable. 2. The user needs the TFCS to support the use of output devices that have higher channel density than the commonly deployed field output devices.				
Source for Justification:				
1. UN ID 4.3.1.4 2. UN ID 4.3.4.1				
Requirement Text (Comments/Changes)				
Guidance: Expandable means can add capacity later.				
Related Design Elements				
1. Figure 2 Typical Output Functionality (Note 1) 2. Figure 3 Typical Output Functionality (Note 1) 3. 6.1.6 Output Functions				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.44		Future Subsystems		
Requirement Text				
The TFCS shall reserve the Subsystem module addresses for future use shown in Section 4.7.14.7.1 of the cabinet V1 standard.				
Justification for the Requirement:				
1. The user needs the TFCS to be extensible.				
Source for Justification:				
1. UN ID 4.3.1.5				
Requirement Text (Comments/Changes)				
Guidance: Extensible relates to “future proofing”, meaning the interfaces can accommodate future local needs that are unknown at design.				
Related Design Elements				
1. 9.1SB#1 Communications Protocol (RESERVED fields)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the ‘Requirement Text’ section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.15		Relocating Elements		
Requirement Text The TFCS' CS, IA, OA, IOA, FTA, SBA, MA, CPBA and SA shall be able to be located in various positions within the TFCS.				
Justification for the Requirement: 1. The user needs the TFCS to be space-efficient.				
Source for Justification: 1. UN ID 4.3.1.6				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2, Note 1 2. Figure 3, Note 1				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.20		24/7 Operation		
Requirement Text				
The TFCS shall operate 24 hours a day and seven days a week without user intervention.				
Justification for the Requirement:				
1. The user needs the TFCS to be designed for reliable and continuous operation without user intervention.				
Source for Justification:				
1. UN ID 4.3.1.7				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Section 10.6Derating				
2. The parts used to create the elements of the TFCS shall not be operated at greater than 80 percent of their rated values unless otherwise indicated within this standard.				
3. ENVIRONMENTAL AND TESTING REQUIREMENTS				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.45		MTBF		
Requirement Text				
If published, MTBF shall be calculated using the latest revision of MIL standard 217 and published with the calculation criteria.				
Justification for the Requirement:				
1. The user needs the TFCS to be designed for reliable and continuous operation without user intervention.				
Source for Justification:				
1. UN ID 4.3.1.7				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 10.5Mean Time Between Failure (MTBF)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.46		Design Life		
Requirement Text				
Design Life shall be stated in terms of MTBF.				
Justification for the Requirement:				
1. The user needs the TFCS to be designed for reliable and continuous operation without user intervention.				
Source for Justification:				
1. UN ID 4.3.1.7				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 10.5Mean Time Between Failure (MTBF)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.24		Derating		
Requirement Text				
The parts used to create the elements of the TFCS shall not be operated at greater than 80 percent of their rated values unless otherwise indicated within this standard.				
Justification for the Requirement:				
1. The user needs the TFCS to be designed for reliable and continuous operation without user intervention.				
Source for Justification:				
1. UN ID 4.3.1.7				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 10.6Derating				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement					
Requirement ID	Requirement Title				
5.12.1	Corrosion Resistance				
Requirement Text The TFCS shall be constructed of materials that are inherently corrosion resistant or have been treated so as to be corrosion resistant.					
Justification for the Requirement: 1. The user needs the TFCS housing to be made of materials that are resistant to rust and corrosion.					
Source for Justification: 1. UN ID 4.3.10.2					
Requirement Text (Comments/Changes) Guidance:					
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the					

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement					
Requirement ID	Requirement Title				
5.12.2	Aluminum Sheet				
Requirement Text Aluminum sheets used in the construction of the TFCS shall be Type 3003-H14 or Type 5052-H32 ASTM Designation B209 aluminum alloy.					
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to be made of materials that are resistant to rust and corrosion.					
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.2					
Requirement Text (Comments/Changes) Guidance:					
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement					
Requirement ID	Requirement Title				
5.12.3	Aluminum Sheet Plating				
Requirement Text Aluminum sheeting used in the construction of the TFCS internal subassemblies shall be treated using clear chromate conversion per MIL-C-5541, Class 3.					
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to be made of materials that are resistant to rust and corrosion.					
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.2					
Requirement Text (Comments/Changes) Guidance:					
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.4		Stainless Steel Sheet		
Requirement Text Stainless steel sheets used in the construction of the TFCS shall be annealed or one-quarter-hard complying with ASTM Designation A666 for Type 304, Grades A or B, stainless steel sheet.				
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to be made of materials that are resistant to rust and corrosion.				
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.2				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.5		Cold Rolled Steel		
Requirement Text Cold rolled steel sheet, rod, bar, and extruded used in the construction of the TFCS shall be Type 1018/1020.				
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to be made of materials that are resistant to rust and corrosion.				
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.2				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.6		Cold Rolled Steel Plating		
Requirement Text Cold rolled steel used in the construction of the TFCS shall be plated using either cadmium plating meeting the requirements of Federal Specification QQ-P-416C, Type 2 Class I or zinc plating meeting the requirements of ASTM B633-85 Type II SC4.				
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to be made of materials that are resistant to rust and corrosion.				
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.2				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.7		Stainless Steel Hardware		
Requirement Text				
All bolts, nuts, washers, screws, hinges and hinge pins used in the construction of the TFCS shall be stainless steel unless otherwise specified.				
Justification for the Requirement:				
1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to be made of materials that are resistant to rust and corrosion.				
Source for Justification:				
1. UN ID 4.3.1.8 2. UN ID 4.3.10.2				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.8		Doors		
Requirement Text The TFCS cabinet designs shall include door(s) with appropriate door frames, seals, gaskets, strikers, stops and three-point locking mechanisms.				
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to include one or more doors to support easy user access to subsystems and assemblies.				
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.3				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2, Note 1, cabinet shell and packaging is not controlled by the standard. 2. Figure 3, Note 1, cabinet shell and packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement					
Requirement ID	Requirement Title				
5.12.9	Ventilation				
Requirement Text The TFCS cabinet housings shall be designed with thermostatic ventilation in the event that the internal cabinet temperature exceeds the pre-programmed temperature. Incoming air shall be filtered.					
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to protect its subsystems and assemblies from heat, wind, snow, dust and rain.					
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.4					
Requirement Text (Comments/Changes) Guidance:					
Related Design Elements 1. Figure 2, Note 1, housings and packaging is not controlled by the standard. 2. Figure 3, Note 1, housings and packaging is not controlled by the standard.					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the					

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement				
Requirement ID		Requirement Title		
5.12.10		Cabinet Thickness		
Requirement Text The TFCS cabinet housing shall be fabricated of 0.125 inch minimum thickness aluminum sheet.				
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to protect its subsystems and assemblies from heat, wind, snow, dust and rain.				
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.4				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2, Note 1, housing and packaging is not controlled by the standard. 2. Figure 3, Note 1, housing and packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.11		Rodent Resistance		
Requirement Text The TFCS cabinet housings shall be designed to reduce or eliminate any openings which might permit rodents or insects to enter the cabinet.				
Justification for the Requirement: 1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS to protect its subsystems and assemblies from small animals and insects.				
Source for Justification: 1. UN ID 4.3.1.8 2. UN ID 4.3.10.5				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2, Note 1, housing and packaging is not controlled by the standard. 2. Figure 3, Note 1, housing and packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.12		High Security Design		
Requirement Text				
The TFCS shall support a high-security cabinet design which permits the usage of 3/16" sheet aluminum for the cabinet housing.				
Justification for the Requirement:				
1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to be resistant to vandalism or theft.				
Source for Justification:				
1. UN ID 4.3.1.8 2. UN ID 4.3.10.6				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.13		Cabinet Surface Presentation		
Requirement Text				
The TFCS surfaces shall be prepared per Paragraph 5.12.13				
Justification for the Requirement:				
1. The user needs the TFCS to be constructed using quality standards for workmanship, electronic design and manufacturing. 2. The user needs the TFCS housing to have an external finish that is resistant to graffiti and bill posting.				
Source for Justification:				
1. UN ID 4.3.1.8 2. UN ID 4.3.10.7				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.1.26		Operating Ambient Temperature		
Requirement Text				
The TFCS's shall have an operating ambient temperature range from -37 degrees Celsius to +74 degrees Celsius.				
Justification for the Requirement:				
1. The user needs the TFCS to operate under extreme hot, cold and humid environmental conditions.				
Source for Justification:				
1. UN ID 4.3.1.9				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 11.5.4.1.1Ambient Temperature				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement					
Requirement ID		Requirement Title			
5.1.1.25		Storage Temperature Range			
Requirement Text					
The TFCS shall be capable of withstanding an ambient storage temperature range of -45 degrees Celsius to +85 degrees Celsius.					
Justification for the Requirement:					
1. The user needs the TFCS to operate under extreme hot, cold and humid environmental conditions.					
Source for Justification:					
1. UN ID 4.3.1.9					
Requirement Text (Comments/Changes)					
Guidance:					
Related Design Elements					
1. 11.5.4.1.1Ambient Temperature					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?		X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.1.27		Operational Relative Humidity		
Requirement Text				
The TFCS operating shall be capable of operation with a relative humidity shall not exceeding 95 percent non-condensing over the temperature range of -37 degrees Celsius to +74 degrees Celsius.				
Justification for the Requirement:				
1. The user needs the TFCS to operate under extreme hot, cold and humid environmental conditions.				
Source for Justification:				
1. UN ID 4.3.1.9				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 11.5.4.1.2Humidity				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement					
Requirement ID		Requirement Title			
5.1.35		Electronic Emissions			
Requirement Text The TFCS shall adhere to the Code of Federal Regulations, Title 47 – Telecommunication, Chapter 1 – Federal Communications Commission, Part 15.					
Justification for the Requirement: 1. The user needs the TFCS to have limited electronic emissions that cause radio frequency interference (RFI) and electromagnetic interference (EMI).					
Source for Justification: 1. UN ID 4.3.1.10					
Requirement Text (Comments/Changes) Guidance: Requires further investigation as to the FCC requirements for roadside electrical cabinet equipment. Need to define meaning of “adhere” in this context.					
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard. 3. 11.10Electronic Emissions					
Requirement Criteria		Yes		No	
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?		X		
Note: An answer of no requires a comment or change in the Comments/Change field of the ‘Requirement Text’ section above.					

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.34		Electrostatic Discharge (ESD)		
Requirement Text The TFCS shall adhere to applicable sections of International Electrotechnical Commission (IEC) Standard 61000-4-2:2008 “Electromagnetic compatibility (EMC) – Part 4-2: Testing and Measurement Techniques – Electrostatic Discharge Immunity Test.”				
Justification for the Requirement: 1. The user needs the TFCS to be resistant to electrostatic discharge (ESD).				
Source for Justification: 1. UN ID 4.3.1.19				
Requirement Text (Comments/Changes) Guidance: Investigate further, specifically, what was done in the DMS standards, what compliance level is required, and who certifies compliance to the standard. Also need to specify the term “adhere to” and whether installed or as an assembly out of the box. Need to be specific.				
Related Design Elements 1. Figure 2, Note 1, packaging is not controlled by the standard. 2. Figure 3, Note 1, packaging is not controlled by the standard. 3. 11.11Electrostatic Discharge				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the				

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement				
Requirement ID		Requirement Title		
5.1.31		Noise Level		
Requirement Text				
The TFCS shall have no element that emits an audible noise level exceeding a peak level of 55 dBA when measured with all doors open at a distance of one meter away from the TFCS.				
Justification for the Requirement:				
1. The user needs the TFCS to have limited audible noise.				
Source for Justification:				
1. UN ID 4.3.1.11				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2, Note 1, packaging is not controlled by the standard.				
2. Figure 3, Note 1, packaging is not controlled by the standard.				
Audible noise is attenuated of the enclosure packaging materials and door openings				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the				

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement				
Requirement ID		Requirement Title		
5.1.32		Shock		
Requirement Text				
The elements of the TFCS shall suffer neither permanent mechanical deformation nor any damage that renders the unit inoperable, when subjected to a shock of 10 g applied in each of three mutually perpendicular planes.				
Justification for the Requirement:				
1. The user needs the TFCS to withstand vibration and shock.				
Source for Justification:				
1. UN ID 4.3.1.12				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 11.9 Shock (Impact) Test (DAT)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.33		Vibration		
Requirement Text				
The elements of the TFCS shall continue their defined functions and maintain physical integrity when subjected to a vibration of 5 to 30 Hz up to 0.5 g applied in each of three mutually perpendicular planes.				
Justification for the Requirement:				
1. The user needs the TFCS to withstand vibration and shock.				
Source for Justification:				
1. UN ID 4.3.1.12				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 11.8 Vibration				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.38		No Sharp Edges		
Requirement Text				
The TFCS shall have all sharp edges and corners rounded and free of burrs.				
Justification for the Requirement:				
1. The user needs the TFCS to be safe for use by field personnel.				
Source for Justification:				
1. UN ID 4.3.1.15				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2, Note 1, packaging is not controlled by the standard.				
2. Figure 3, Note 1, packaging is not controlled by the standard.				
Finishing of the enclosure packaging materials				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.39		NEC Best Practices		
Requirement Text				
The TFCS shall adhere to the best practices of the National Electrical Code (NEC) for wire sizing, ampacity and user safety.				
Justification for the Requirement:				
1. The user needs the TFCS to be safe for use by field personnel.				
Source for Justification:				
1. UN ID 4.3.1.15				
Requirement Text (Comments/Changes)				
Guidance: Investigate verification methods. For example, inspection of wire sizes and materials or testing.				
Related Design Elements				
1. Section 12WIRE REQUIREMENTS				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.29		Power Factor		
Requirement Text				
The TFCS shall maintain an overall power factor (PF) of 0.8 or greater, as measured at the service entry, when operating from AC power and isolated from the Application Computer for any such measurement.				
Justification for the Requirement:				
1. The user needs the TFCS to be energy efficient.				
Source for Justification:				
1. UN ID 4.3.1.16				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.4.2AC Line Input				
2. 6.4.11.1General				
3. 6.4.12.1General				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.30		Power Supply Efficiency		
Requirement Text				
The TFCS's power supplies shall be designed to operate with an average efficiency of 80 percent or greater at load.				
Justification for the Requirement:				
1. The user needs the TFCS to be energy efficient.				
Source for Justification:				
1. UN ID 4.3.1.16				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.4.2AC Line Input				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.40		No Liquid Mercury		
Requirement Text				
The TFCS shall contain no elements that use liquid mercury.				
Justification for the Requirement:				
1. The user needs the TFCS to be designed so that it uses no devices containing liquid mercury.				
Source for Justification:				
1. UN ID 4.3.1.17				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 10.1Mercury				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID	Requirement Title			
5.5.5	High Density Inputs			
Requirement Text				
The TFCS shall provide a high-density input assembly capable of directly supporting 12 current-technology four-channel NEMA TS-2 detector modules using half-width front panels in each single-width slot.				
Justification for the Requirement:				
1. The user needs the TFCS to be of a design that reduces the time required for maintenance personnel to perform maintenance actions in the field.				
Source for Justification:				
1. UN ID 4.3.1.20				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2 "Typical Input Functionality" (Note 1)				
2. Figure 3 "Typical Input Functionality" (Note 1)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement	
Requirement ID	Requirement Title
5.7.1.7	Externally Accessible V and I
Requirement Text Load Switch voltage and current values shall be externally accessible	
Justification for the Requirement: 1. The user needs the TFCS to be designed to support diagnostic testing.	
Source for Justification: 1. UN ID 4.3.1.18	
Requirement Text (Comments/Changes) Guidance: Accessible not as test points, but rather accessible digital data representing the V I.	
Related Design Elements 1. 8.3.3Electrical 2. SB#1 / SB#2 Interface shall consist of the following interface links conforming to the requirements of the Electronic Industries Association EIA-485 Standard for Electrical Characteristics of Generators and Receivers for use in Balanced Digital Multipoint Systems, dated April 1983: • SB#1 of two interface links (SB#1 TXD ± and SB#1 RXD ±) • SB#2 of two interface links (SB#2 TXD ± and SB#2 RXD ±) • LINE SYNC of one interface link (LINE SYNC ±) • NRESET of one interface link (NRESET ±) • POWER DOWN of one interface link (POWERDOWN ±) • Where differences occur between the EIA-485 standard and this document, this document shall govern •	

- SB#1 / SB#2 Interface shall also include the following power conductors:
- +5 VDC ISO positive bias for Terminator Unit depicted in
- Figure 5

ISO GND negative bias for Terminator Unit depicted in
Figure 5

EQ GND Equipment Ground

3. SB#3 Interface

4. Table 66: SB#3 Type 129 Response

Requirement Criteria		Yes		No	
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?			X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement				
Requirement ID		Requirement Title		
5.13.1		Diagnostic Display Local Display		
Requirement Text				
The TFCS shall contain a Diagnostic Display Unit (DDU) which supports local display of both historical and current cabinet status and log data collected by the monitoring subsystem.				
Justification for the Requirement:				
1. The user needs the TFCS to be of a design that reduces the time required for maintenance personnel to perform maintenance actions in the field.				
Source for Justification:				
1. UN ID 4.3.1.20				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.5Model 2220 Auxiliary Display Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.10		Utilizes a Monitor Assembly		
Requirement Text				
The TFCS shall utilize a monitor assembly (MA) to monitor the operation of the TFCS and provide diagnostic information.				
Justification for the Requirement:				
1. The user needs the TFCS to be of a design that reduces the time required for maintenance personnel to perform maintenance actions in the field.				
Source for Justification:				
1. UN ID 4.3.1.20				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.3Model 2212 Cabinet Monitor Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.2		Commonly-Deployed Input Devices		
Requirement Text The TFCS's Input Assembly shall be compatible with field input devices that are commonly deployed such as inductive loops, pedestrian detectors, vehicle preemption devices, and transit signal priority devices that utilize a "contact closure" type of interface.				
Justification for the Requirement: 1. The user needs the TFCS to support the use of commonly deployed field sensors.				
Source for Justification: 1. UN ID 4.3.2.1				
Requirement Text (Comments/Changes) Guidance: Sensors external to the cabinet				
Related Design Elements 1. 6.6Sensor Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID	Requirement Title			
5.3.3	NEMA TS 2 Four-Channel Detector Modules			
Requirement Text The TFCS's Input Assembly shall be able to utilize current-technology four-channel NEMA TS 2 detector modules with half-width front panels in each single-width slot.				
Justification for the Requirement: 1. The user needs the TFCS to support the use of commonly deployed field sensors. 2. The user needs the TFCS to support input devices that have higher channel density than commonly deployed field input devices.				
Source for Justification: 1. UN ID 4.3.2.1 2. UN ID 4.3.2.2				
Requirement Text (Comments/Changes) Guidance: See NEMA TS 2-2003, including output and status for each channel.				
Related Design Elements 1. 6.6Sensor Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.4		Four-Channel Isolator Modules		
Requirement Text				
The TFCS input assembly shall provide an option for the use of four-channel isolator modules.				
Justification for the Requirement:				
1. The user needs the TFCS to support the use of commonly deployed field sensors.				
Source for Justification:				
1. UN ID 4.3.2.1				
Requirement Text (Comments/Changes)				
Guidance: See NEMA TS 2-2003				
Related Design Elements				
1. 6.6Sensor Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.2.1.1		Controller Subsystem Communications Interfaces		
Requirement Text				
The TFCS's Controller Subsystem shall receive field input data from the Input Assembly or Input/Output Assembly via the serial bus.				
Justification for the Requirement:				
1. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard. 2. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification:				
1. UN ID 4.3.3.1 2. UN ID 4.3.11.2				
Requirement Text (Comments/Changes)				
Guidance: Broken out from compound requirement 5.2				
Related Design Elements				
1. 9.1.6.4Poll Raw Inputs Data				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.2.1.2		Receives Detector Status Information		
Requirement Text				
The TFCS's Controller Subsystem shall receive detector status information from the input assembly or Input/Output Assembly via the serial bus, at a minimum open, shorted and changes in inductance by more than 25%.				
Justification for the Requirement:				
1. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard. 2. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification:				
1. UN ID 4.3.3.1 2. UN ID 4.3.11.2				
Requirement Text (Comments/Changes)				
Guidance: Broken out from compound requirement 5.2				
Related Design Elements				
1. Table 20 shows the SIU pins for detector Status that is reported to the CU via SB#1.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.2.1.3		Sends Field Output Commands		
Requirement Text				
The TFCS's Controller Subsystem shall send field output commands to the output assembly or Input/Output Assembly.				
Justification for the Requirement:				
1. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard. 2. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification:				
1. UN ID 4.3.3.1 2. UN ID 4.3.11.2				
Requirement Text (Comments/Changes)				
Guidance: Broken out from compound requirement 5.2				
Related Design Elements				
1. Table 46: Set Outputs Command				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.2.1.4		Sends Switch Pack Driver Information to the Monitor Assembly		
Requirement Text The TFCS's Controller Subsystem shall send the switch pack driver information (the state of the switch packs commanded by the Controller Subsystem) to the Monitor Assembly.				
Justification for the Requirement: 1. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard. 2. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification: 1. UN ID 4.3.3.1 2. UN ID 4.3.11.2				
Requirement Text (Comments/Changes) Guidance: Broken out from compound requirement 5.2				
Related Design Elements 1. Table 54: CMU Type 67 Command				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.2.1.5		Sends Flash Command to the Monitor Assembly		
Requirement Text				
The TFCS's Controller Subsystem shall send a flash command to the Monitor Assembly.				
Justification for the Requirement:				
1. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard. 2. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification:				
1. UN ID 4.3.3.1 2. UN ID 4.3.11.2				
Requirement Text (Comments/Changes)				
Guidance: Broken out from compound requirement 5.2				
Related Design Elements				
1. Table 52 Type 62 Command				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement					
Requirement ID		Requirement Title			
5.2.1.6		Receives CMU Programming Information from Monitor Assembly			
Requirement Text The TFCS's Controller Subsystem Monitor Subsystem shall receive CMU programming information from the Monitor Assembly.					
Justification for the Requirement: 1. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard. 2. The user needs the TFCS to provide uniform internal interfaces and protocols.					
Source for Justification: 1. UN ID 4.3.3.1 2. UN ID 4.3.11.2					
Requirement Text (Comments/Changes) Guidance: Broken out from compound requirement 5.2					
Related Design Elements 1. Table 54: CMU Type 67 Command					
Requirement Criteria		Yes		No	
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?			X	
Note: An answer of no requires a comment or change in the					

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement				
Requirement ID		Requirement Title		
5.2.1.7		Signal Control Priority		
Requirement Text				
The TFCS shall provide an interface for signal control priority.				
Justification for the Requirement:				
3. The user needs the TFCS to have an application computer that supports the cabinet interfaces defined in this standard. 4. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification:				
3. UN ID 4.3.3.1 4. UN ID 4.3.11.2				
Requirement Text (Comments/Changes)				
Guidance: Broken out from compound requirement 5.2				
Related Design Elements				
1. 6.6.1General provides an interface for Preemptor or Isolator for Priority input.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement			
Requirement ID		Requirement Title	
5.2.3		Intersection Control Applications	
Requirement Text The TFCS Controller Subsystem shall support Intersection Control Applications developed for the ITS Cabinet V1 without modifications.			
Justification for the Requirement: 1. The user needs the TFCS to have configurations suitable for intersection control applications.			
Source for Justification: 1. UN ID 4.3.3.2			
Requirement Text (Comments/Changes) Guidance:			
Related Design Elements 1. 6.1Model 2218 Serial Interface Unit (SIU) 2. 6.2Model 2202 High Density Switch Pack / Flasher Unit (HDSP/FU) 3. 6.2.3.1Field Signal Control Inputs 4. 6.2.6.5Field Signal Control Input 5. 6.3Model 2212 Cabinet Monitor Unit 6. 6.6Sensor Unit 7. 8.3.1.1Serial Bus 1 8. 8.7.2.2Flash Programming Blocks 9. 8.7.2.3Flash Transfer Relay 10. 8.7.3Field Wire Sense 11. 8.7.5Field Wire Interface			
Requirement Criteria		Yes	No
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	
3	Is the requirement unambiguous?	X	
4	Is the requirement feasible?	X	

5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?			X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement			
Requirement ID		Requirement Title	
5.2.4		Ramp Meter Control Applications	
Requirement Text The TFCS Controller Subsystem shall support Ramp Meter Applications developed for the ITS Cabinet V1 without modifications.			
Justification for the Requirement: 1. The user needs the TFCS to have configurations suitable for ramp metering applications.			
Source for Justification: 1. UN ID 4.3.3.3			
Requirement Text (Comments/Changes) Guidance:			
Related Design Elements 1. 6.1Model 2218 Serial Interface Unit (SIU) 2. 6.2Model 2202 High Density Switch Pack / Flasher Unit (HDSP/FU) 3. 6.2.3.1Field Signal Control Inputs 4. 6.2.6.5Field Signal Control Input 5. 6.3Model 2212 Cabinet Monitor Unit 6. 6.6Sensor Unit 7. 8.3.1.1Serial Bus 1 8. 8.7.2.2Flash Programming Blocks 9. 8.7.2.3Flash Transfer Relay 10. 8.7.3Field Wire Sense 11. 8.7.5Field Wire Interface			
Requirement Criteria		Yes	No
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	
3	Is the requirement unambiguous?	X	
4	Is the requirement feasible?	X	

5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?			X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.1.1		Sends Field Input Data		
Requirement Text The TFCS's Input Assembly shall send field input data to the Controller Subsystem via the Serial Bus.				
Justification for the Requirement: 1. The user needs the TFCS to have configurations suitable for roadside data collection applications.				
Source for Justification: 1. UN ID 4.3.3.4				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. 9.1.6.4Poll Raw Inputs Data 2. 9.1.6.5Poll Filtered Input Data				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.3.1.2		Sends Detector Status Information		
Requirement Text				
The TFCS's Input Assembly shall send detector status information to the Controller Subsystem via the Serial Bus.				
Justification for the Requirement:				
1. The user needs the TFCS to have configurations suitable for roadside data collection applications.				
Source for Justification:				
1. UN ID 4.3.3.4				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2, Note 2				
2. Figure 3, Note 2				
3. 6.1.1General				
4. Table 20: INPUT SIU 1 to IN SLOT Signal Mapping				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.4.3.1		Two Output Channels per HDSP120		
Requirement Text				
The HDSP120 shall have two output channels.				
Justification for the Requirement:				
1. The user needs the TFCS to support the use of output devices that have higher channel density than the commonly deployed field output devices.				
Source for Justification:				
1. UN ID 4.3.4.1				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.2Model 2202 High Density Switch Pack / Flasher Unit (HDSP/FU)				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.4.10		Two Output Channels per HDSP120		
Requirement Text				
TFCS shall support the Six Pack and Twelve Pack regular density output files of the ITS Cabinet V1.				
Justification for the Requirement:				
1. The user needs the TFCS to support the use of output devices that have higher channel density than the commonly deployed field output devices.				
Source for Justification:				
1. UN ID 4.3.4.1				
Requirement Text (Comments/Changes)				
Guidance: Requirement deleted during Walkthrough.				
Related Design Elements				
1. Requirement deleted during walkthrough				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.4.3.3		Amperes per HDSP120 Channel		
Requirement Text				
The SP120 shall be able to support output devices up to 2 Amps per channel.				
Justification for the Requirement:				
1. The user needs the TFCS to support the use of output devices that have higher channel density than the commonly deployed field output devices.				
Source for Justification:				
1. UN ID 4.3.4.1				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.2.4.1Field Signal Outputs				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.7.3		CMU Display Unit		
Requirement Text				
The TFCS shall contain a CMU Display Unit (CDU) which provides a user display for the CMU.				
Justification for the Requirement:				
1. The user needs the TFCS to provide a configurable capability to independently monitor each field display to ensure that the actual state of the field display matches the state of its corresponding application computer output, and indicates an anomaly when they do not match. 2. The user needs the TFCS to provide a display for the maintenance personnel to view the malfunction, anomaly, or status information. 3. The user needs the TFCS to provide system reports and logs that include the output of system monitoring.				
Source for Justification:				
1. UN ID 4.3.6.3 2. UN ID 4.3.6.5 3. UN ID 4.3.8.1				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.5Model 2220 Auxiliary Display Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement			
Requirement ID		Requirement Title	
5.4.4		Low Voltage Switch Pack Modules	
Requirement Text The following power limitations shall apply to the Low Power / Low Voltage cabinet option - Less than 30 Volts RMS - Less than 42 Volts Peak - Less than 60 Volts DC			
Justification for the Requirement: 1. The user needs the TFCS to provide for field wiring that is at voltage and current levels below those dangerous to humans.			
Source for Justification: 1. UN ID 4.3.4.2			
Requirement Text (Comments/Changes) Guidance:			
Related Design Elements 1. 6.2.2.2Field Signal Voltage Sense Inputs 2. 6.2.4.1Field Signal Outputs 3. 6.2.5.1.3LV+MAINS 4. 6.2.5.1.4LV+SIGNAL 5. 6.2.6.7.2Flasher Channel Status Thresholds (HDFU-LV mode) 6. 6.3.5.13.1HDFU Flasher Thresholds 7. 6.3.5.14CMU Power Failure 8. 6.3.6.22212-LV Field Signal Voltage Inputs 9. 6.3.7.3Main Contactor Secondary Status 10. 6.3.8.1MAINS Power 11. 6.3.10.1Power			
Requirement Criteria		Yes	No
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	
3	Is the requirement unambiguous?	X	

4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?		X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					
Verify the Requirement					
Requirement ID	Requirement Title				
5.6.3	Channel-Programmable Flasher Outputs				
Requirement Text					
The TFCS' Field Termination Assembly shall provide the means for selecting the specific flasher output (red, yellow, or no indication) used for each channel's display when in flashing operation.					
Justification for the Requirement:					
1. The user needs the TFCS to support configurable continued operation based on the specifics of the element failures.					
Source for Justification:					
1. UN ID 4.3.5					
Requirement Text (Comments/Changes)					
Guidance:					
Related Design Elements					
1. Figure 21: Termination Interface for 2 Switch Pack Channels					
2. 8.7.2.2Flash Programming Blocks					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			

4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?			X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					
Verify the Requirement					
Requirement ID	Requirement Title				
5.7.4.1	Four Fused Output Channels				
Requirement Text					
The TFCS's monitor assembly shall provide for a flasher unit with four fused output channels.					
Justification for the Requirement:					
1. The user needs the TFCS to support configurable continued operation based on the specifics of the element failures.					
Source for Justification:					
1. UN ID 4.3.5					
Requirement Text (Comments/Changes)					
Guidance:					
Related Design Elements					
1. 6.2.1General					
2. 6.3.5.13Flasher Unit Output Failed					
Standard states "shall be limited to..." meaning by a fuse or limiting device not specified by the standard, plus ability to identify failure.					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			

5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement				
Requirement ID		Requirement Title		
5.7.2		Cabinet Monitor Unit		
Requirement Text				
The TFCS' Monitor Assembly shall provide for a Cabinet Monitor Unit (CMU) with minimum functionality of ITS Cabinet V1 CMU				
Justification for the Requirement:				
1. The user needs the TFCS to provide a status monitoring capability. 2. The user needs the TFCS to provide a configurable field output monitoring capability. 3. The user needs the TFCS to be capable of a configurable response to each malfunction or anomaly detected as part of system monitoring. 4. The user needs the TFCS to have a non-volatile method of maintaining the system reports and logs.				
Source for Justification:				
1. UN ID 4.3.6.1 2. UN ID 4.3.6.2 3. UN ID 4.3.6.6 4. UN ID 4.3.8.3				
Requirement Text (Comments/Changes)				
Guidance: Monitors system, bus and specific conditions. Fault and report coverage must be at least the same as the V1 CMU.				
Related Design Elements				
1. 6.3Model 2212 Cabinet Monitor Unit				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the				

Comments/Change field of the 'Requirement Text' section above.
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Verify the Requirement				
Requirement ID		Requirement Title		
5.4.1.3		Send Voltage Levels to Monitor Assembly (AC Voltage)		
Requirement Text				
The TFCS's Output Assembly shall send the voltage level per output channel to the Monitor Assembly via the Monitor Bus.				
Justification for the Requirement:				
1. The user needs the TFCS to provide a configurable field output monitoring capability.				
Source for Justification:				
1. UN ID 4.3.6.2				
Requirement Text (Comments/Changes)				
Guidance: For both High Voltage and Low Voltage operation.				
Related Design Elements				
1. 9.3.8.1Voltage / Current Encoding				
2. 9.3.8.22202 HDSP/FU Status				
3. Table 66: SB#3 Type 129 Response				
4. Table 68: SB#3 Type 130 Response				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement	
Requirement ID	Requirement Title
5.1.3.1	Serial Bus Communications Architecture
Requirement Text The TFCS shall utilize a Serial Bus (SB) communications architecture for communication between the elements of the TFCS.	
Justification for the Requirement: 1. The user needs the TFCS to monitor internal communications. 2. The user needs the TFCS to have an internal communications capability that can be used to communicate between its subsystems and assemblies.	
Source for Justification: 1. UN ID 4.3.6.4 2. UN ID 4.3.11.1	
Requirement Text (Comments/Changes) Guidance: SB2 has a defined method to detect communications corruption and loss of communications.	
Related Design Elements 1. 8.3SB#1 / SB#2 Interface 2. 8.3.3Electrical 3. SB#1 / SB#2 Interface shall consist of the following interface links conforming to the requirements of the Electronic Industries Association EIA-485 Standard for Electrical Characteristics of Generators and Receivers for use in Balanced Digital Multipoint Systems, dated April 1983: • SB#1 of two interface links (SB#1 TXD ± and SB#1 RXD ±) • SB#2 of two interface links (SB#2 TXD ± and SB#2 RXD ±) • LINE SYNC of one interface link (LINE SYNC ±) • NRESET of one interface link (NRESET ±) • POWER DOWN of one interface link (POWERDOWN ±) • Where differences occur between the EIA-485 standard and this document, this document shall govern • • SB#1 / SB#2 Interface shall also include the following power conductors: • +5 VDC ISO positive bias for Terminator Unit depicted in • Figure 5 • ISO GND negative bias for Terminator Unit depicted in • Figure 5 • EQ GND Equipment Ground 4. SB#3 Interface	

Requirement Criteria		Yes	No
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	
3	Is the requirement unambiguous?	X	
4	Is the requirement feasible?	X	
5	Is the requirement verifiable?	X	
		Insp.	Anal. Test Demo.
6	If verifiable, by which method?	X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.			
Verify the Requirement			
Requirement ID	Requirement Title		
5.7.1.6	Ethernet Communications		
Requirement Text			
CMU shall have an Ethernet interface			
Justification for the Requirement:			
1. The user needs the TFCS to provide a display for the maintenance personnel to view the malfunction, anomaly, or status information. 2. The user needs the TFCS to be capable of providing system reports locally or to a remote system using a standardized interface and protocol.			
Source for Justification:			
1. UN ID 4.3.6.5 2. UN ID 4.3.8.2			
Requirement Text (Comments/Changes)			
Guidance:			
Related Design Elements			
1. 6.3.13Terminal Port			
Requirement Criteria		Yes	No
1	Is the justification/basis for the requirement clear and valid?	X	
2	Is the requirement well-formed?	X	

3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?			X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.					

Verify the Requirement				
Requirement ID	Requirement Title			
5.7.1.1	Receives Switch Pack Driver Information from the Controller Subsystem			
Requirement Text The TFCS's Monitor Assembly shall receive switch pack driver information from the Controller Subsystem.				
Justification for the Requirement: 1. The user needs the TFCS to be capable of a configurable response to each malfunction or anomaly detected as part of system monitoring.				
Source for Justification: 1. UN ID 4.3.6.6				
Requirement Text (Comments/Changes) Guidance: The state of the switch packs commanded by the controller subsystem.				
Related Design Elements 1. 9.1.7.3Type 67 Command – Switch Pack Drivers 2. Table 54: CMU Type 67 Command				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.9.7		Service Power Input Options		
Requirement Text				
The TFCS shall provide manufacturing options to operate on different nominal service power sources.				
Justification for the Requirement:				
1. The user needs the TFCS to be accept a nominal service power of 120 volts alternating current (VAC) at 60 hertz (Hz).				
Source for Justification:				
1. UN ID 4.3.1.14				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. 6.4Cabinet Power Supply Requirements, High Voltage Versions				
2. Figure 2: High Level Functional Block Diagram, High Voltage Version				
3. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.9.7.2		High Voltage Service Assembly		
Requirement Text				
The TFCS shall include a Service Assembly option for 120 VAC nominal service power input.				
Justification for the Requirement:				
1. The user needs the TFCS to be accept a nominal service power of 120 volts alternating current (VAC) at 60 hertz (Hz).				
Source for Justification:				
1. UN ID 4.3.1.14				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.10.1		Clean Power Input		
Requirement Text				
The TFCS' Cabinet Power Supply shall accept clean power from clean power bus assembly.				
Justification for the Requirement:				
1. The user needs the TFCS to distribute clean power within the cabinet, i.e., power that is protected against surges and spikes and is filtered to regulate electrical noise.				
Source for Justification:				
1. UN ID 4.3.7.1				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.47		24 VDC Output Power		
Requirement Text				
The TFCS shall distribute 24 VDC output power to subassemblies				
Justification for the Requirement:				
1. The user needs the TFCS to distribute output power externally to field displays.				
Source for Justification:				
1. UN ID 4.3.7.2				
Requirement Text (Comments/Changes)				
Guidance: 12 VDC output power is optional. Standard will include procurement guidance for users that have equipment powered by 12 VDC.				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version				
2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version				
3. 6.4.4Outputs				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.10.4		24 VDC Output Power		
Requirement Text				
The TFCS' Cabinet Power Supply shall provide 24 VDC for the elements of the TFCS.				
Justification for the Requirement:				
1. The user needs the TFCS to convert power for use by internal devices. .				
Source for Justification:				
1. UN ID 4.3.7.3				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version				
2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version				
3. 6.4.4Outputs				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID		Requirement Title		
5.10.5		12 VDC Output Power		
Requirement Text				
The TFCS' Cabinet Power Supply shall provide 12 VDC for the elements of the TFCS as an option				
Justification for the Requirement:				
1. The user needs the TFCS to convert power for use by internal devices. .				
Source for Justification:				
1. UN ID 4.3.7.3				
Requirement Text (Comments/Changes)				
Guidance: 12 VDC output power is optional per procurement specifications. Standard will include procurement guidance for users that have equipment powered by 12 VDC.				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version				
2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version				
3. 6.4.4Outputs				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.9.7.2		Low Voltage Service Assembly		
Requirement Text				
The TFCS shall include a Service Assembly option for 48 VDC nominal service power input.				
Justification for the Requirement:				
1. The user needs the TFCS to convert power for use by internal devices. .				
Source for Justification:				
1. UN ID 4.3.7.3				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.9.6.1		Backup Power Subsystem Interface		
Requirement Text				
The TFCS shall include standardized connector and signals from the standby source to the TFCS DC power source.				
Justification for the Requirement:				
1. The user needs the TFCS to convert power for use by internal devices. 2. The user needs the TFCS to provide a method for continued operations when there are service power interruptions.				
Source for Justification:				
1. UN ID 4.3.7.3 2. UN ID 4.3.7.4				
Requirement Text (Comments/Changes)				
Guidance: Broken out from 5.9.6 compound requirement				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version. 2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version.				
Standard controls the names and electrical characteristics of signals, but not the form factor of the backup system				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.9.6.2		Backup Power Subsystem Switchover		
Requirement Text				
The TFCS shall provide switching to and from standby power in the event of a main power failure.				
Justification for the Requirement:				
1. The user needs the TFCS to convert power for use by internal devices. 2. The user needs the TFCS to provide a method for continued operations when there are service power interruptions.				
Source for Justification:				
1. UN ID 4.3.7.3 2. UN ID 4.3.7.4				
Requirement Text (Comments/Changes)				
Guidance: Broken out from 5.9.6 compound requirement				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version. 2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version. Standard controls the names and electrical characteristics of signals, but not the form factor of the backup system.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test Demo.
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID	Requirement Title			
5.12.17	External Antenna			
Requirement Text The TFCS shall provide, for all types of cabinet housings, a method to mount an external antenna without invalidating the certification.				
Justification for the Requirement: 1. The user needs the TFCS to accommodate a variety of communications equipment and media for communications external to the system.				
Source for Justification: 1. UN ID 4.3.9.1				
Requirement Text (Comments/Changes) Guidance: Anticipate antenna mounting in the cabinet design before submitting for approval.				
Related Design Elements 1. Figure 2: High Level Functional Block Diagram, High Voltage Version 2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version Standard controls the names and electrical characteristics of signals, but not antenna mounting.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.1.17		Legacy Housing		
Requirement Text				
The TFCS shall have overall sizes compatible with legacy cabinet models 342, 346, and 340 from the ITS Cabinet V1 Standard.				
Justification for the Requirement:				
1. The user needs the TFCS housing to have mounting capabilities which accommodate existing transportation industry base, pedestal, and pole mountings.				
Source for Justification:				
1. UN ID 4.3.10.1				
Requirement Text (Comments/Changes)				
Guidance: Interpreted as mounting bolt hole pattern, and within legacy dimensions.				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version, Note 1				
2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version, Note 1				
EIA-310 dimensions are within the legacy dimensions of 342, 346 and 340 sizes of V1 standard				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement					
Requirement ID	Requirement Title				
5.1.18	Small Size Cabinet				
Requirement Text The TFCS shall be available in a single door model 306 small-sized cabinet.					
Justification for the Requirement: 1. The user needs the TFCS housing to have mounting capabilities which accommodate existing transportation industry base, pedestal, and pole mountings.					
Source for Justification: 1. UN ID 4.3.10.1					
Requirement Text (Comments/Changes) Guidance:					
Related Design Elements 1. Figure 2: High Level Functional Block Diagram, High Voltage Version, Note 1 2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version, Note 1 EIA-310 dimensions are within the legacy dimensions of 306 size.					
Requirement Criteria		Yes	No		
1	Is the justification/basis for the requirement clear and valid?	X			
2	Is the requirement well-formed?	X			
3	Is the requirement unambiguous?	X			
4	Is the requirement feasible?	X			
5	Is the requirement verifiable?	X			
		Insp.	Anal.	Test	Demo.
6	If verifiable, by which method?	X			
Note: An answer of no requires a comment or change in the					

Comments/Change field of the 'Requirement Text' section above.
--

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.14		Video Monitor		
Requirement Text				
The TFCS shall provide space and electrical power for a small video monitor as an option.				
Justification for the Requirement:				
1. The user needs the TFCS housing to provide space and electrical connections for a small video monitor.				
Source for Justification:				
1. UN ID 4.3.10.8				
Requirement Text (Comments/Changes)				
Guidance: Possibly handheld				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version, Note 1				
2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version, Note 1				
EIA-310 dimensions allow standard mounting of standard displays.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?		X	

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.15		Backup Power Source		
Requirement Text				
The TFCS design shall include a cabinet configuration that provides internal space for the housing of a backup power source.				
Justification for the Requirement:				
1. The user needs the TFCS to provide internal space for housing batteries commonly used to provide backup power.				
Source for Justification:				
1. UN ID 4.3.10.9				
Requirement Text (Comments/Changes)				
Guidance: Not all backup power sources are batteries. Requirement stated accordingly.				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version, Note 1				
2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version, Note 1				
EIA-310 dimensions allow mounting of standard backup sources. Mechanical configuration is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the				

Comments/Change field of the 'Requirement Text' section above.
--

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.16		Law Enforcement Access		
Requirement Text The TFCS shall provide protected access by law enforcement personnel to the following cabinet functions: • Signals ON/OFF • FLASH / AUTO operation • Manual Control Enable (MCE) • Interval Advance (IA)				
Justification for the Requirement: 1. The user needs the TFCS to provide law enforcement personnel limited access to the TFCS to the extent necessary for direct control of the field location while restricting access to the remainder of the cabinet.				
Source for Justification: 1. UN ID 4.3.10.10				
Requirement Text (Comments/Changes) Guidance:				
Related Design Elements 1. Figure 2: High Level Functional Block Diagram, High Voltage Version, Note 1 2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version, Note 1 3. 8.5.2Form Factor 4. Table 26: Definitions of Terms relating to switches in the Police Panel 5. Table 27: CDC Connector Pin Assignments				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		

Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.

Verify the Requirement				
Requirement ID		Requirement Title		
5.12.18		Limited Enclosure Access		
Requirement Text				
The TFCS shall provide doors with keylocks for installation, removal and service of major assemblies, plus a door with keylocks for access to police panel.				
Justification for the Requirement:				
1. The user needs the TFCS to provide law enforcement personnel limited access to the TFCS to the extent necessary for direct control of the field location while restricting access to the remainder of the cabinet.				
Source for Justification:				
1. UN ID 4.3.10.10				
Requirement Text (Comments/Changes)				
Guidance:				
Related Design Elements				
1. Figure 2: High Level Functional Block Diagram, High Voltage Version, Note 1				
2. Figure 3: High Level Design Functional Block Diagram, Low Voltage Version, Note 1				
EIA-310 dimensions allow mounting of standard backup sources. Mechanical configuration is not controlled by the standard.				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?	X		
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.4.1.1		Receives Field Output Commands		
Requirement Text				
The TFCS's Output Assembly shall send field output commands from the Controller Subsystem via the Serial Bus.				
Justification for the Requirement:				
1. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification:				
1. UN ID 4.3.11.2				
Requirement Text (Comments/Changes)				
Guidance: Broken out from compound requirement 5.4				
Related Design Elements				
1. 9.1.6.7Set Outputs				
2. Table 46: Set Outputs Command				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

Verify the Requirement				
Requirement ID		Requirement Title		
5.4.1.2		Send current levels to the monitor assembly		
Requirement Text				
The TFCS's Output Assembly shall send the level of electrical current per output channel to the Monitor Assembly via the Monitor Bus.				
Justification for the Requirement:				
1. The user needs the TFCS to provide uniform internal interfaces and protocols.				
Source for Justification:				
1. UN ID 4.3.11.2				
Requirement Text (Comments/Changes)				
Guidance: Broken out from compound requirement 5.4				
Related Design Elements				
1. 9.3.8.2.2Type 129 Response – HDSP Status				
2. Table 66: SB#3 Type 129 Response				
3. 9.3.8.2.4Type 130 Response – HDFU Status				
4. Table 68: SB#3 Type 130 Response				
Requirement Criteria		Yes		No
1	Is the justification/basis for the requirement clear and valid?	X		
2	Is the requirement well-formed?	X		
3	Is the requirement unambiguous?	X		
4	Is the requirement feasible?	X		
5	Is the requirement verifiable?	X		
		Insp.	Anal.	Test
6	If verifiable, by which method?			X
Note: An answer of no requires a comment or change in the Comments/Change field of the 'Requirement Text' section above.				

15 NON FUNCTIONAL REQUIREMENTS

To resolve conflicting Requirements, Table 78 of Non-Functional Requirements (NFR) of Cabinet Attributes was developed during the ITS Cabinet Workshop in November 2008. The resulting 22 ITS Cabinet Attributes were included in a survey to rank the importance when resolving conflicting Requirements.

- All of the attributes were considered important but those higher in rank were generally considered more important when compared to those lower in rank.
- On a given project, an attribute that is lower in rank may actually be the most critical to the project such as "external dimensions" for a CBD (central business district) deployment.
- The ranking shown for all of the respondents was not significantly different from the ranking when only voting members of the WG were considered.
- This is not a claim on being representative of the entire industry (note that there were 22 respondents) but it is representative of the ITS Cabinet WG.

Table 78: ATCC Cabinet V2 Attribute Ranking

Rank	Attribute	Definition
1	Reliability	Consistency and stability of performance
2	Operational Safety	Failsafe operation of the cabinet
3	Maintainability	Easy to perform maintenance on the cabinet system
4	Technician Safety	Design is such that the safety of the technician is the priority
5	Interchangeability	Can mix units/components from different vendors
6	Ease of Use	Easy for humans to use the cabinet system
7	Testability	Cabinet system is easy to test
8	Life Expectancy	Cabinet system is designed for long life
9	Initial / Intrinsic Cost	The cost of the cabinet system
10	Modularity	Internal structure is such that there is separation in the functions of the cabinet system and flexibility in the way they are combined
11	Application Scalability	The cabinet system makes sense for simple operations as well as complex ones
12	Expandability	The cabinet system's capability can be expanded
13	Product Life Cost	Anticipated costs over the life of the cabinet system
14	Availability	Ease of acquisition of the components/technology today
15	Versatility	The cabinet system can be used in a variety of ways or modes
16	Physical Security	Intrusion prevention from unauthorized access
17	Ease of Specification	Cabinet system can easily be specified by users
18	Manufacturability	Ease at which the design can be produced
19	Communications Security	Communications within the cabinet design are designed with security in mind
20	Backwards Compatibility	Support for legacy issues / equipment
21	External Dimensions	The size of the cabinet system should be small
22	Internal Dimensions /	Components of the cabinet system should be small

	Compactness	
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16 ATCC POWER SIGNAL NAMING CONVENTIONS

ATCC Power Signal Naming Conventions v3 posted to the ATC Cabinet WG Community on 10/02/2017.

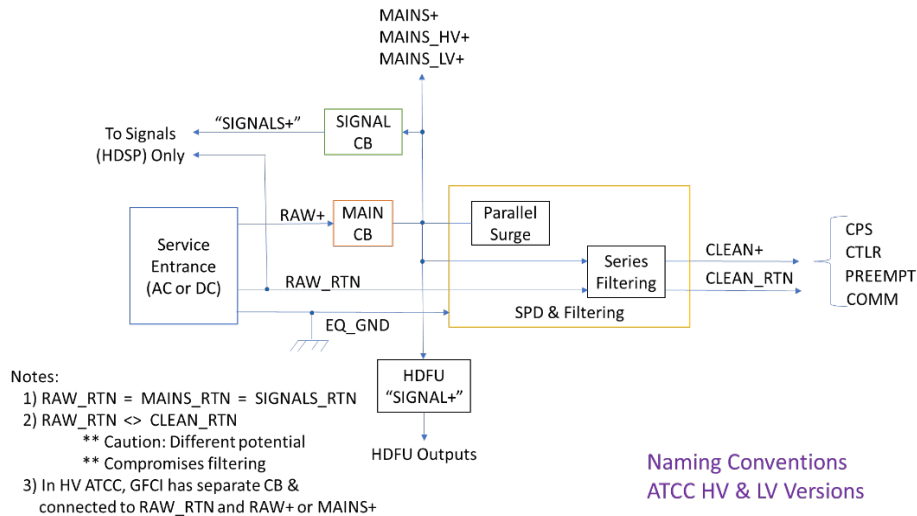


Figure 23: Naming Conventions: ATCC HV & LV Versions

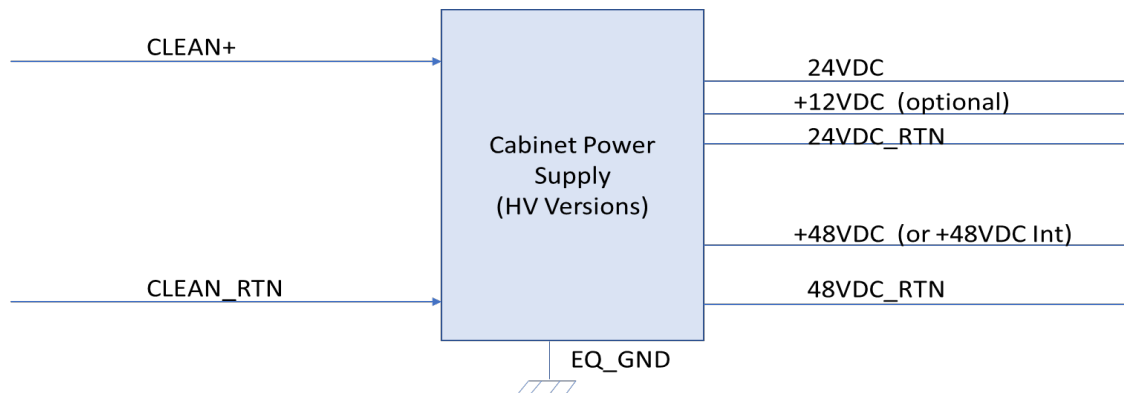
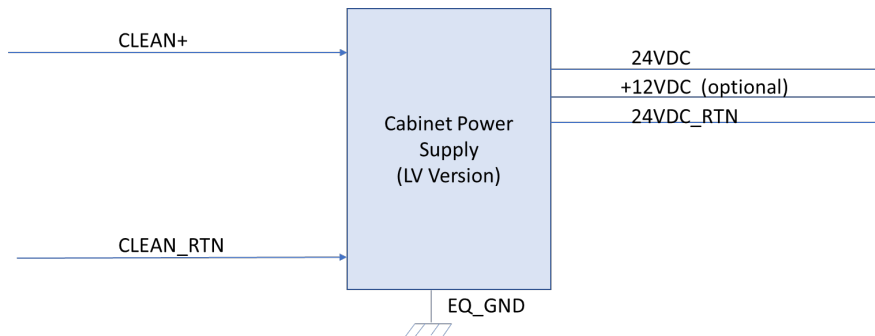


Figure 24: Naming Conventions: ATCC HV Version

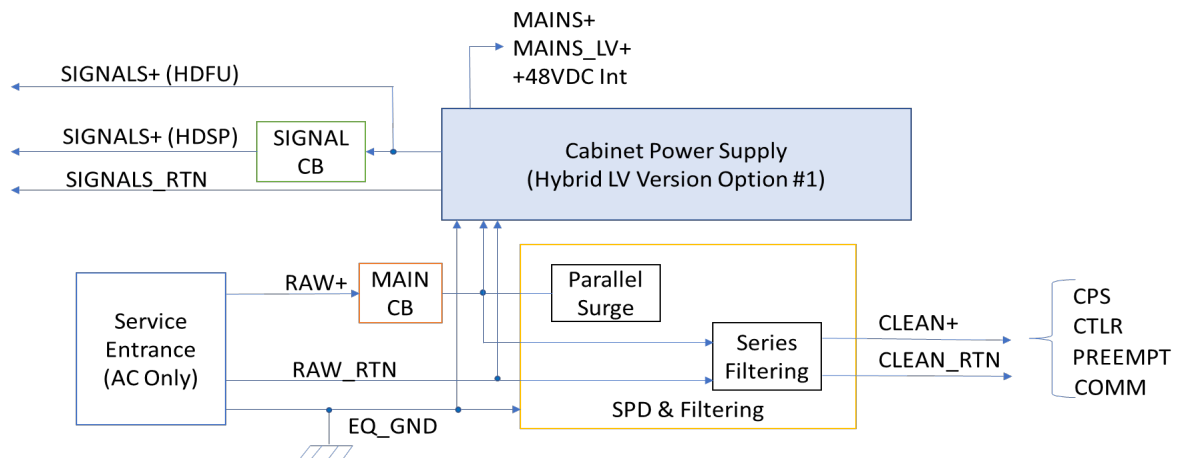


Notes:

- 1) Remember CLEAN+/CLEAN_RTN is 48Vdc
- 2) 24VDC_RTN = 24/12VDC_RTN
- 3) In a LV ATCC, 48VDC for the Police Panel, optional ADU, etc could be provided by MAINS+/RAW_RTN or CLEAN+/CLEAN_RTN
 ** Caution: CLEAN_RTN <> RAW_RTN since not at same potential
- 4) Controller is powered by 48VDC ... issues with no LINESYNC available

Naming Conventions
ATCC LV Version

Figure 25: Naming Conventions: ATCC LV Version

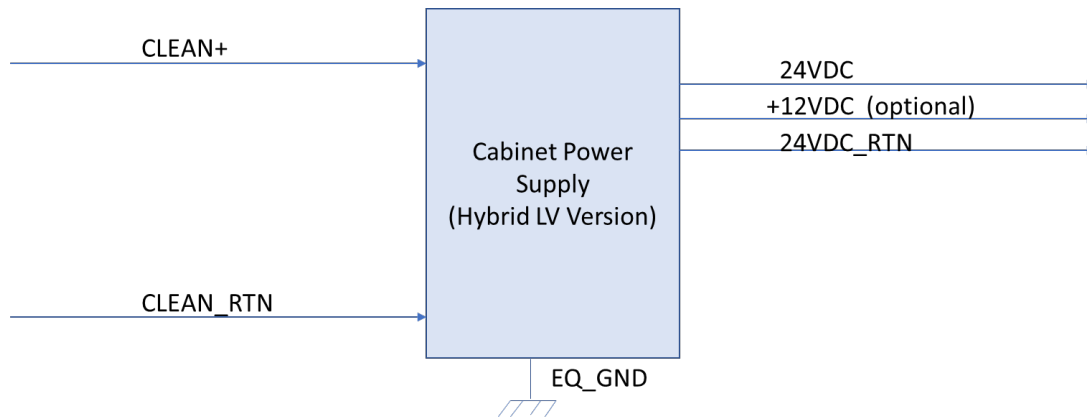


Notes:

- 1) RAW_RTN <> SIGNALS_RTN; SIGNALS_RTN = MAINS_RTN; SIGNALS+(HDFU) <> MAINS+
- 2) RAW_RTN <> CLEAN_RTN
 ** Caution: Different potential, Compromises filtering
- 3) GFCI has separate CB & connected to RAW+/RTN (or after MAIN CB)
- 4) Remember CLEAN+/RTN is 120 Vac

Naming Conventions
ATCC Hybrid LV Version
Option #1
(Page 1 of 2)

Figure 26: Naming Conventions: ATCC Hybrid LV Version, Option #1, Page 1

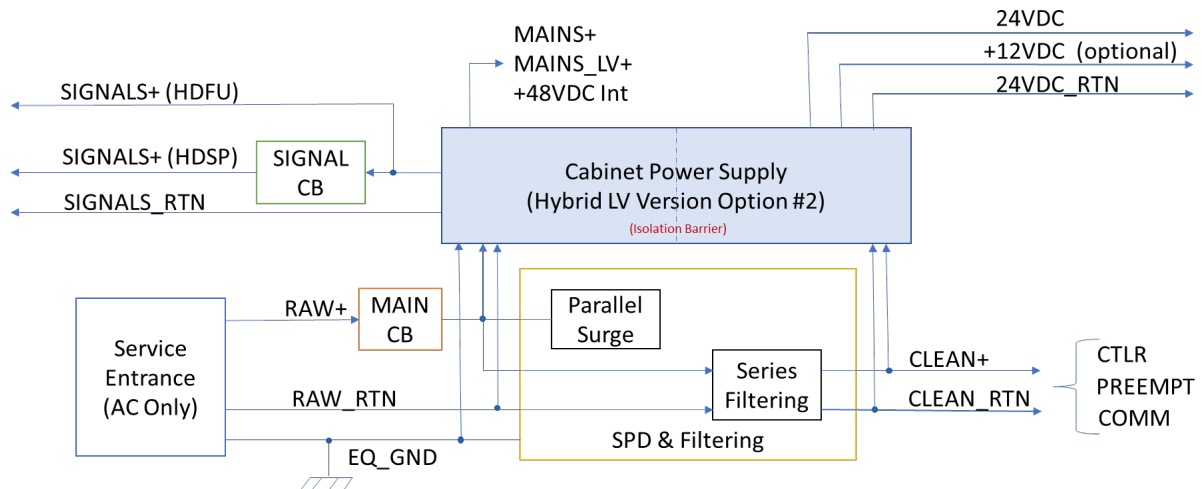


Notes:

- 1) Remember CLEAN_CPS+/CLEAN_RTN is 120 Vac
- 2) 24VDC_RTN = 24/12VDC_RTN
- 3) In Hybrid LV #1, 48VDC for the Police Panel, optional ADU, etc is provided by +48VDC Int/SIGNAL-
- ** Caution: CLEAN_RTN <> RAW_RTN since not at same potential

Naming Conventions
ATCC Hybrid LV Version
Option #1
(page 2 of 2)

Figure 27: Naming Conventions: ATCC Hybrid LV Version, Option #1, Page 2



Notes:

- 1) RAW_RTN <> SIGNALS_RTN; SIGNAL_RTN = MAINS_RTN; SIGNALS+ <> MAINS+
- 2) RAW_RTN <> CLEAN_RTN
- ** Caution: Different potential, Compromises filtering
- 3) GFCI has separate CB & connected to RAW+/RTN (or after MAIN CB)
- 4) 24VDC_RTN = 24/12VDC_RTN
- 5) Controller, Preempt, Comm still have clean AC power

Naming Conventions
ATCC Hybrid LV Version
Option #2
(page 1 of 1)

Figure 28: Naming Conventions: ATCC Hybrid LV Version, Option #2

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